

Artificial Neural Networks

Plan of Attack

Plan of Attack

What we will learn in this section:

- The Neuron
- The Activation Function
- How do Neural Networks work? (example)
- How do Neural Networks learn?
- Gradient Descent
- Stochastic Gradient Descent
- Backpropagation

The Neuron

The Neuron

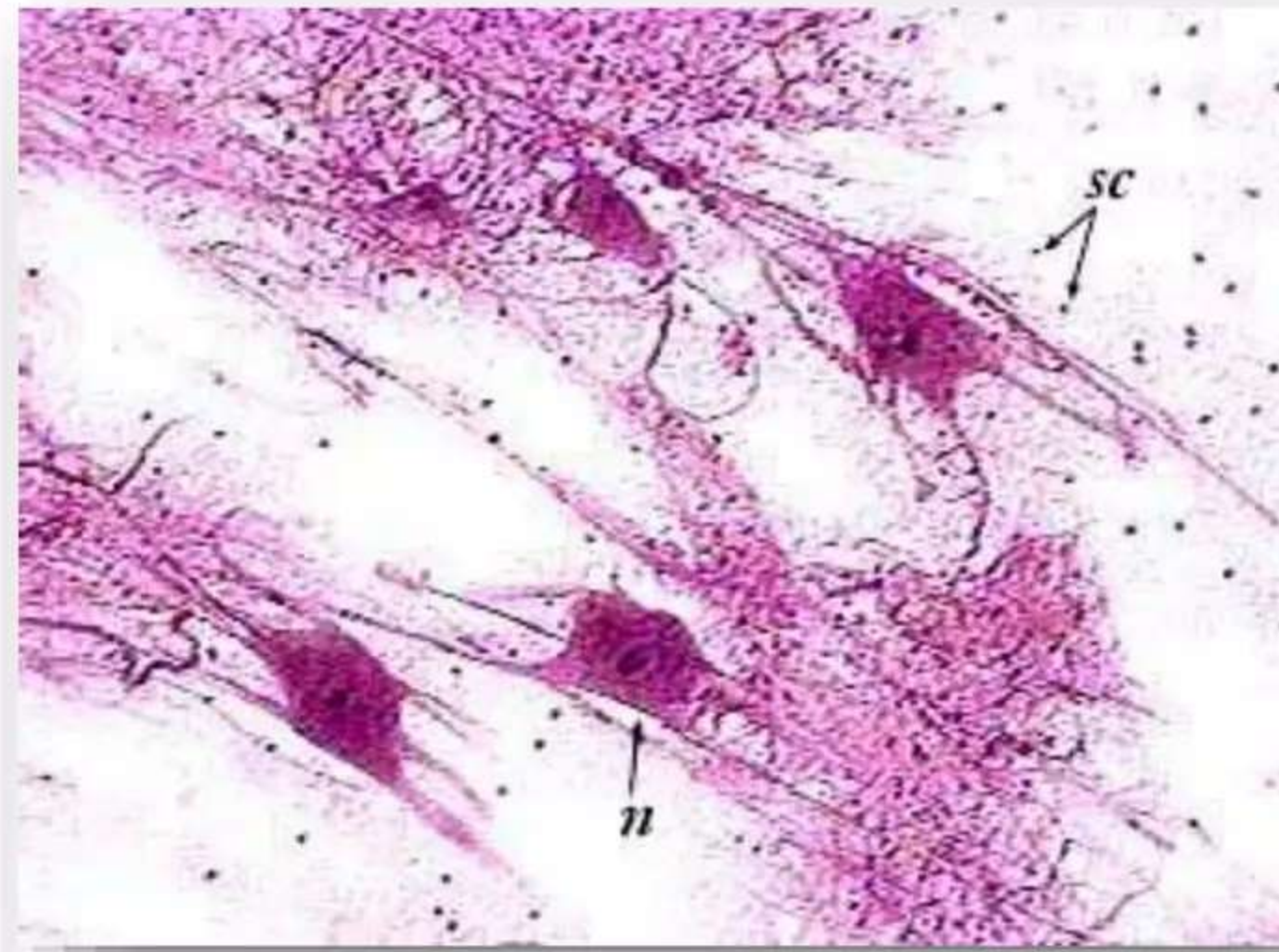
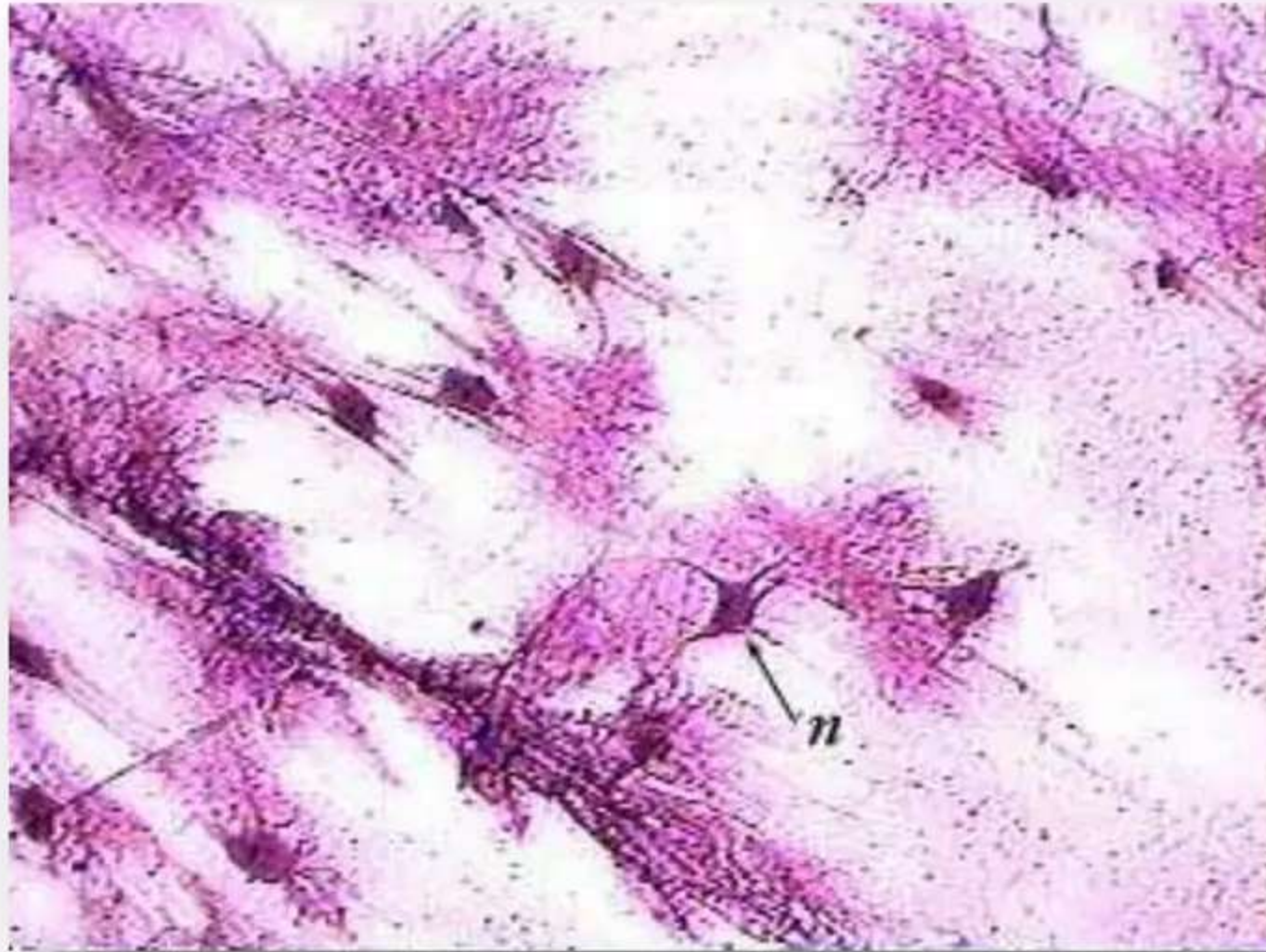


Image Source: www.austincc.edu

The Neuron

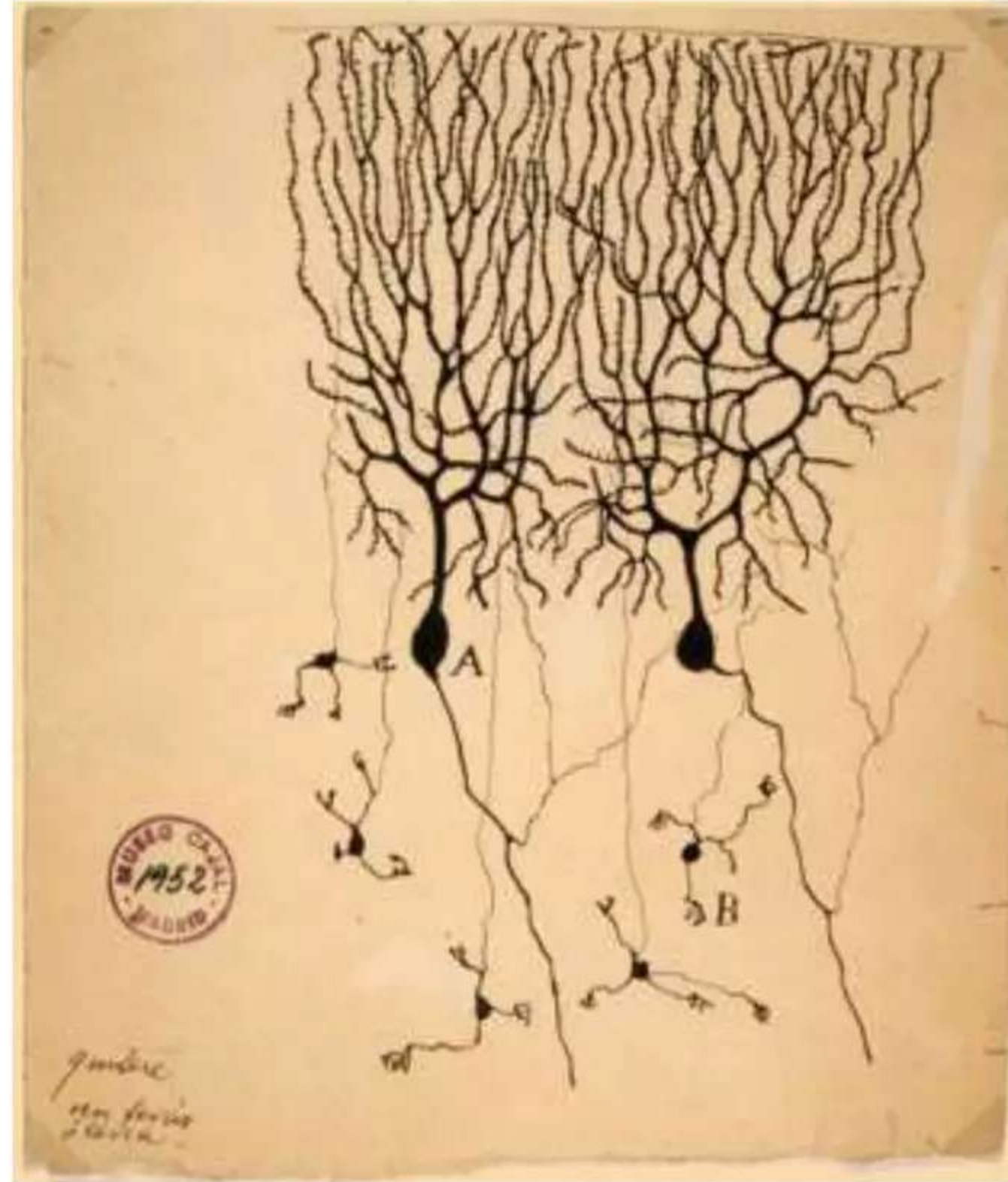


Image Source: Wikipedia

The Neuron

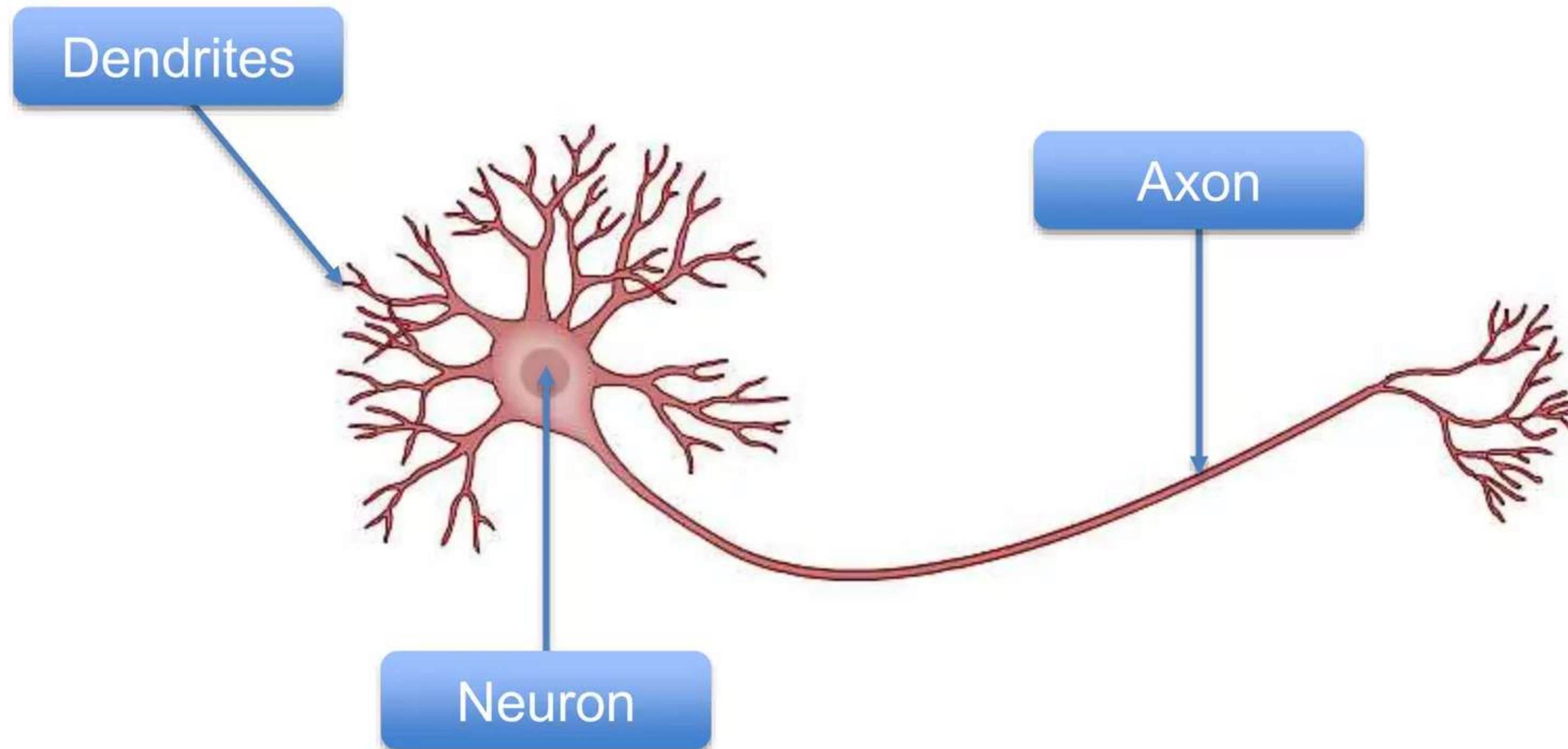


Image Source: Wikipedia

The Neuron

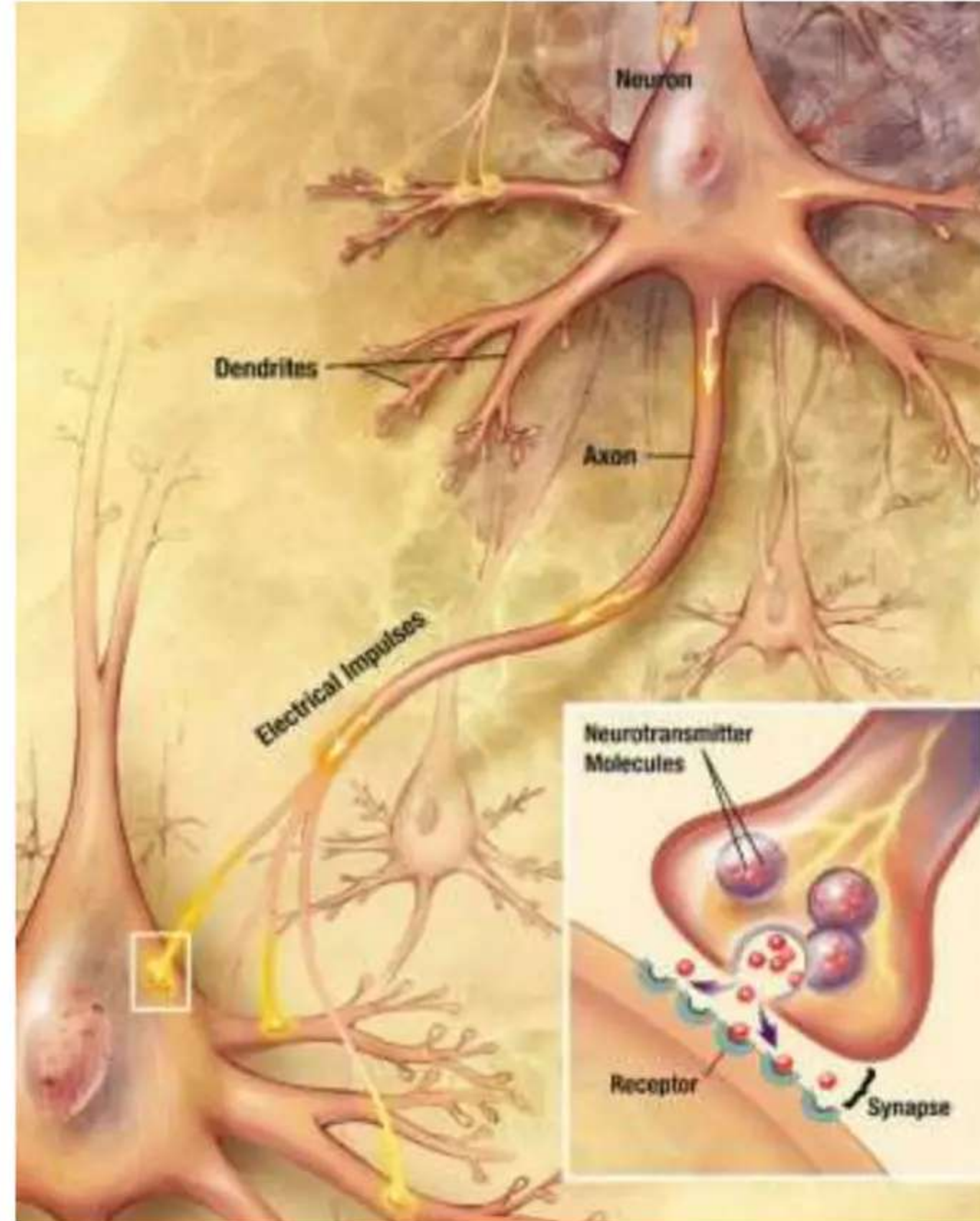
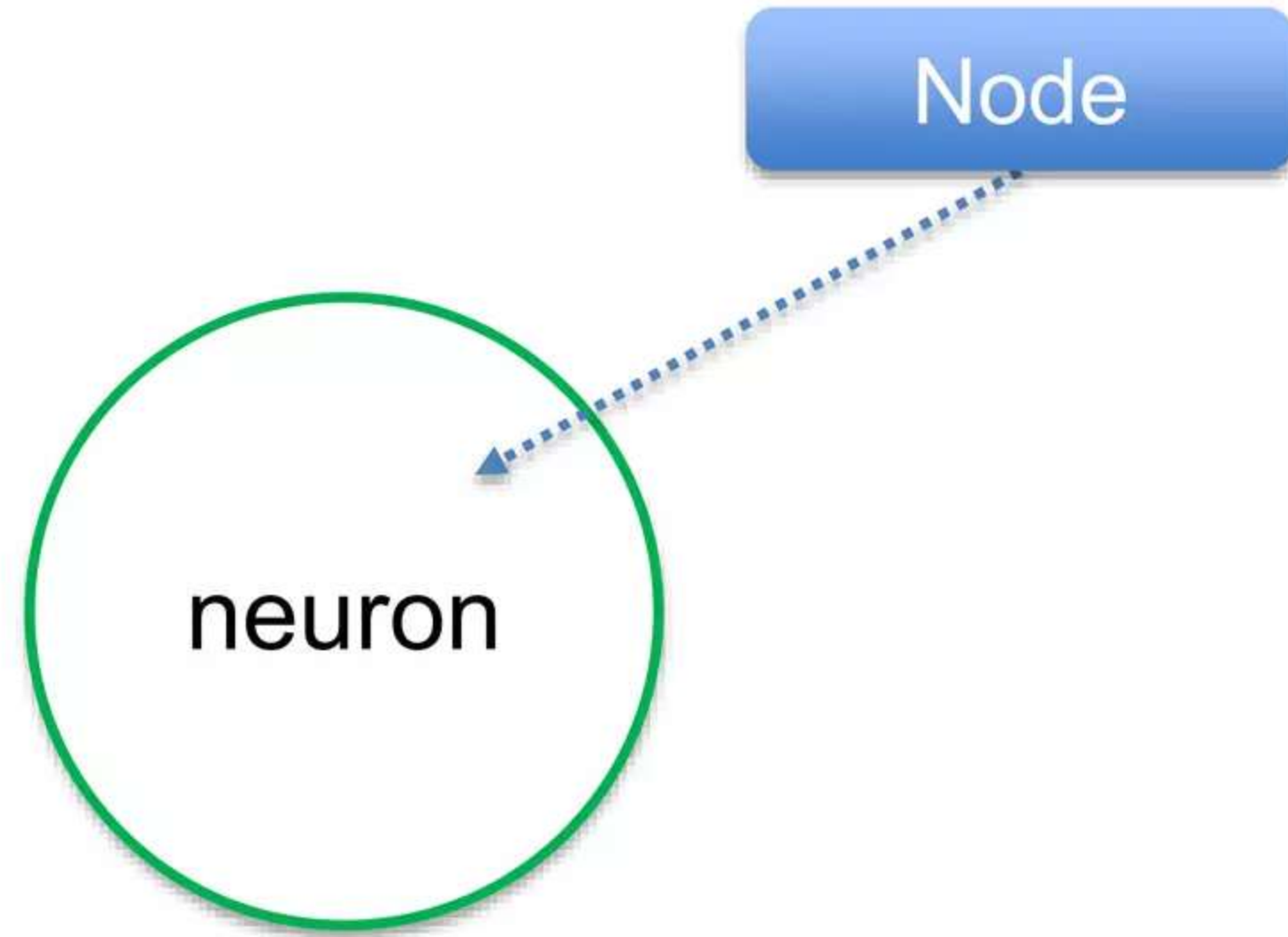
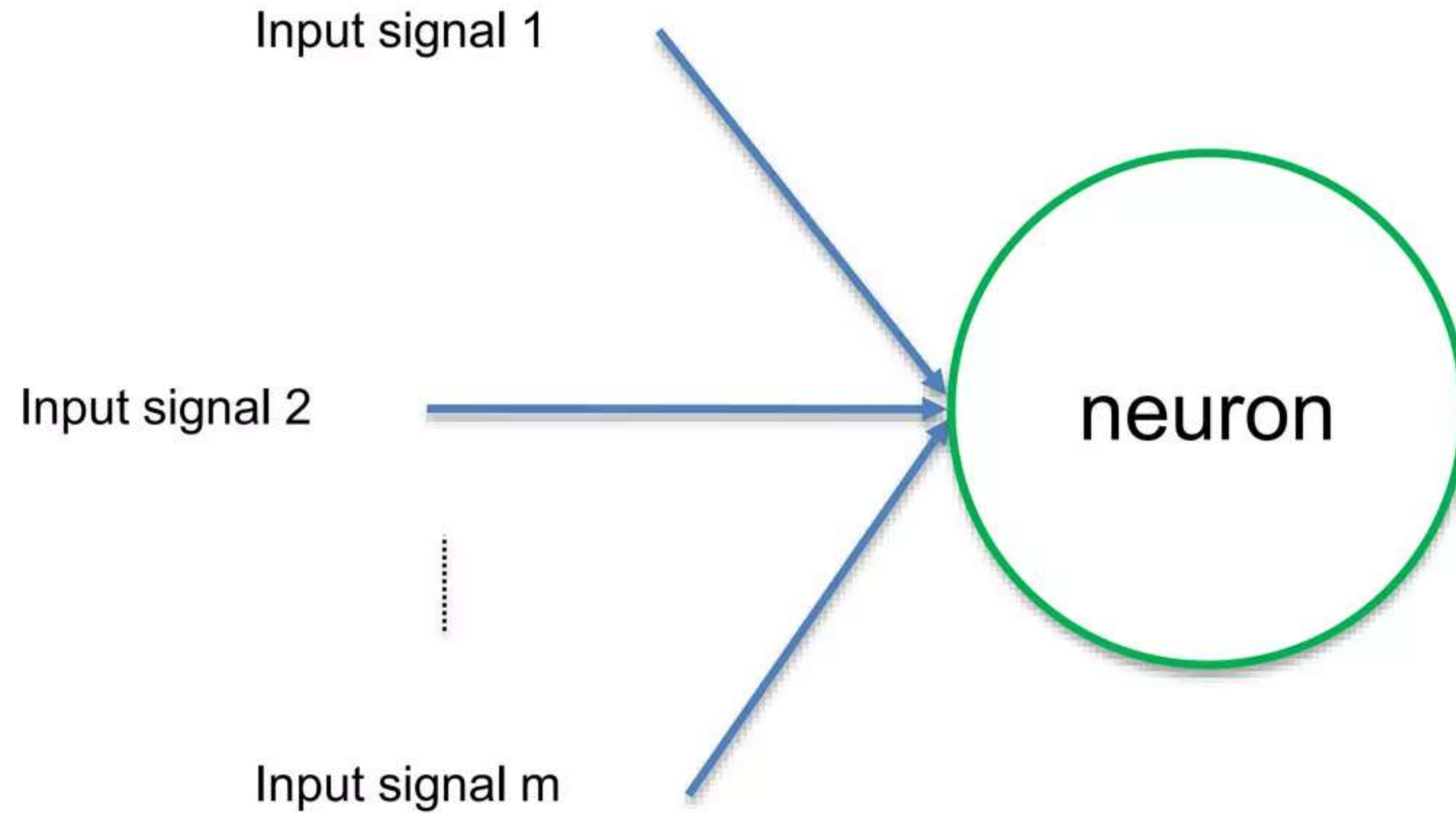


Image Source: Wikipedia

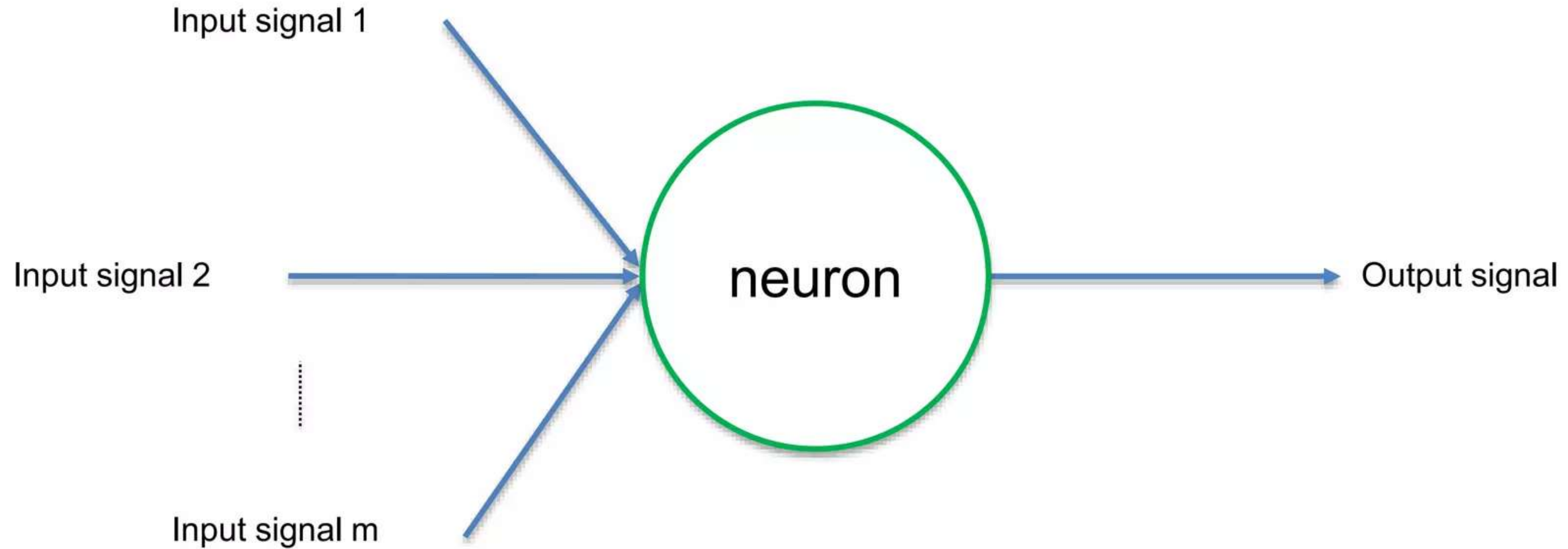
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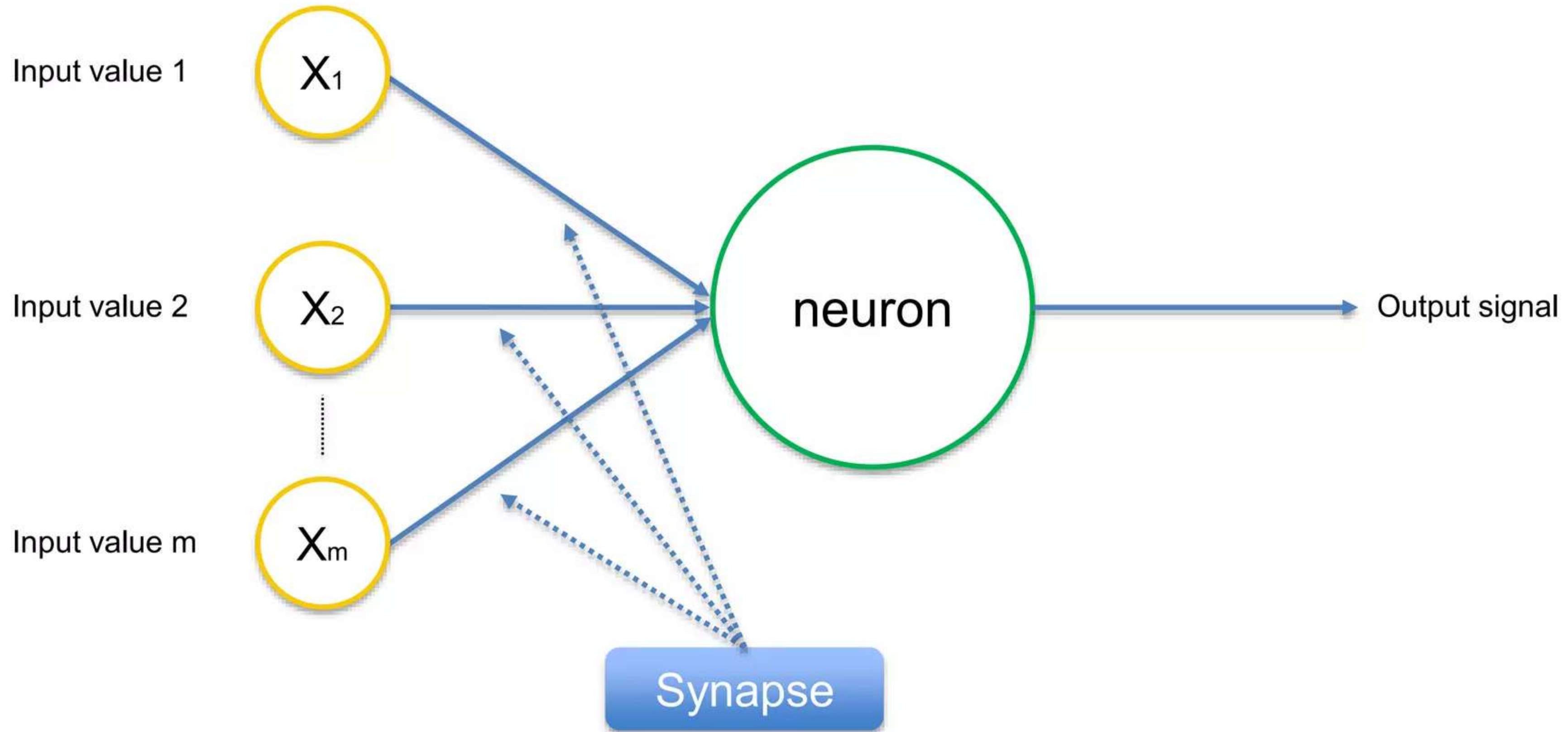
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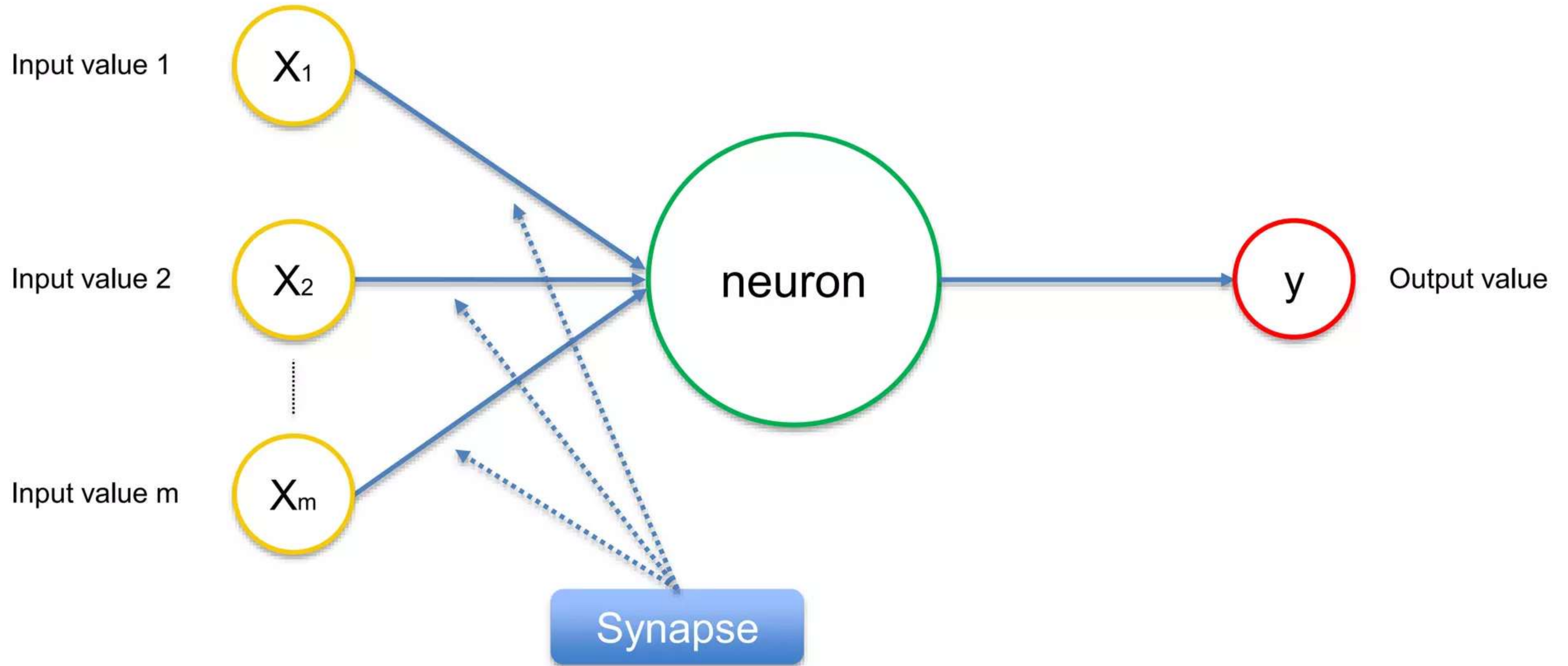
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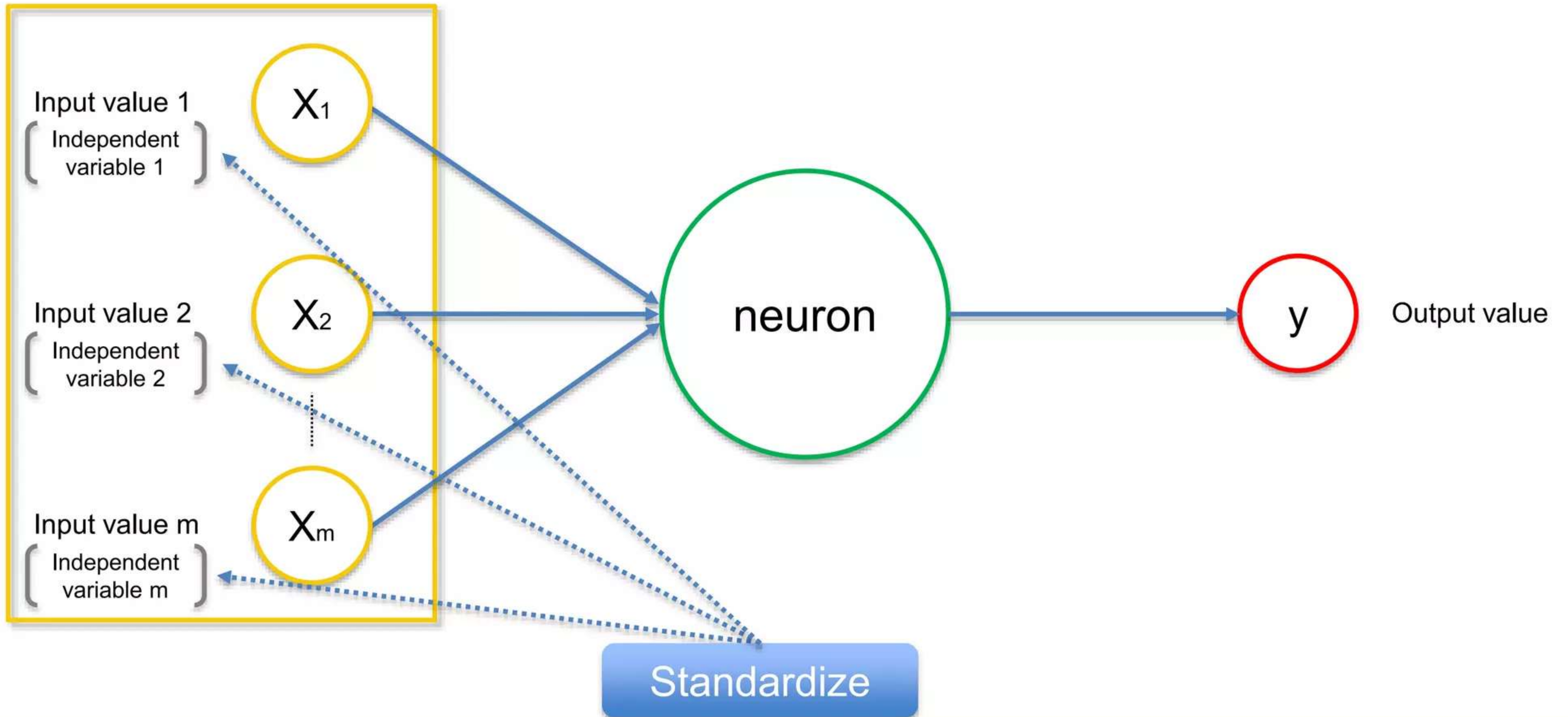
The Neuron



The Neuron



The Neuron



The Neuron

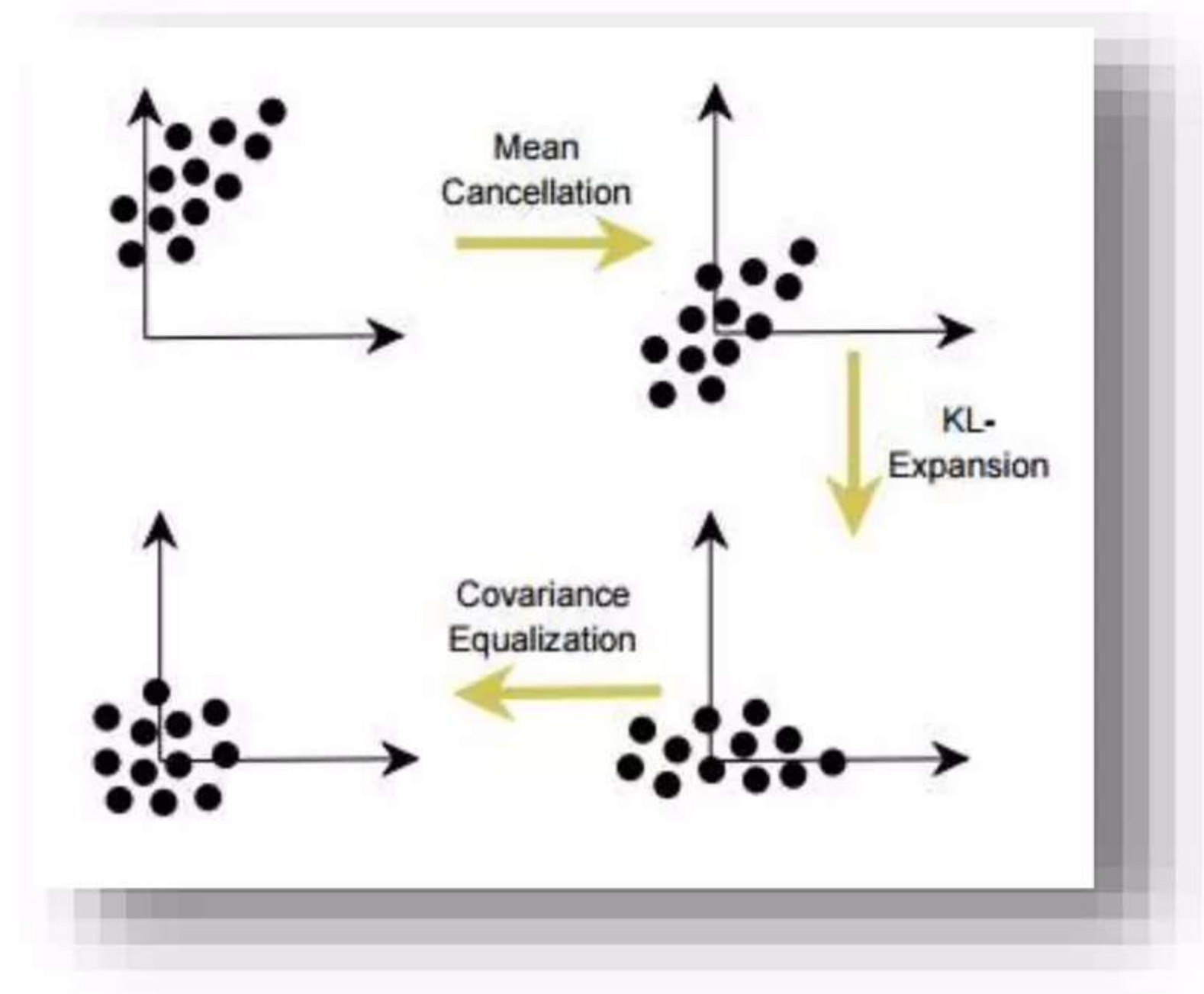
Additional Reading:

Efficient BackProp

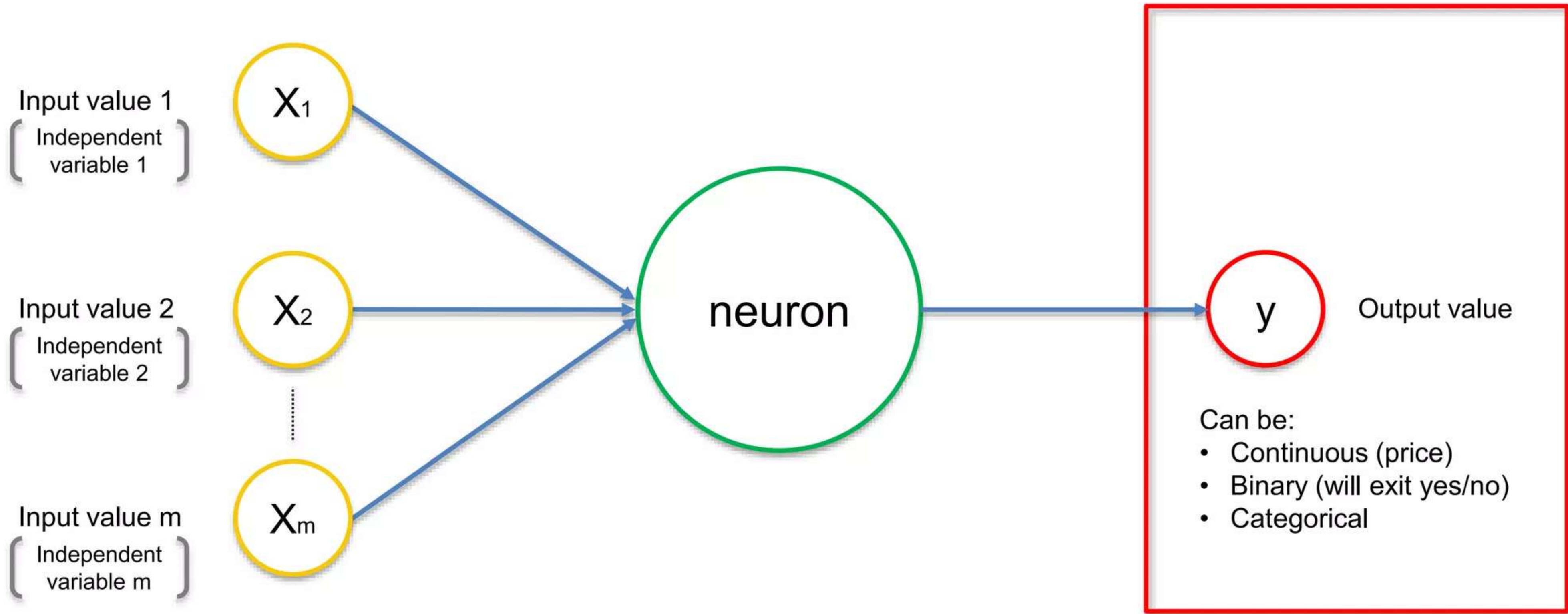
By Yann LeCun et al. (1998)

Link:

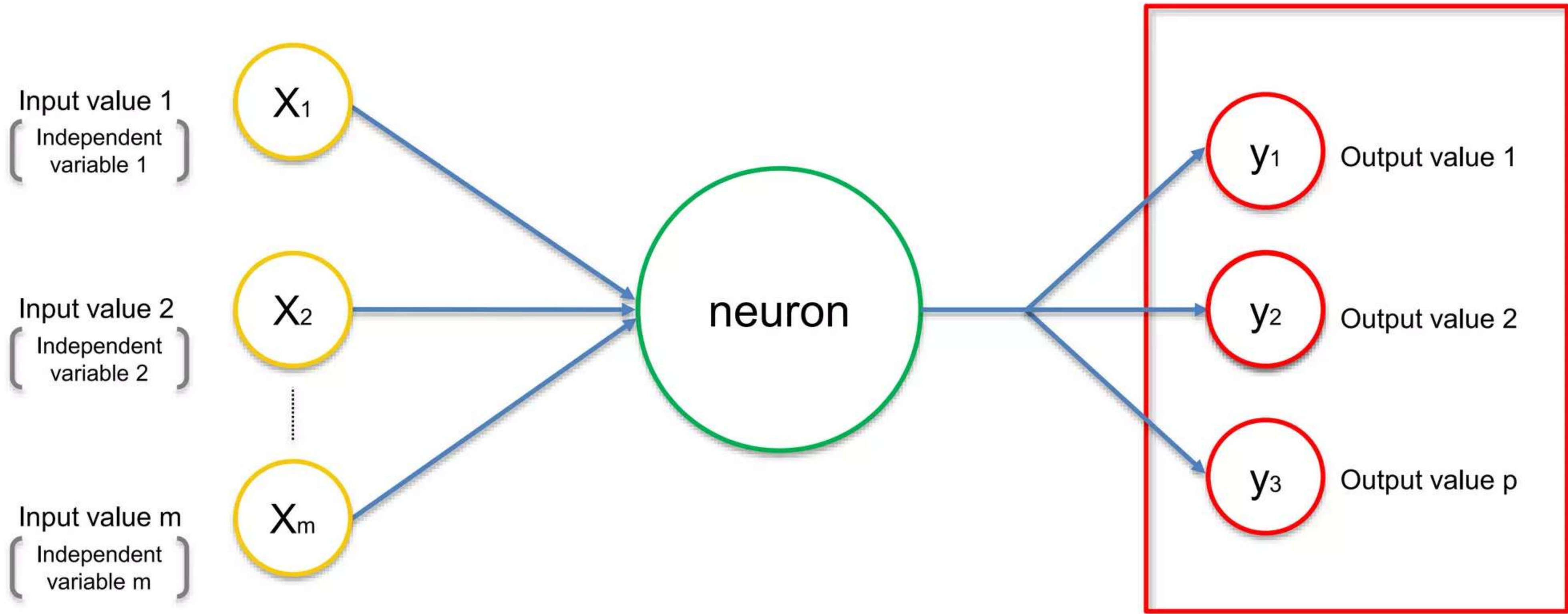
<http://yann.lecun.com/exdb/publis/pdf/lecun-98b.pdf>



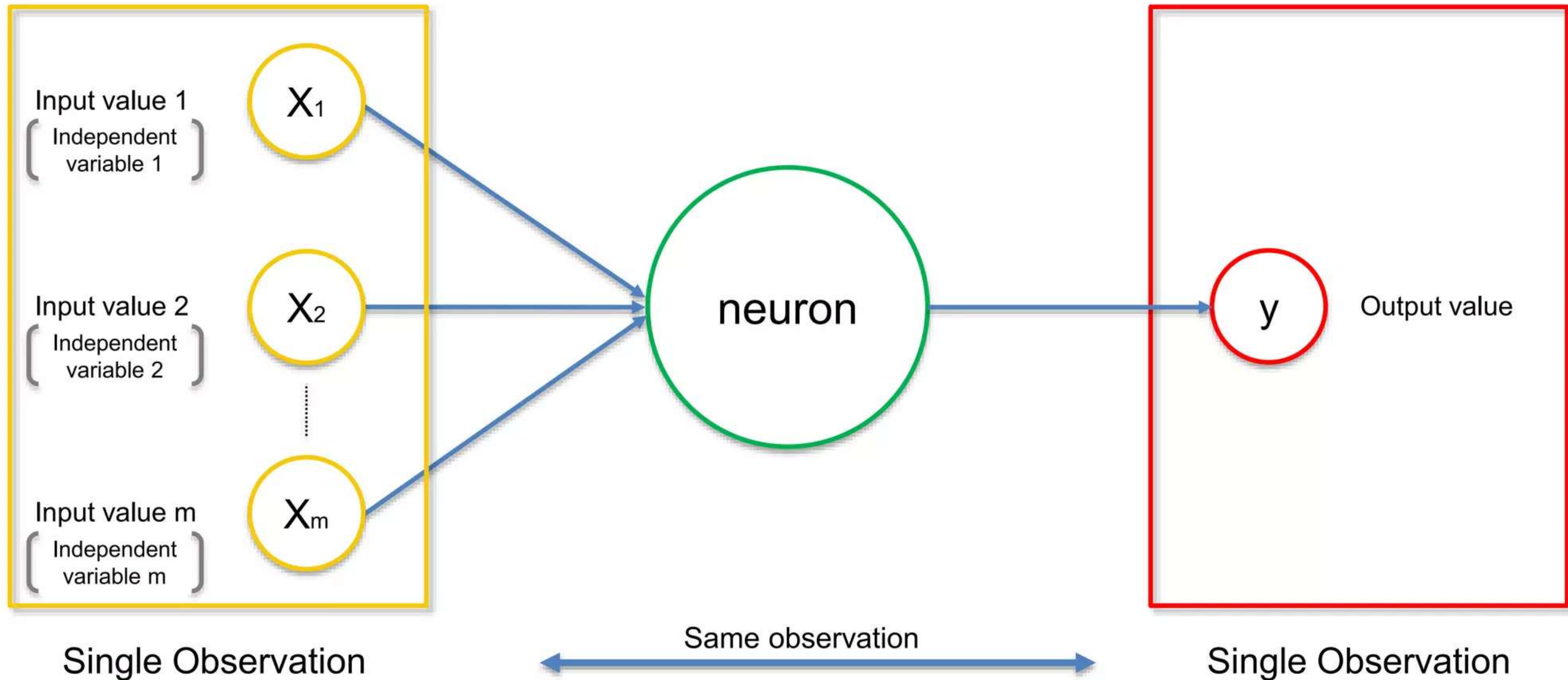
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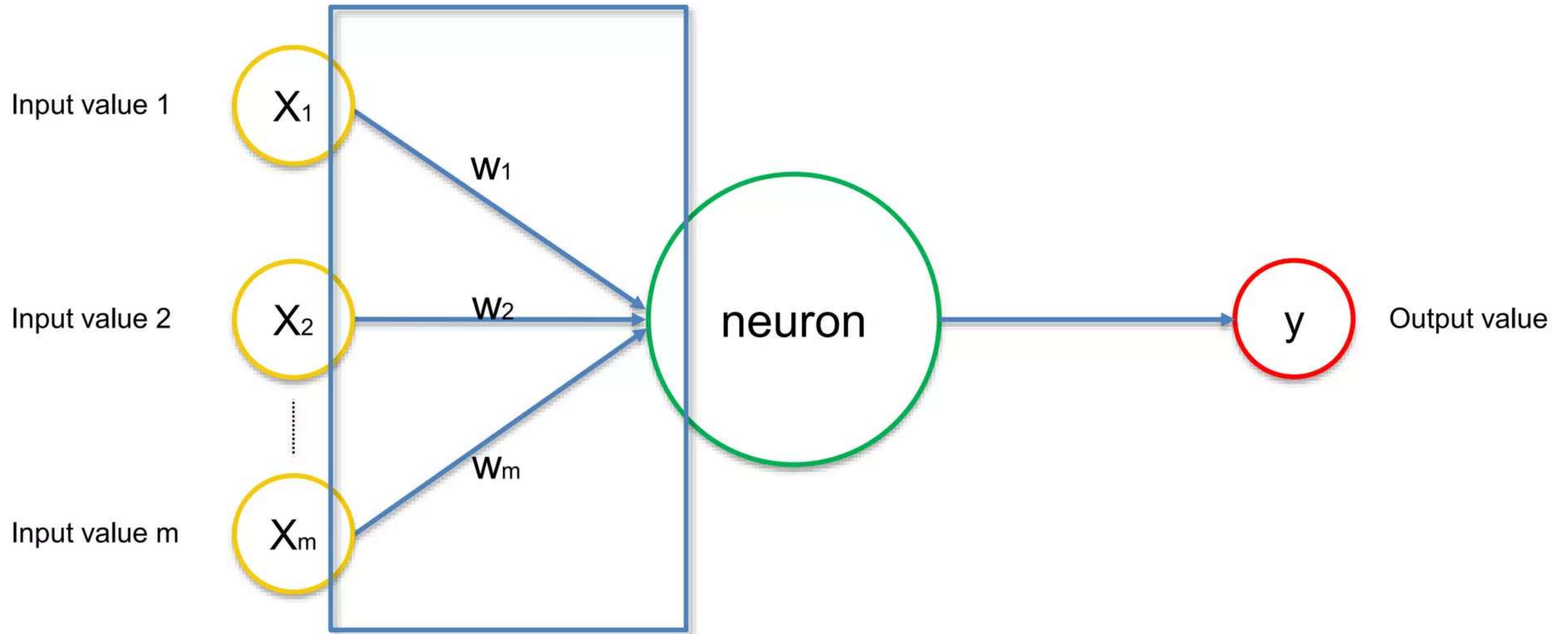
The Neuron



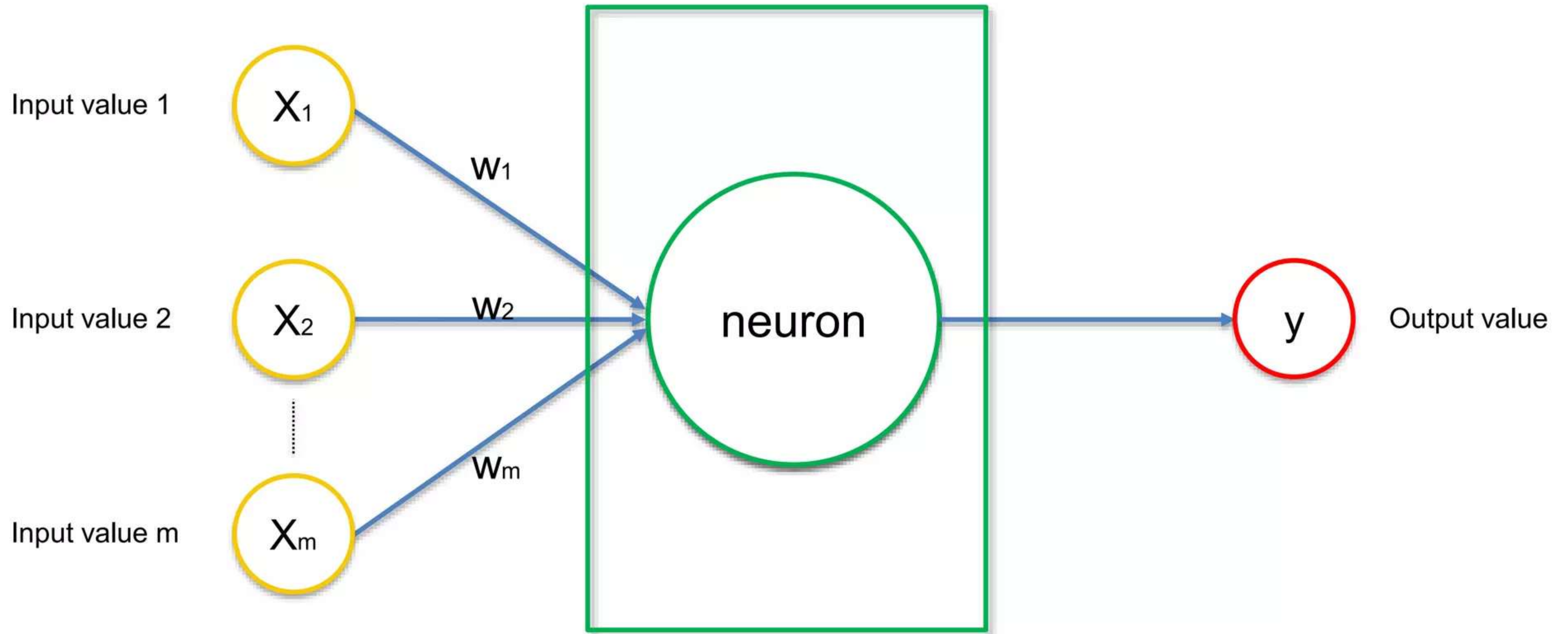
The Neuron



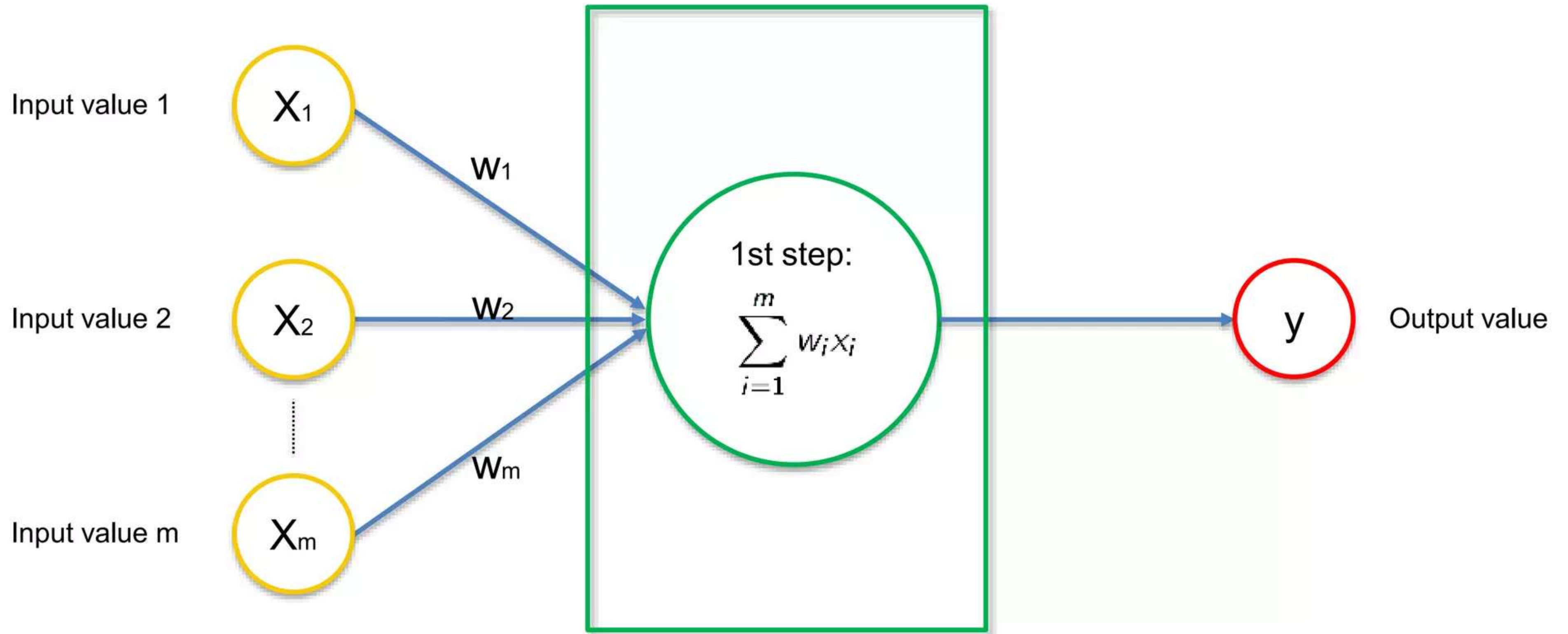
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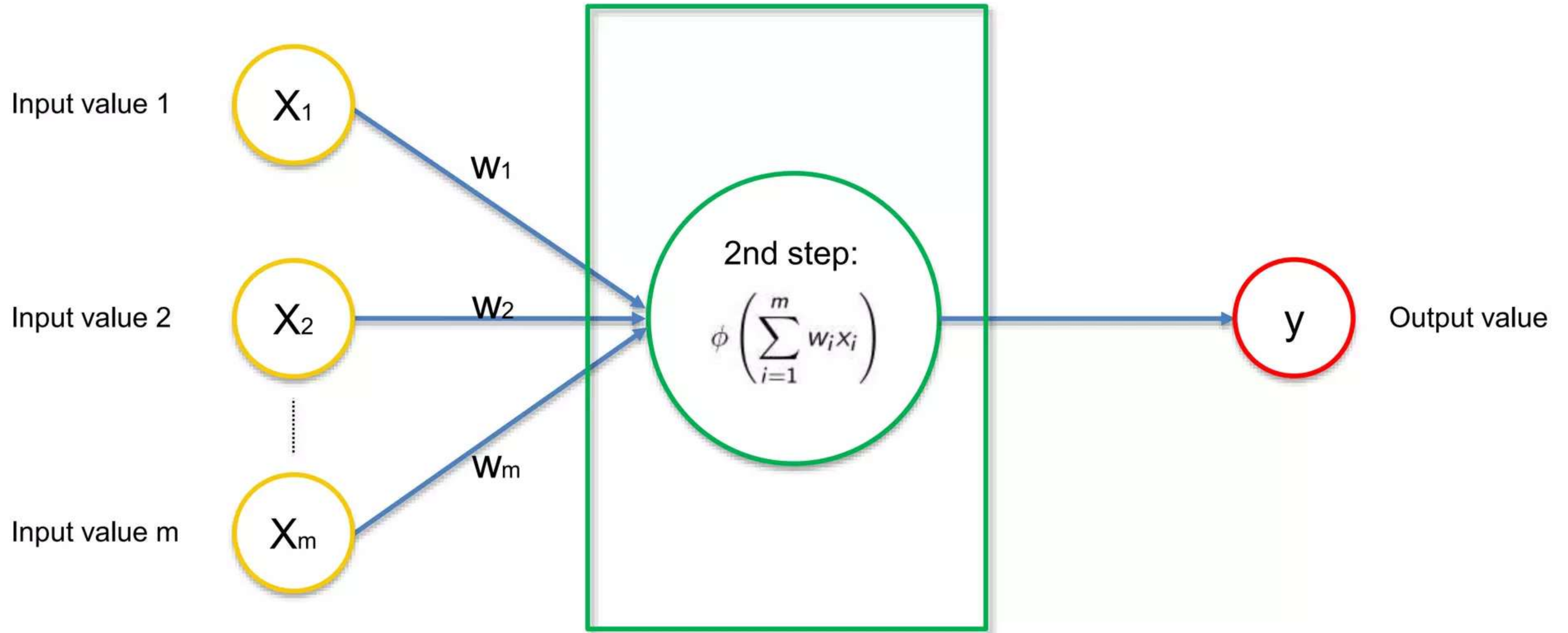
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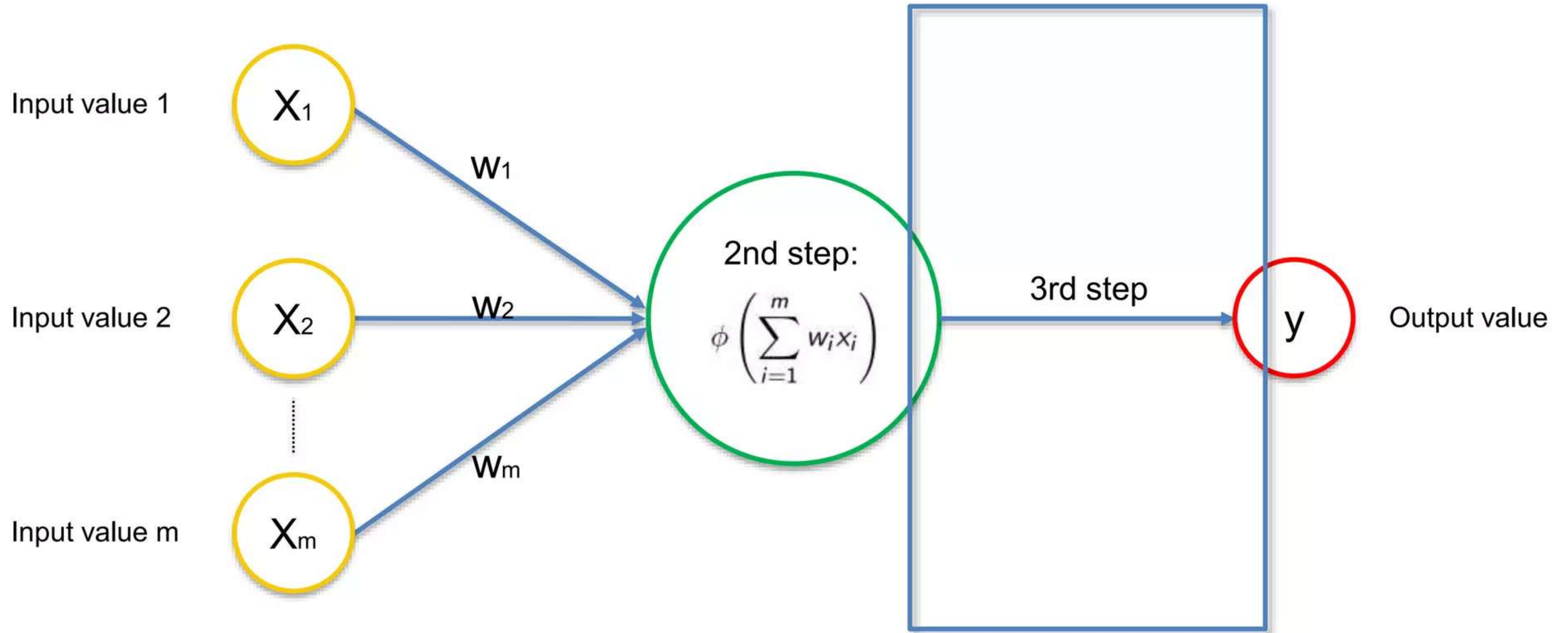
The Neuron



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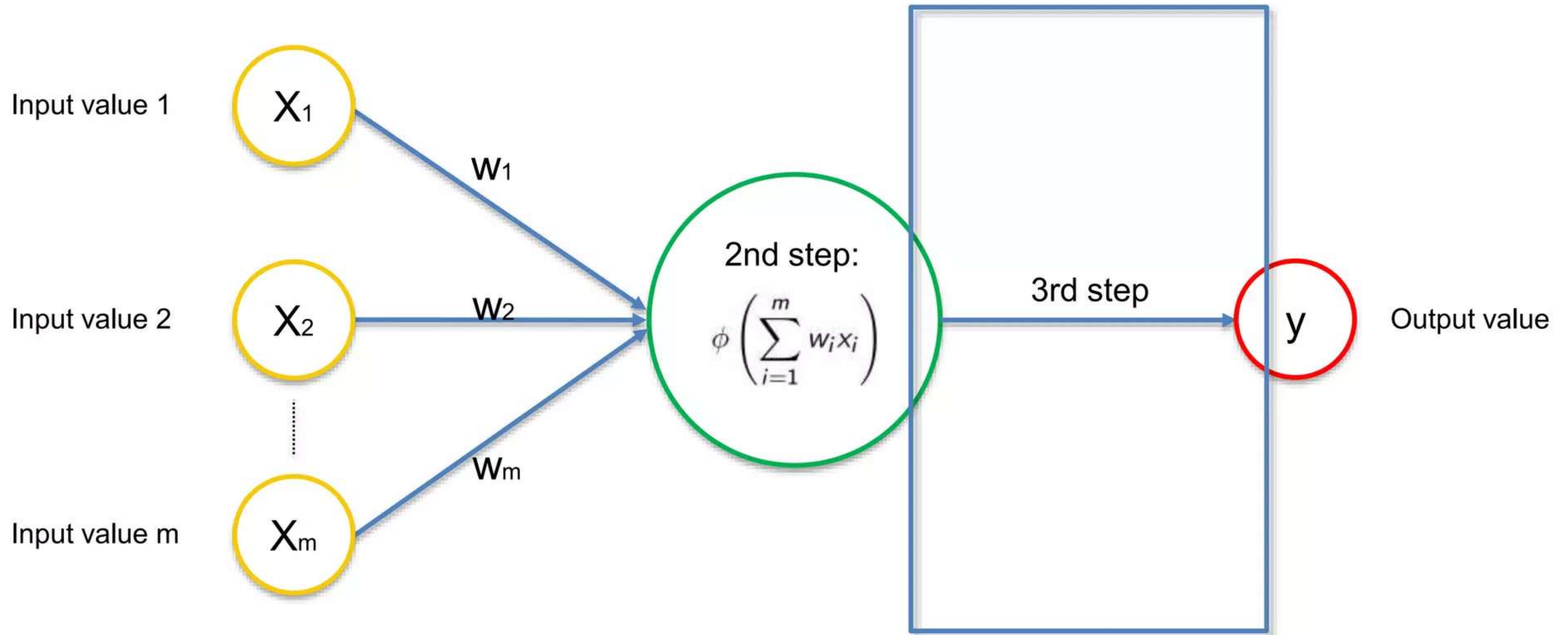


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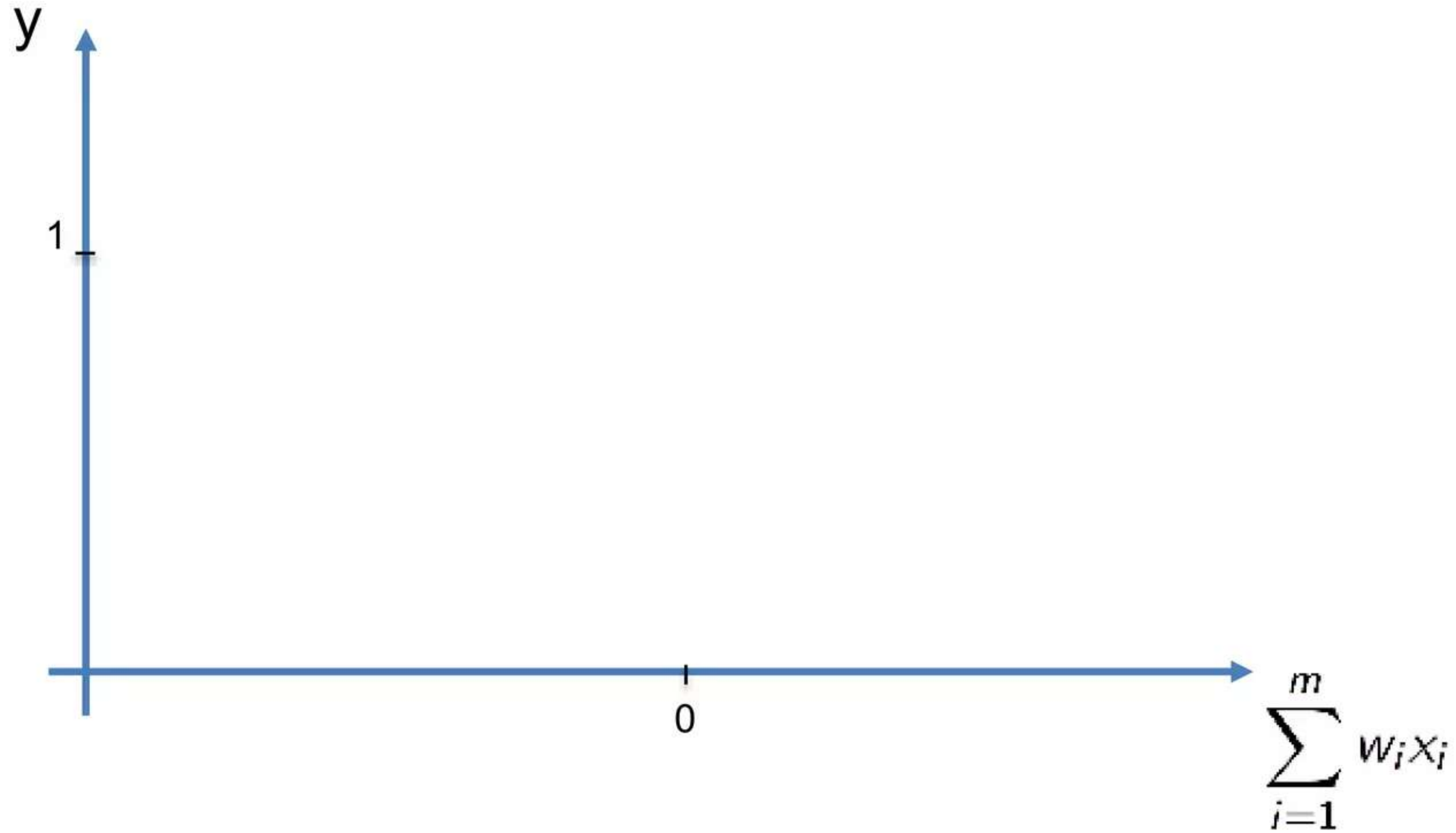


The Activation Function

The Activation Function

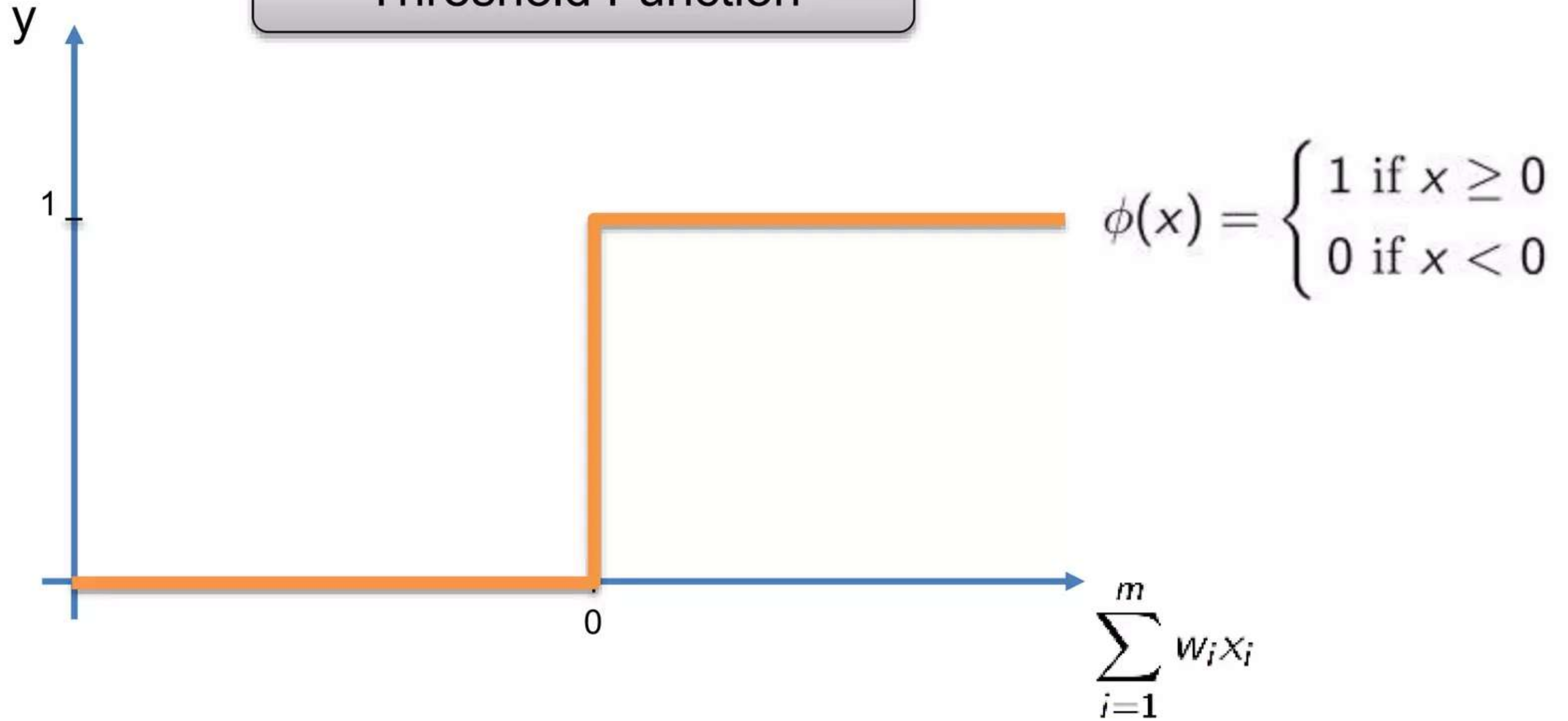


The Activation Function

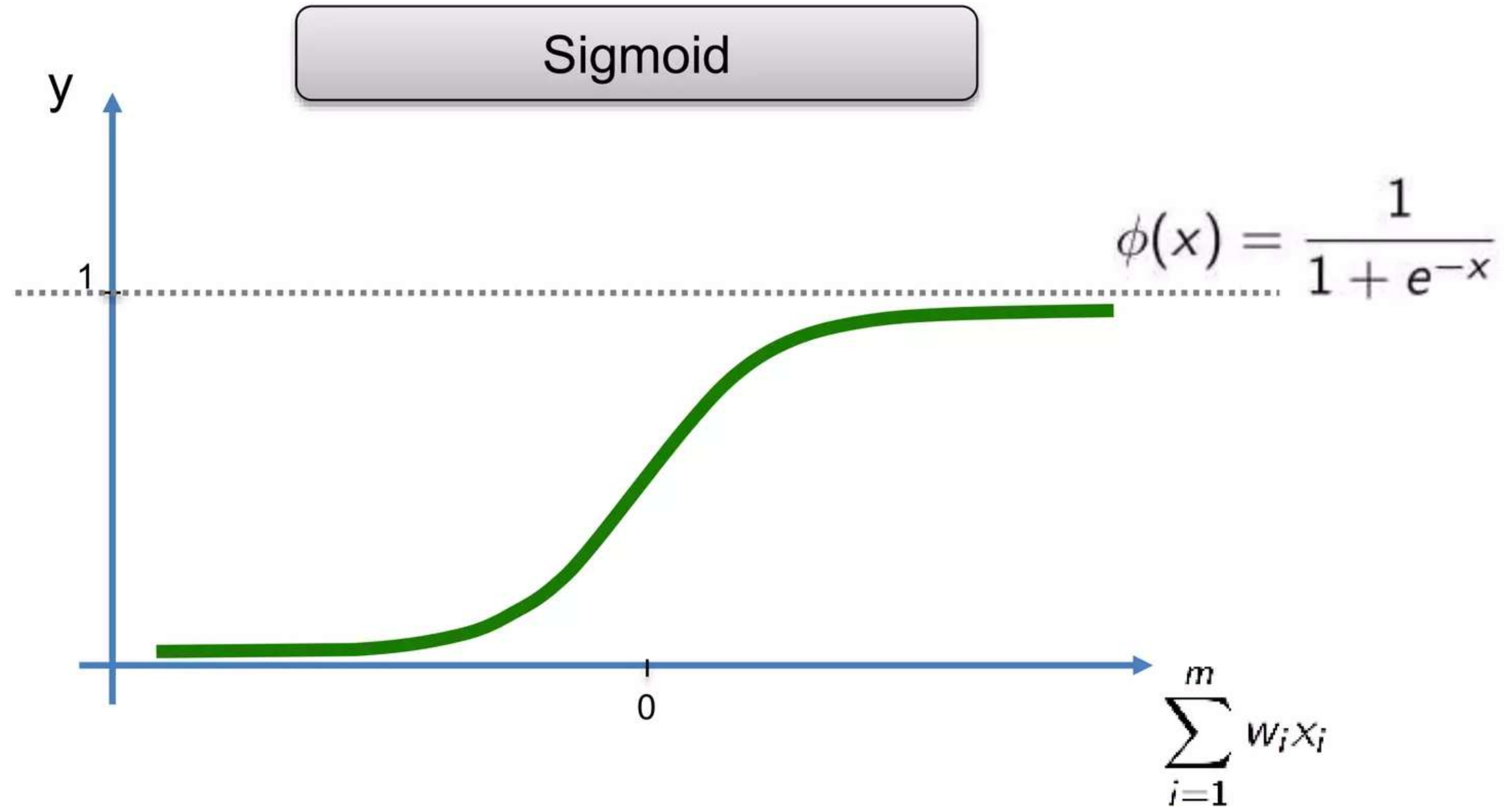


The Activation Function

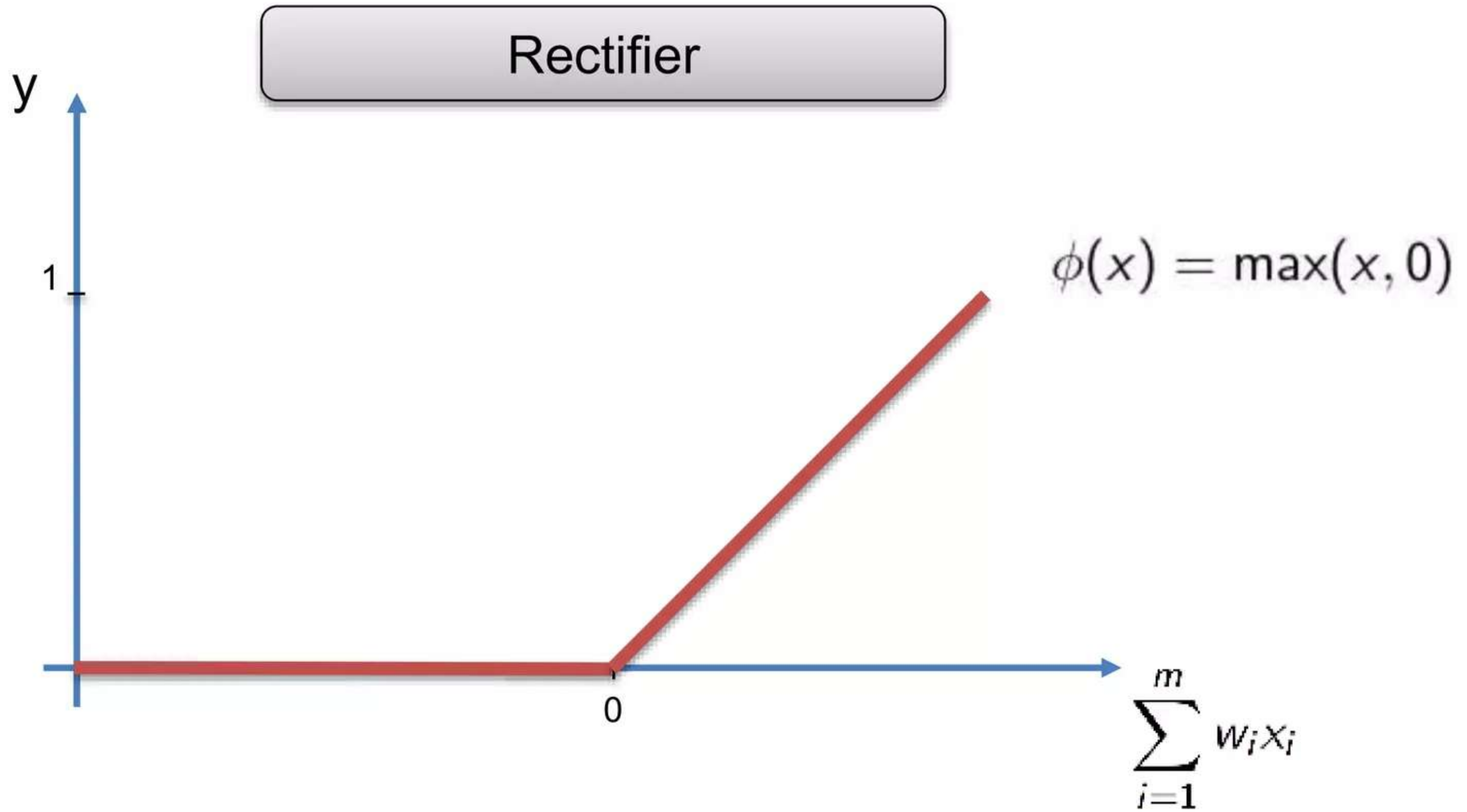
Threshold Function



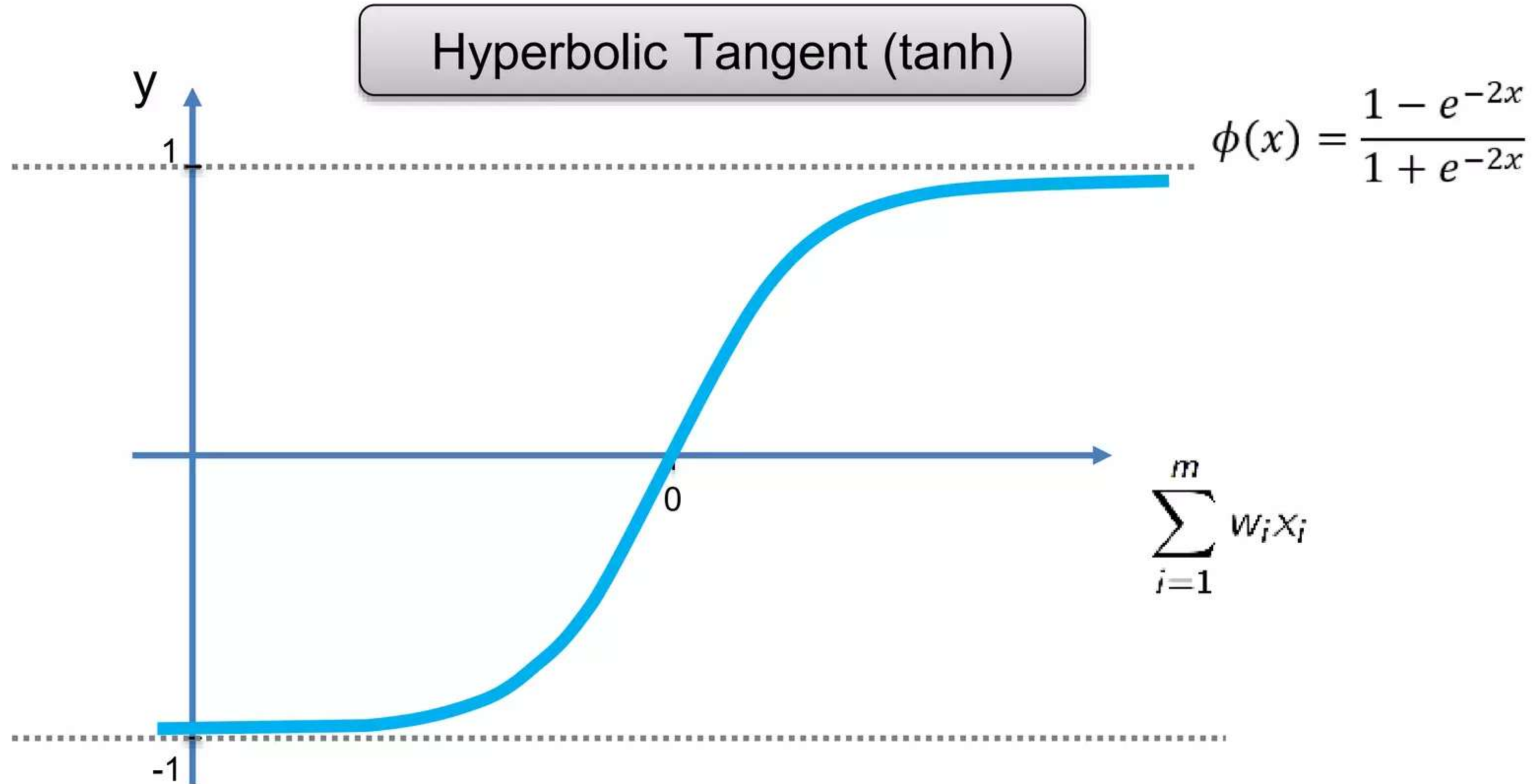
The Activation Function



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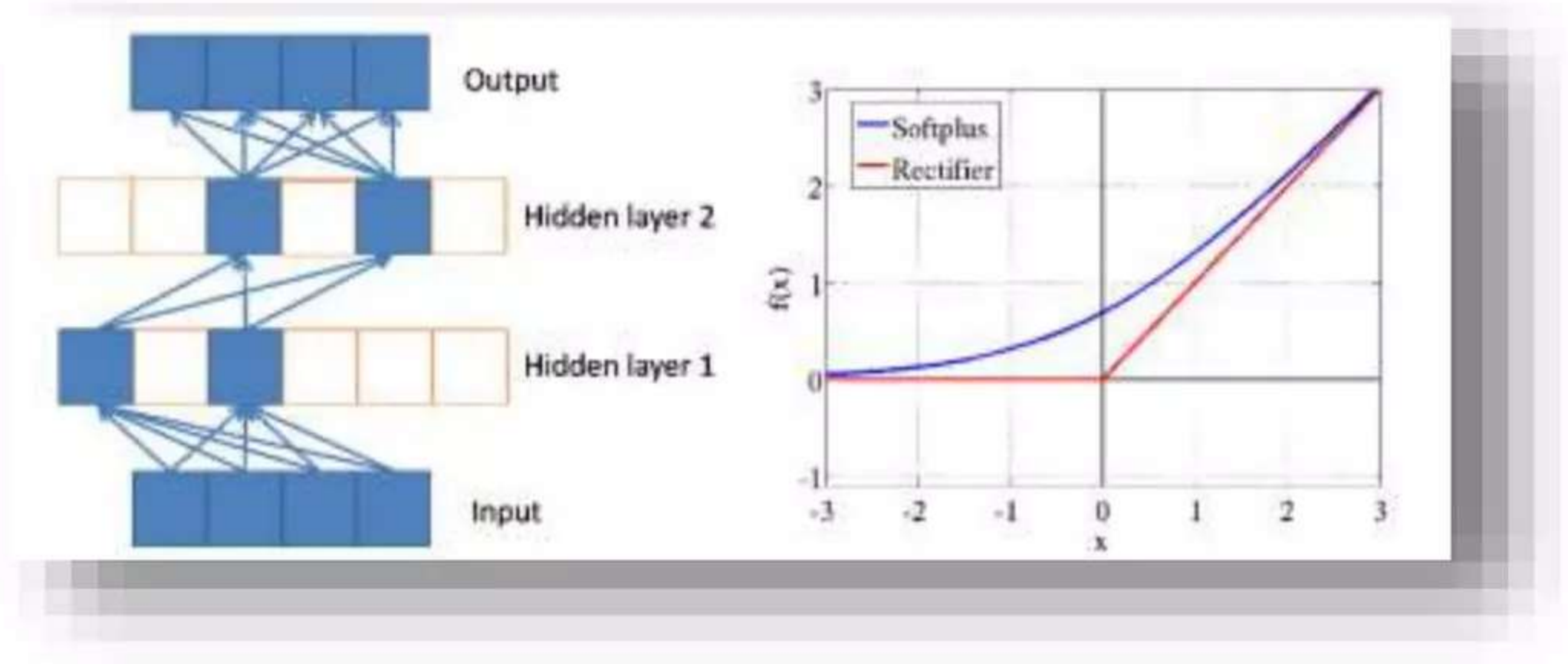


The Activation Function

Additional Reading:

*Deep sparse rectifier
neural networks*

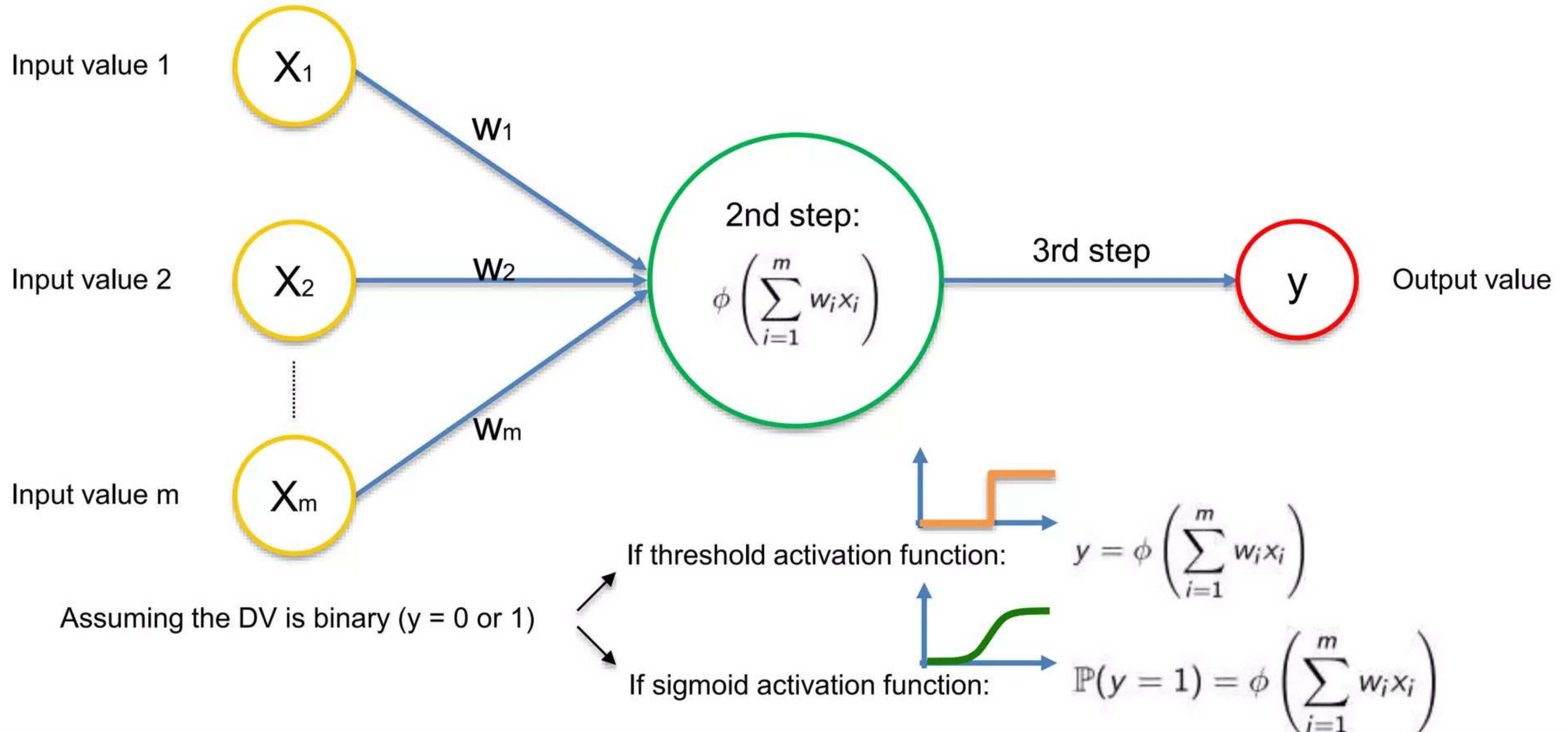
By Xavier Glorot et al. (2011)



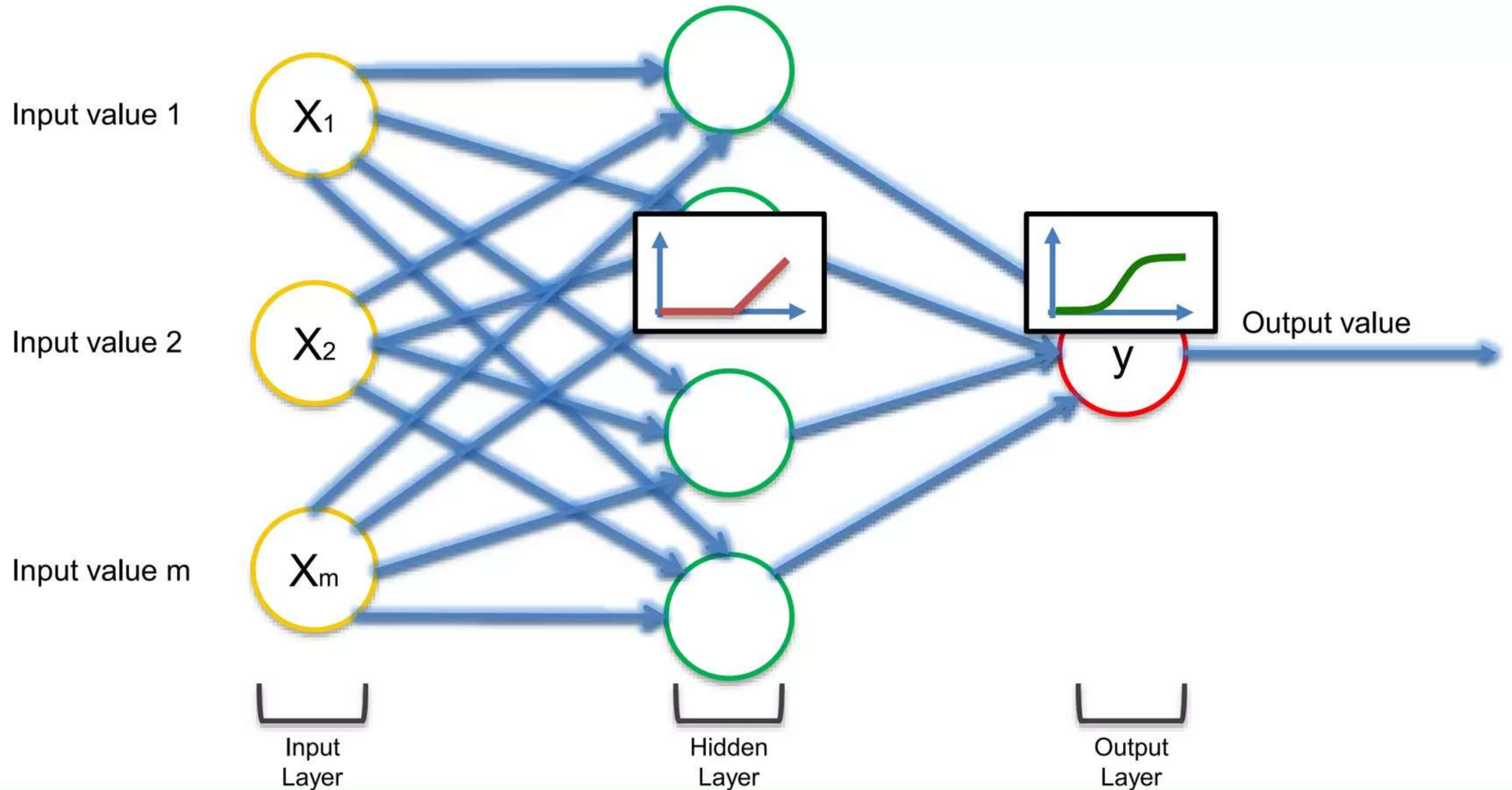
Link:

<http://jmlr.org/proceedings/papers/v15/glorot11a/glorot11a.pdf>

The Activation Function



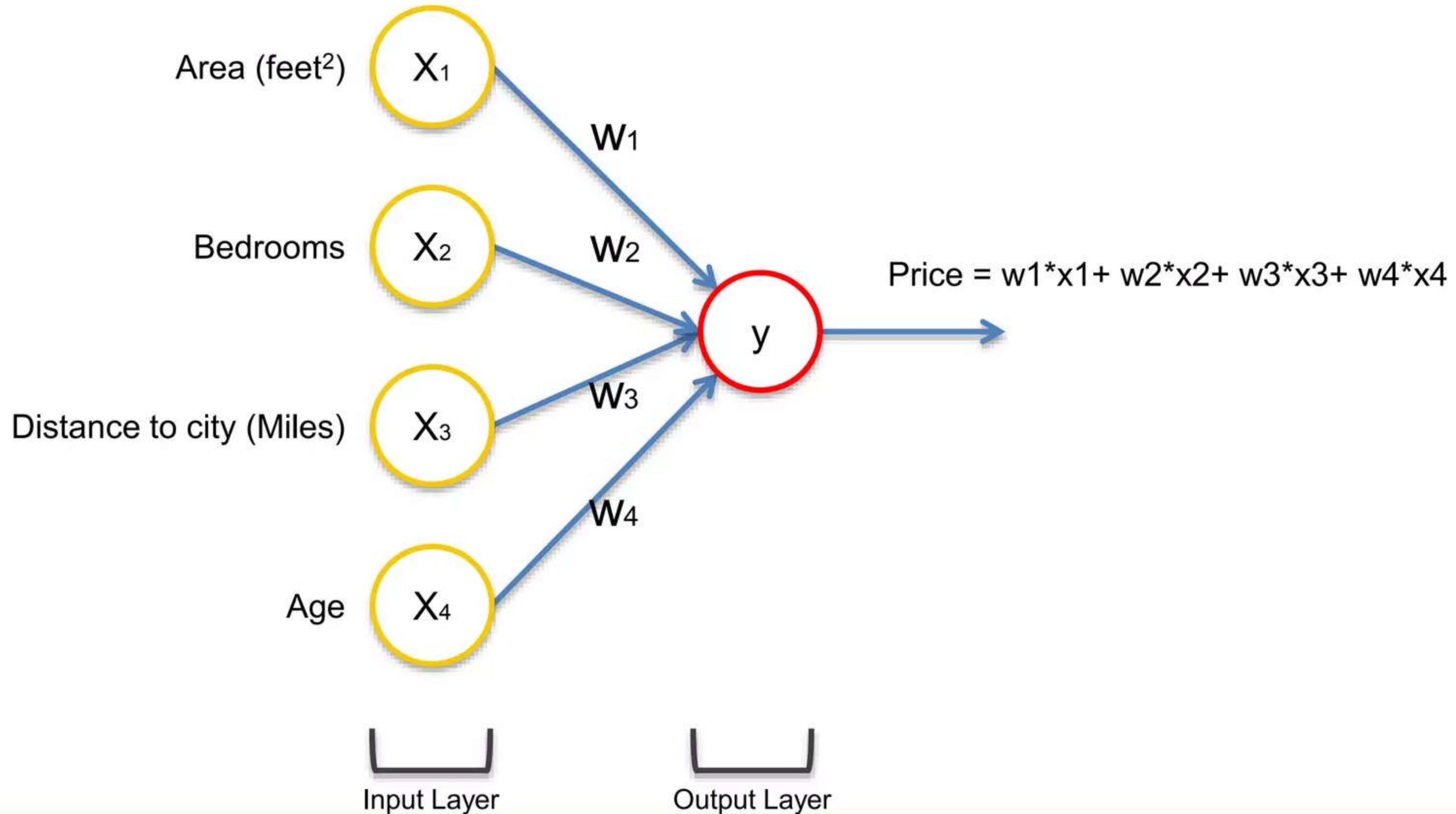
The Activation Function



How do NNs Work?



How Do Neural Networks Work?



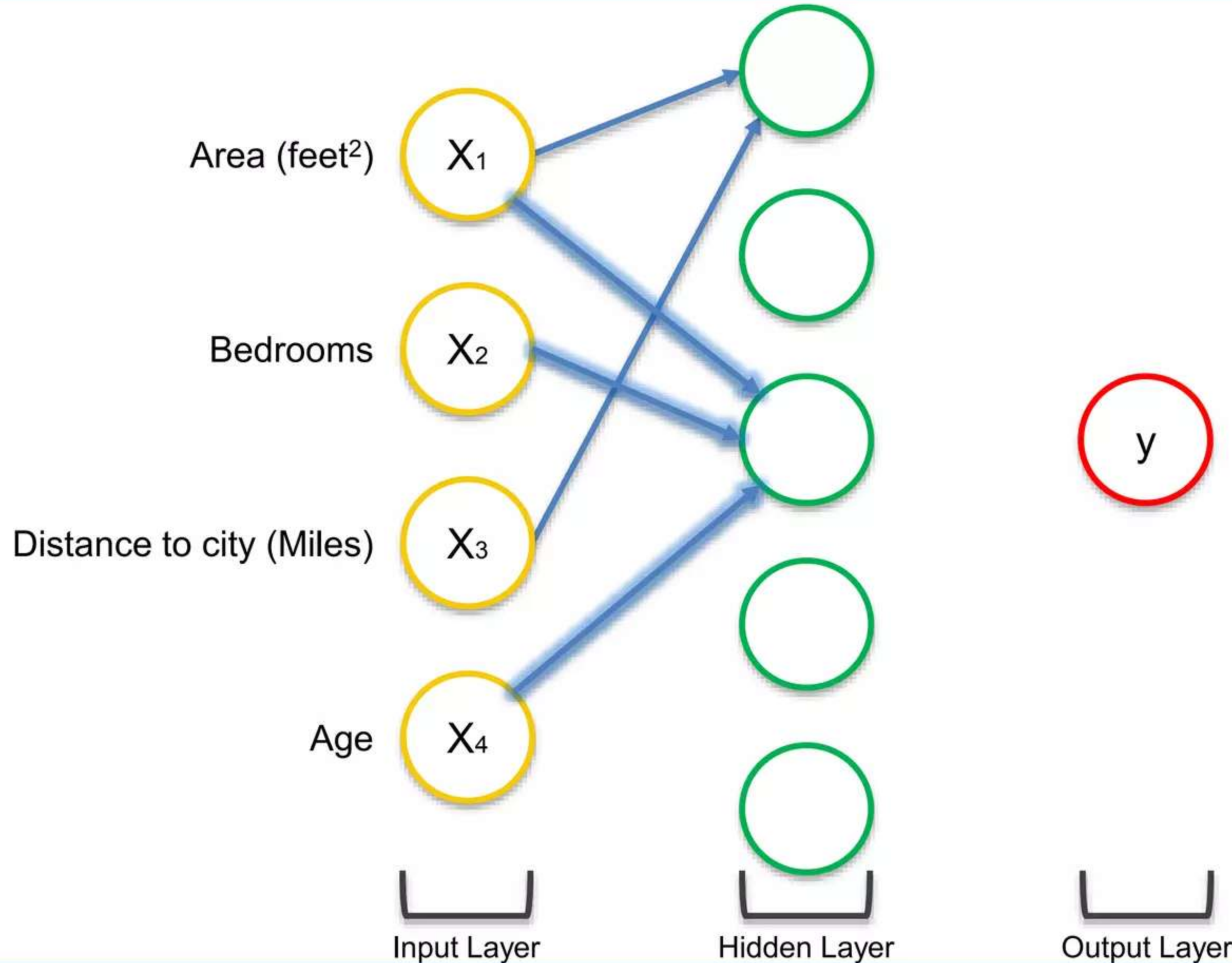
How Do Neural Networks Work?



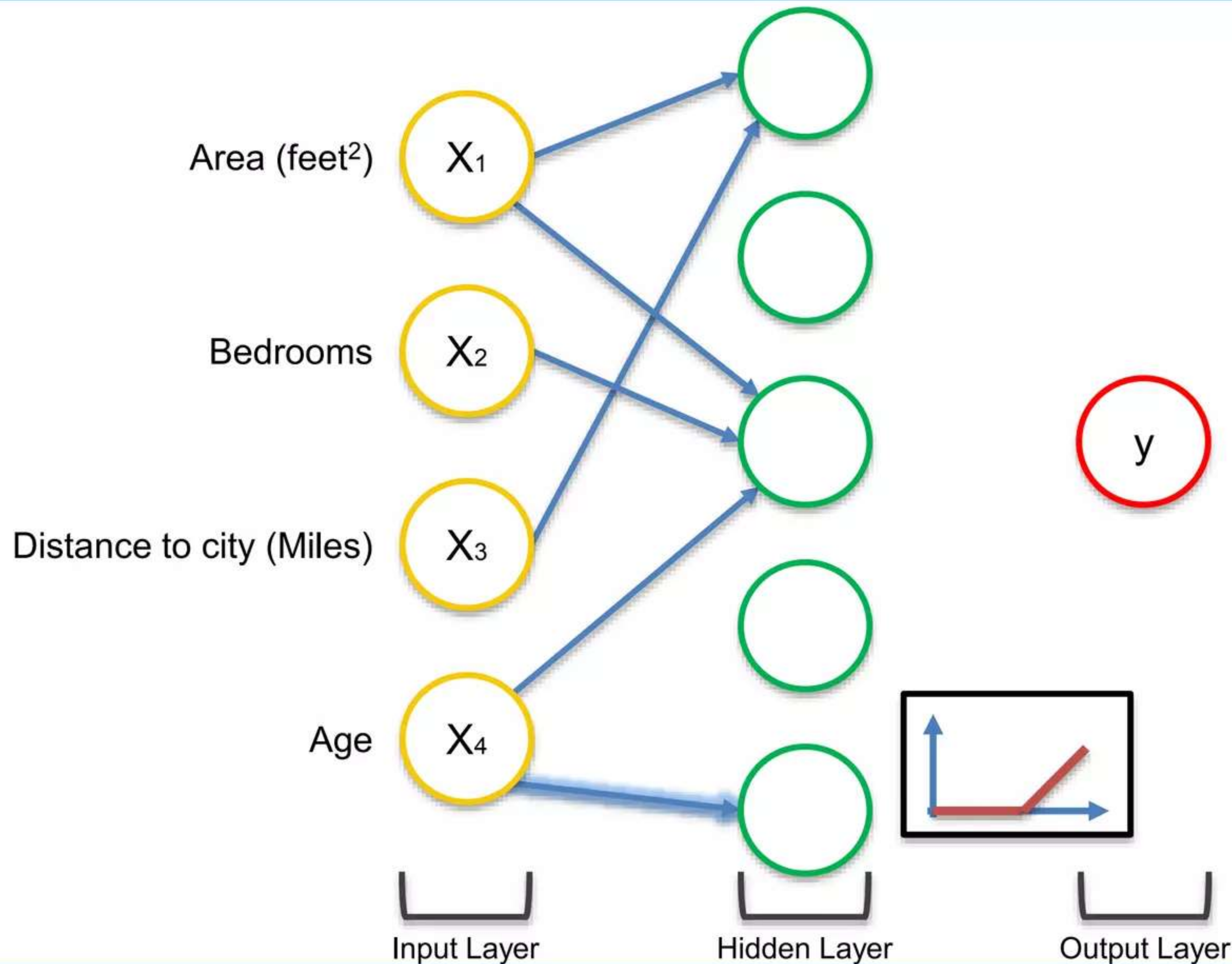
How Do Neural Networks Work?



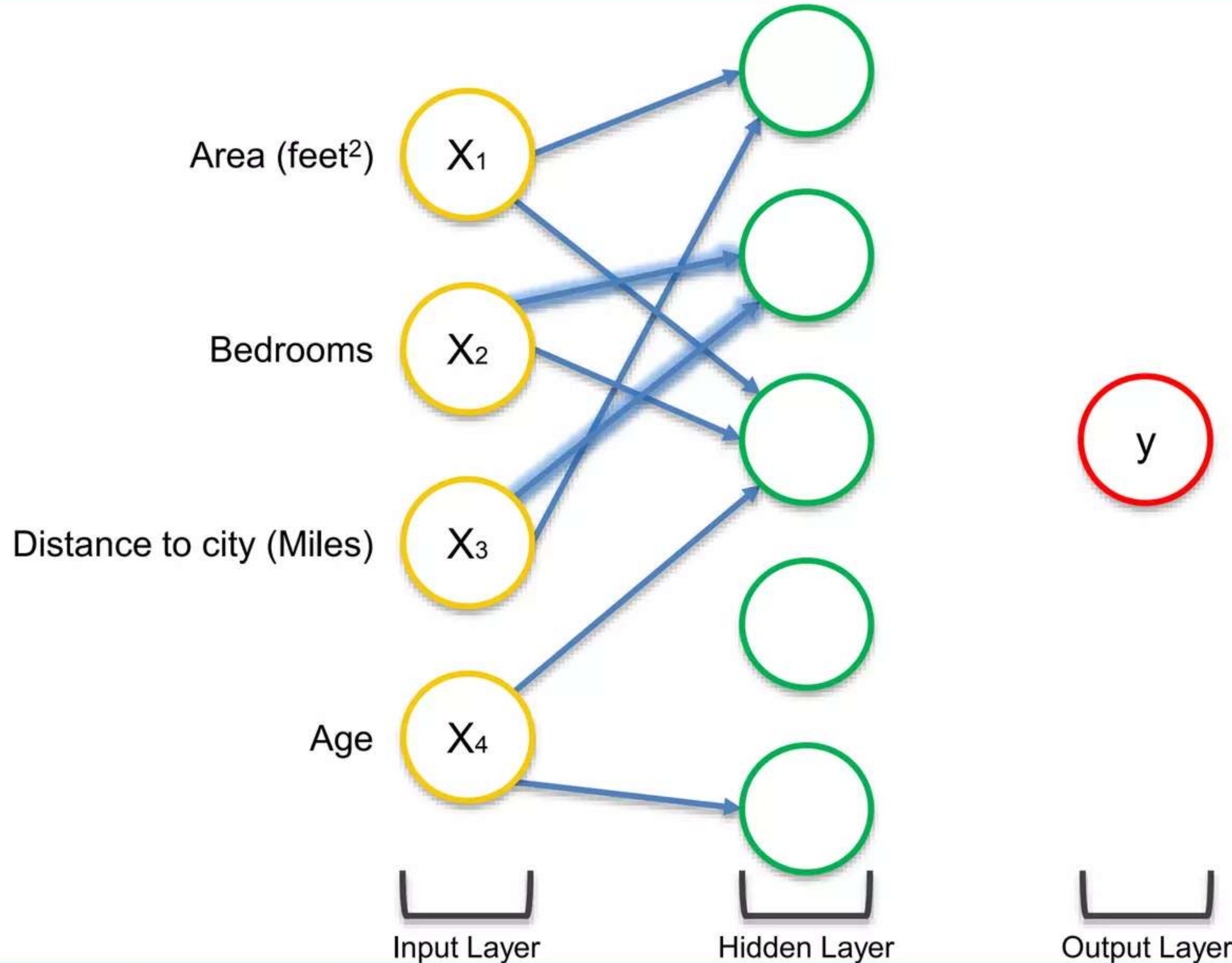
How Do Neural Networks Work?



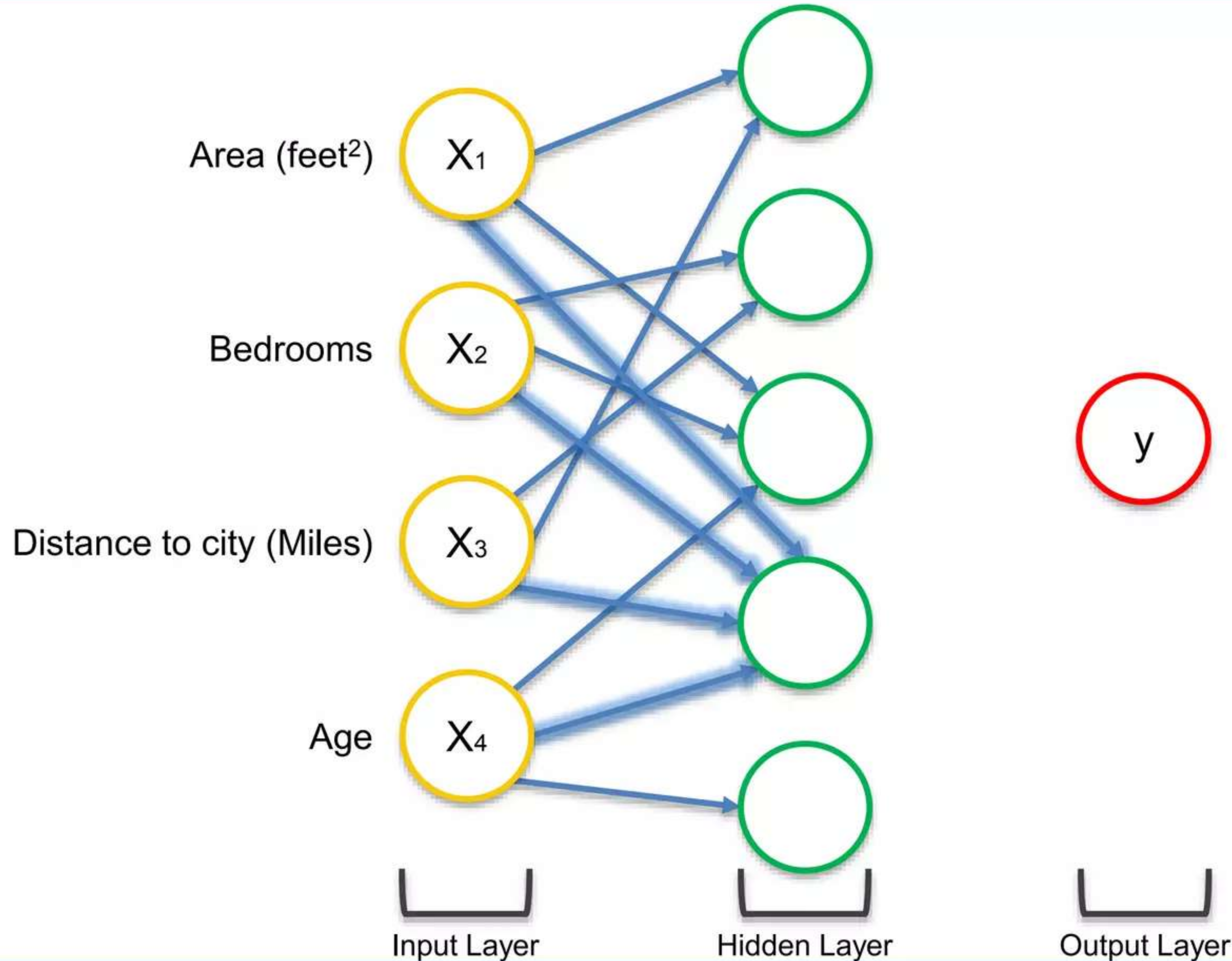
How Do Neural Networks Work?



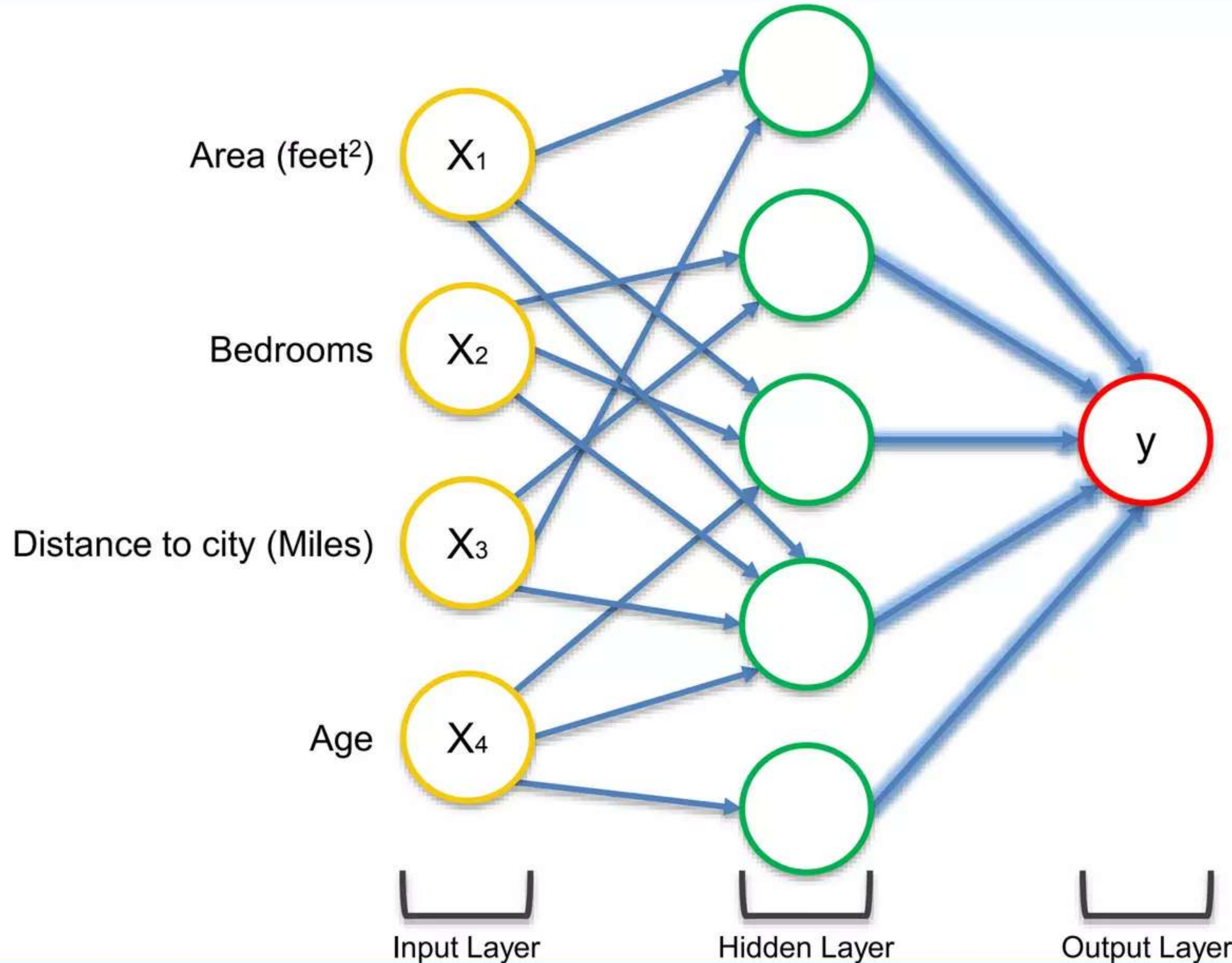
How Do Neural Networks Work?



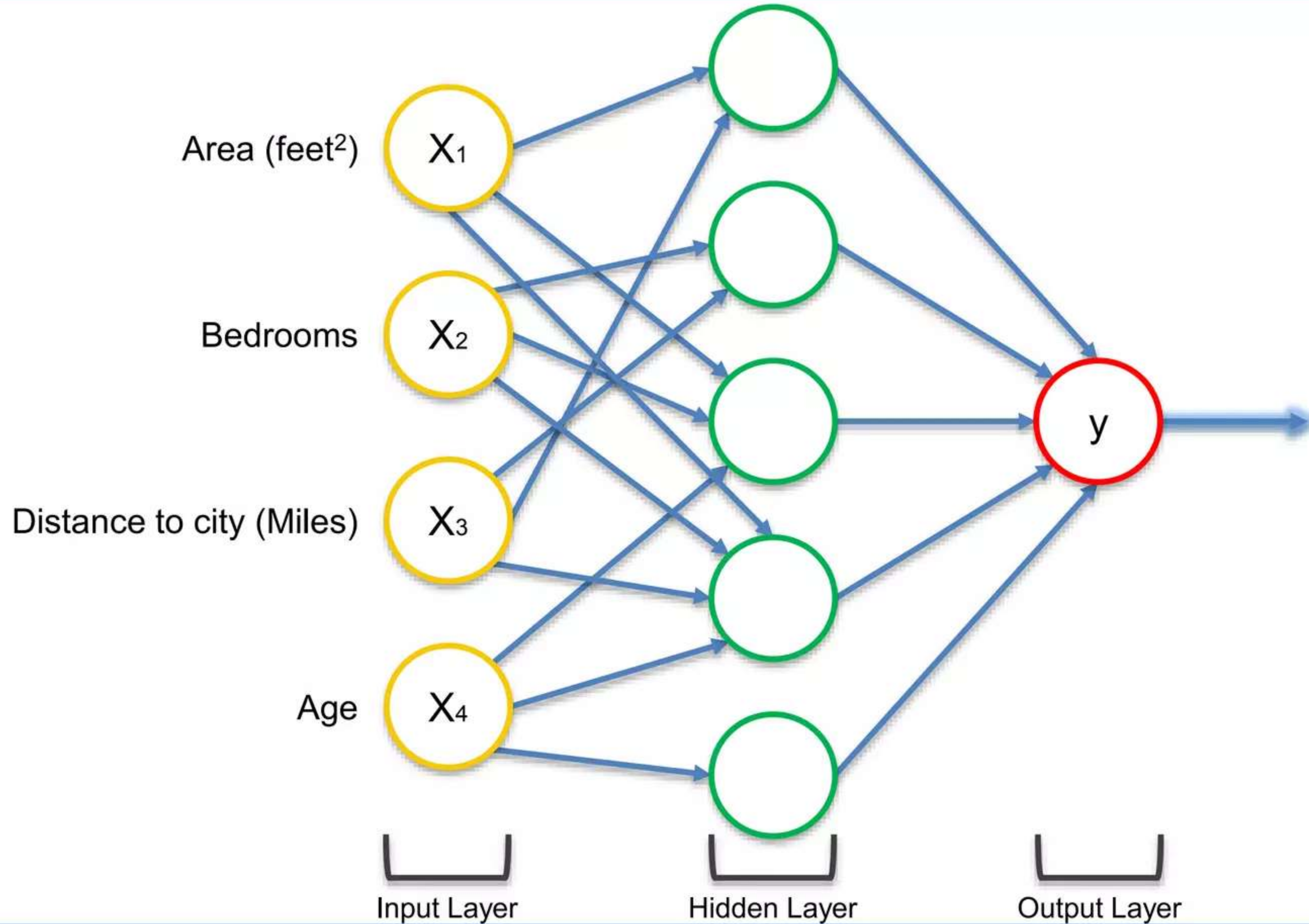
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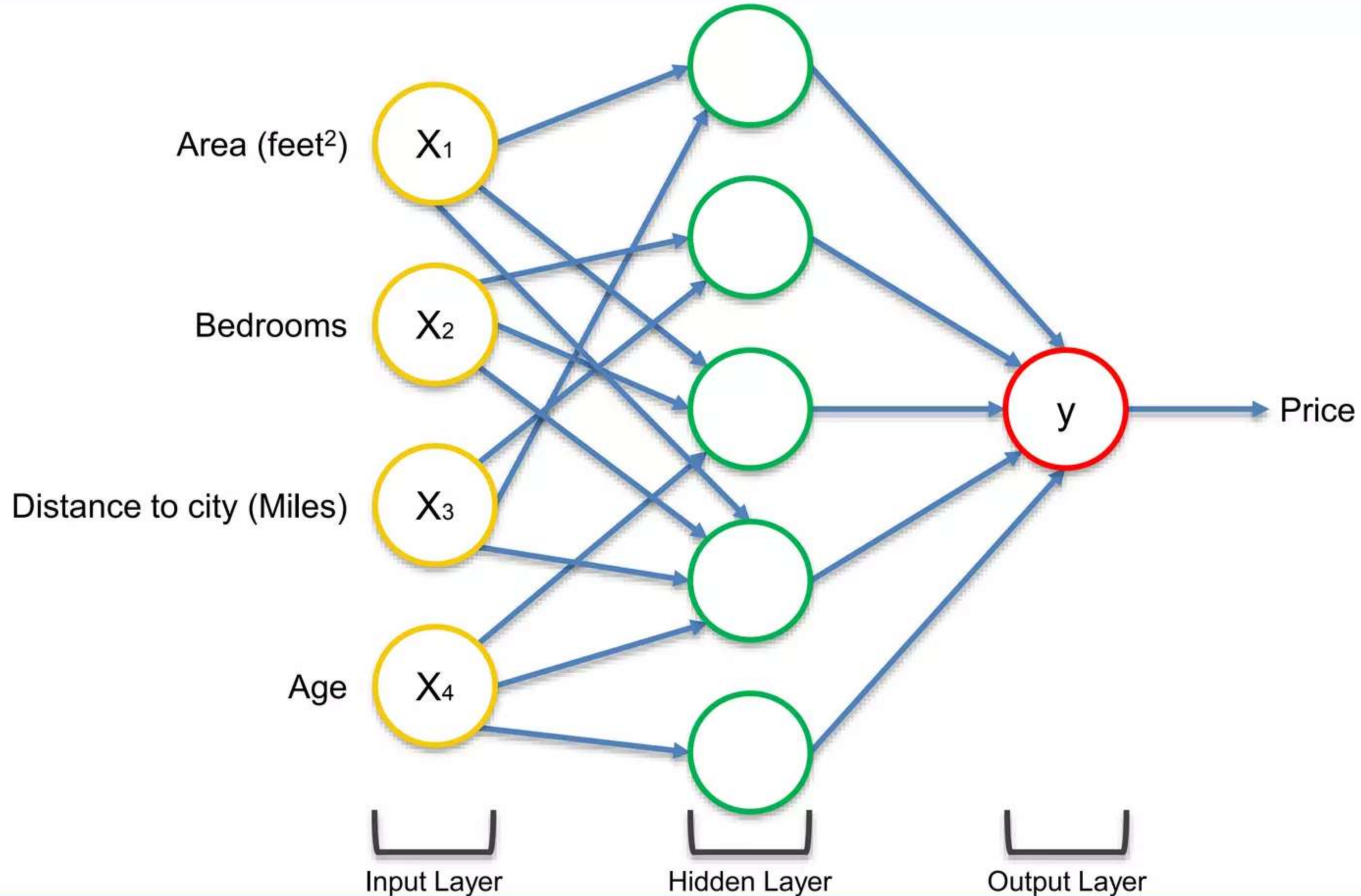
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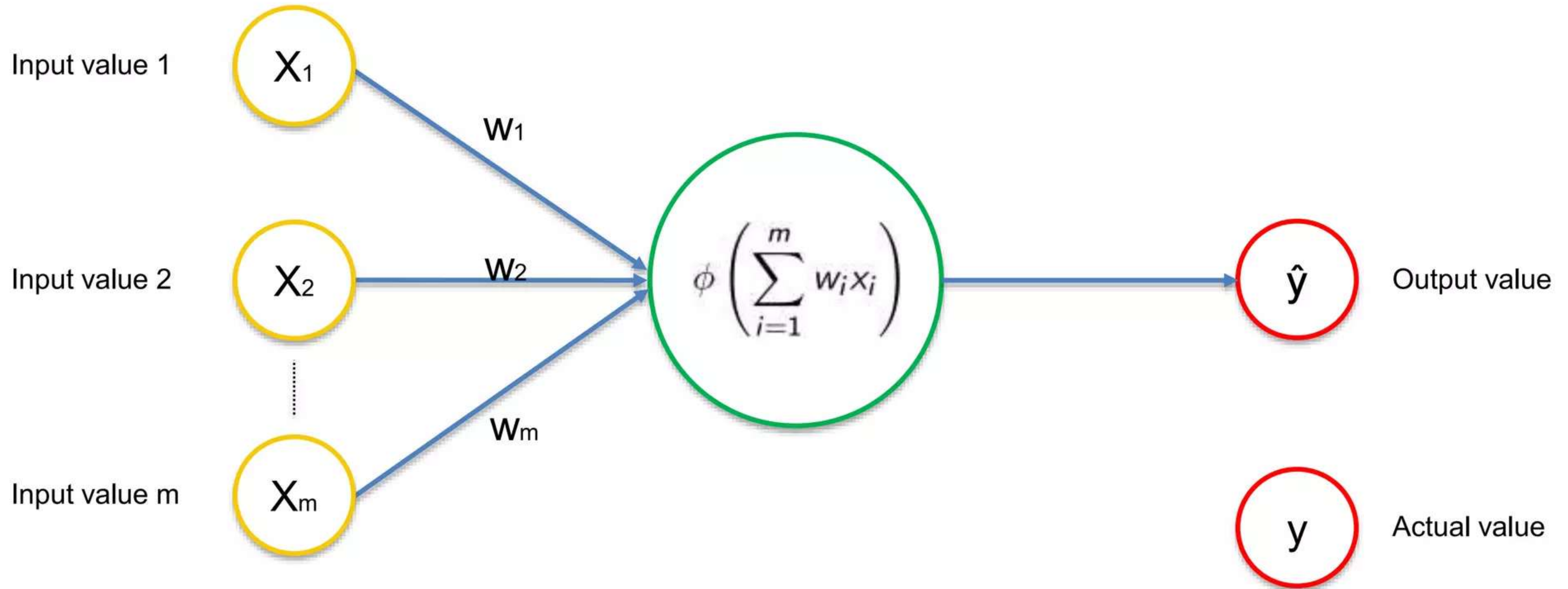


How do NNs Learn?

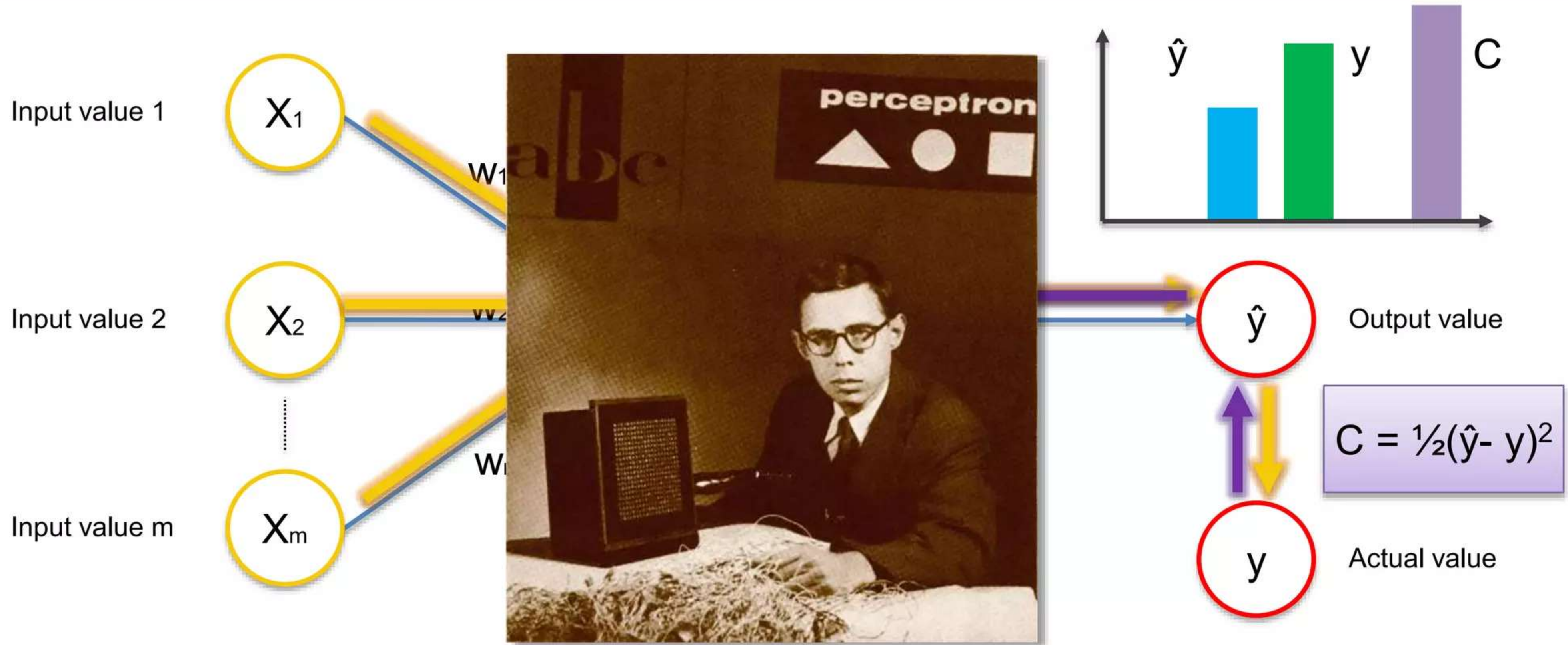
How do Neural Networks learn?



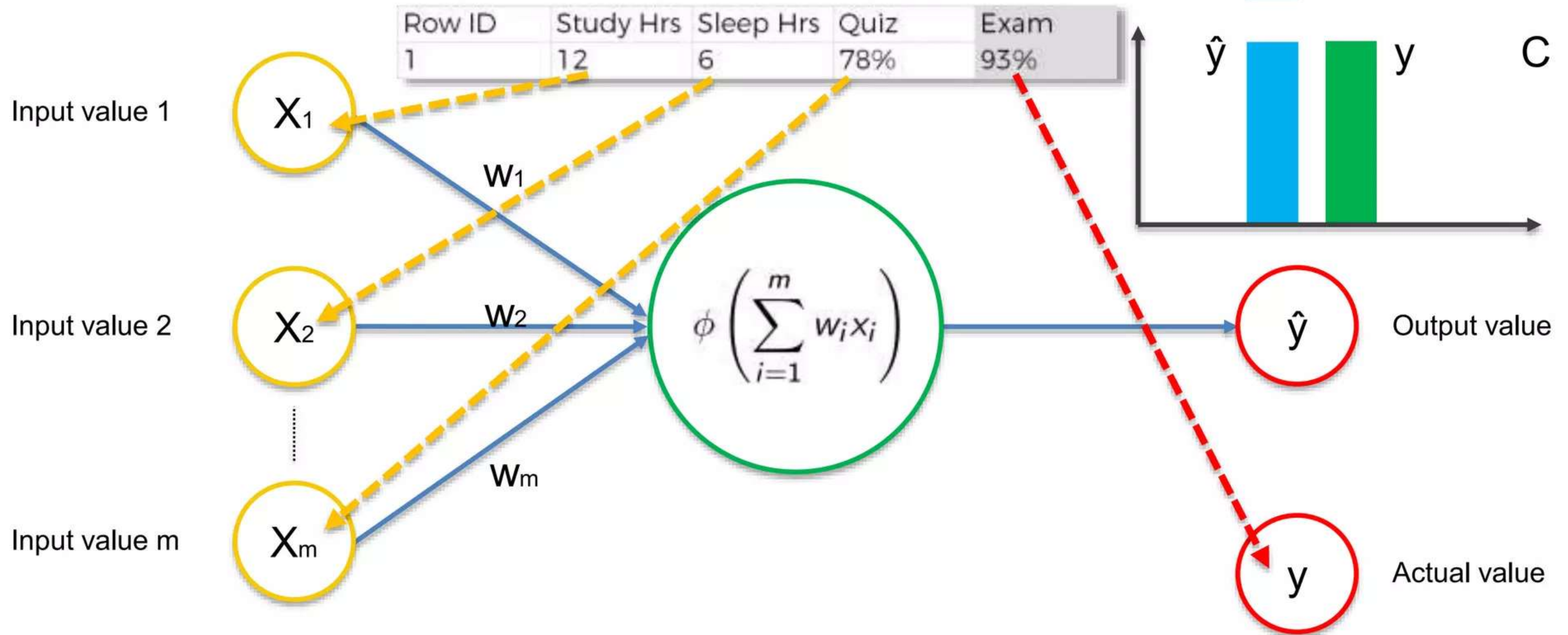
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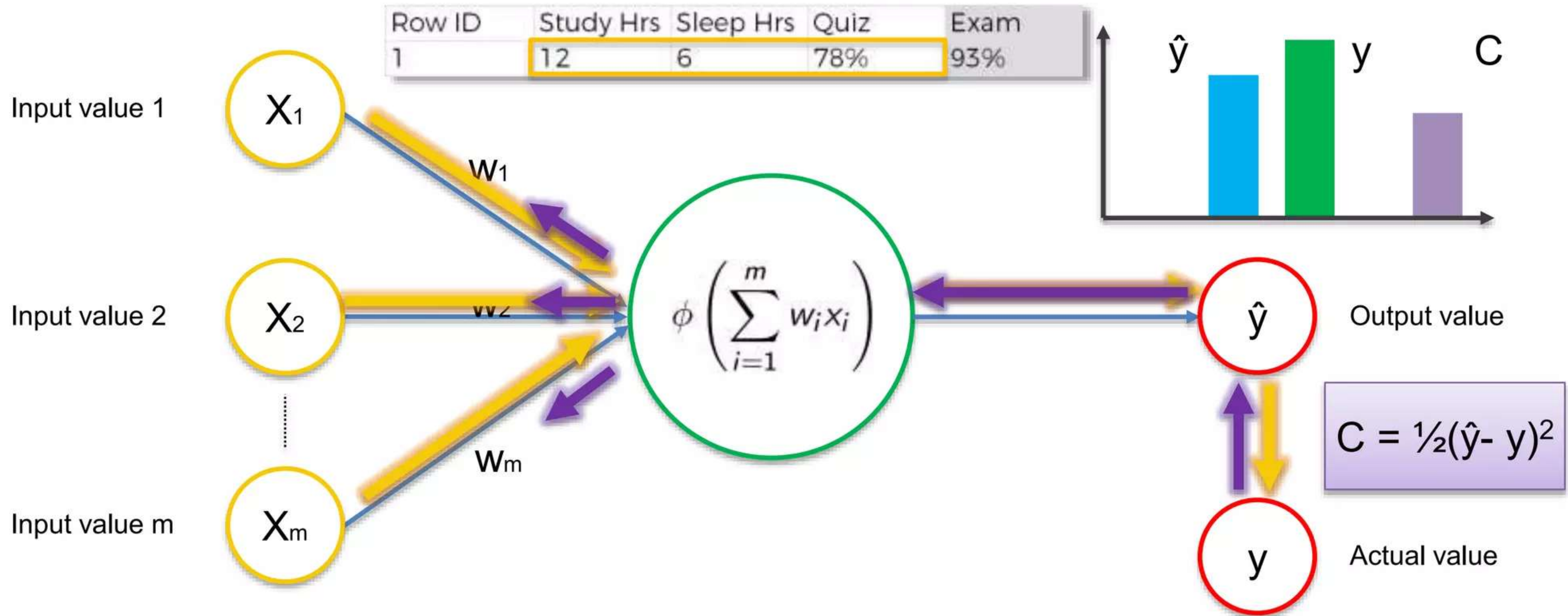
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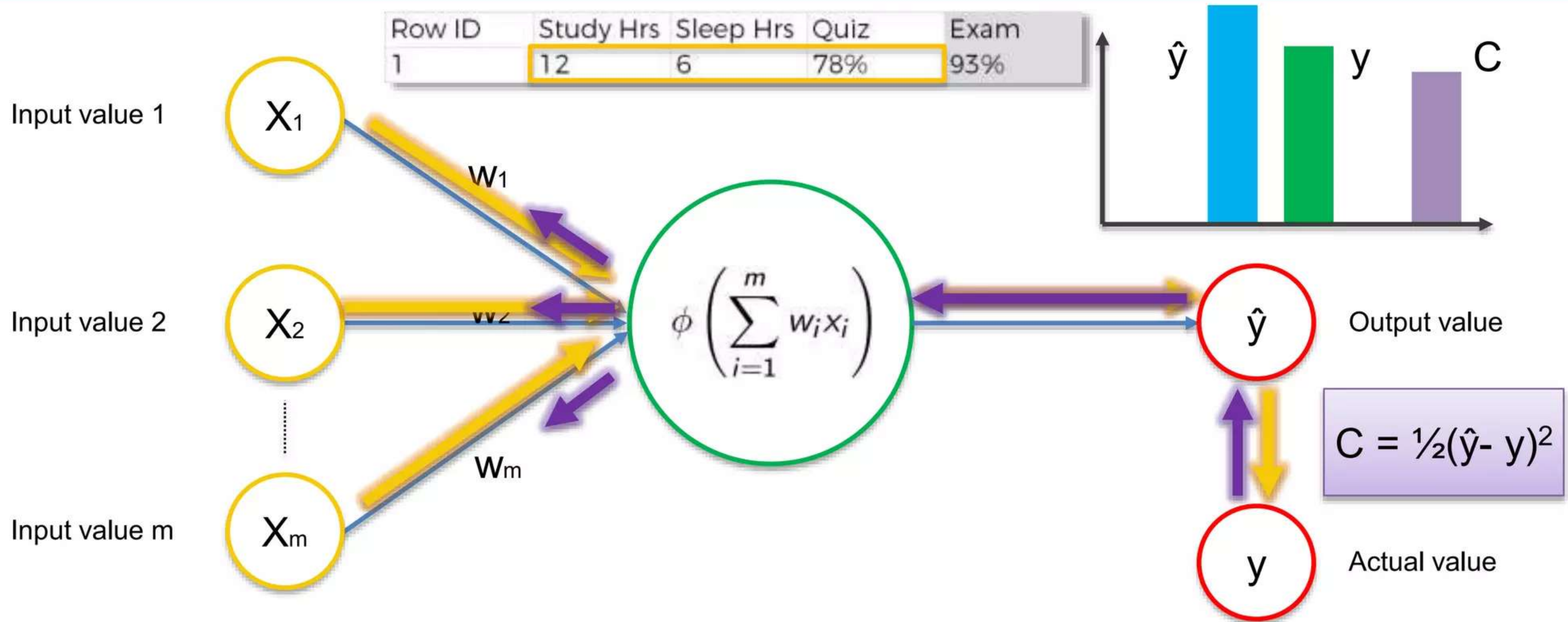
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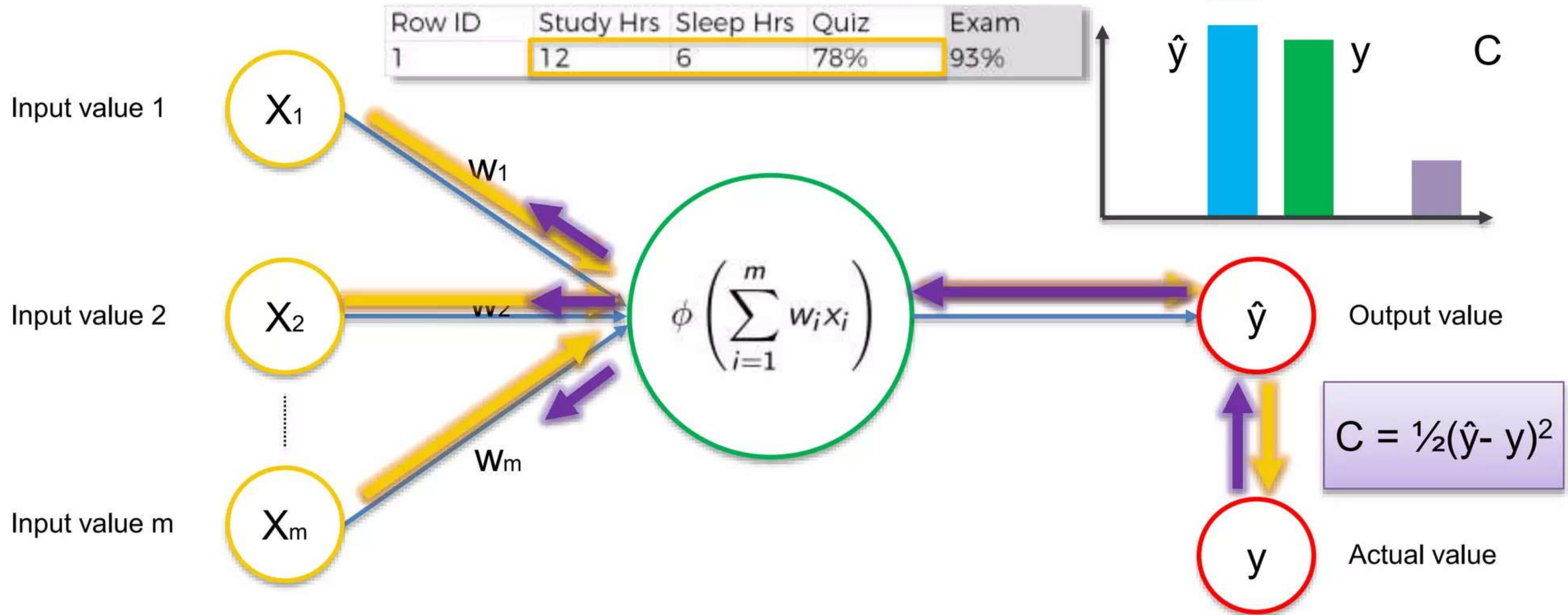
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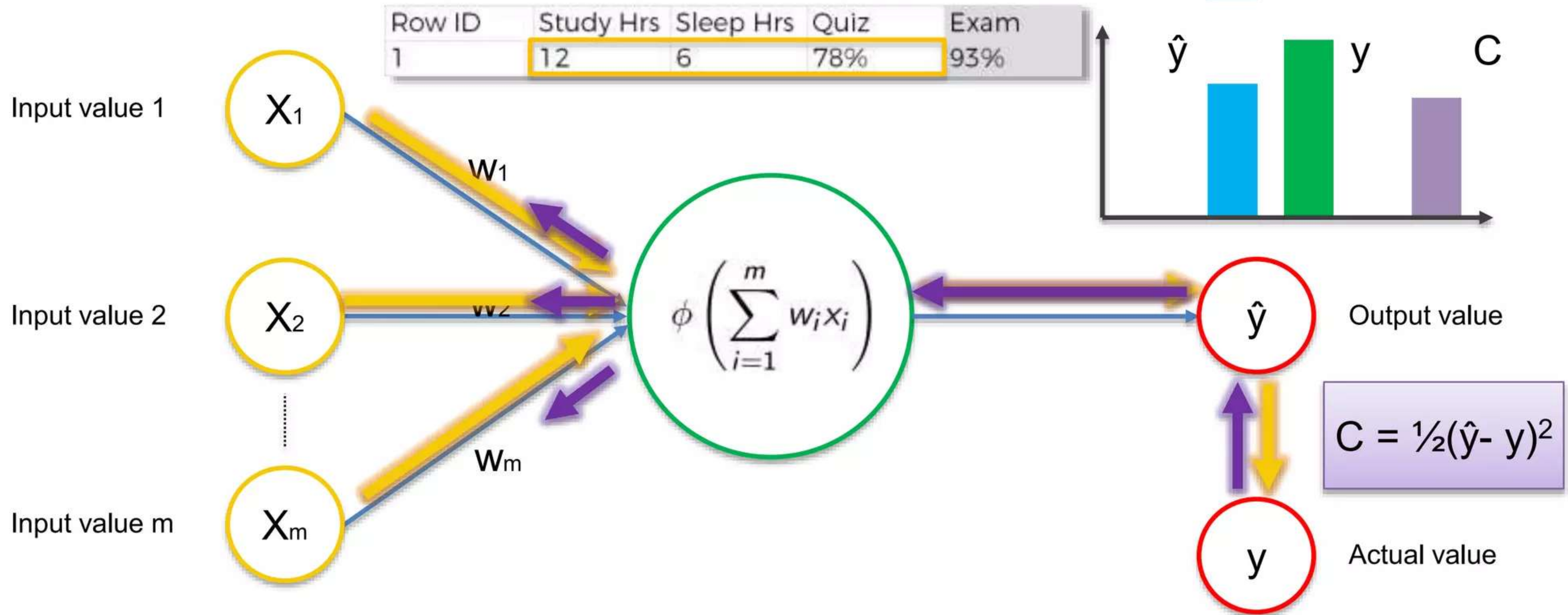
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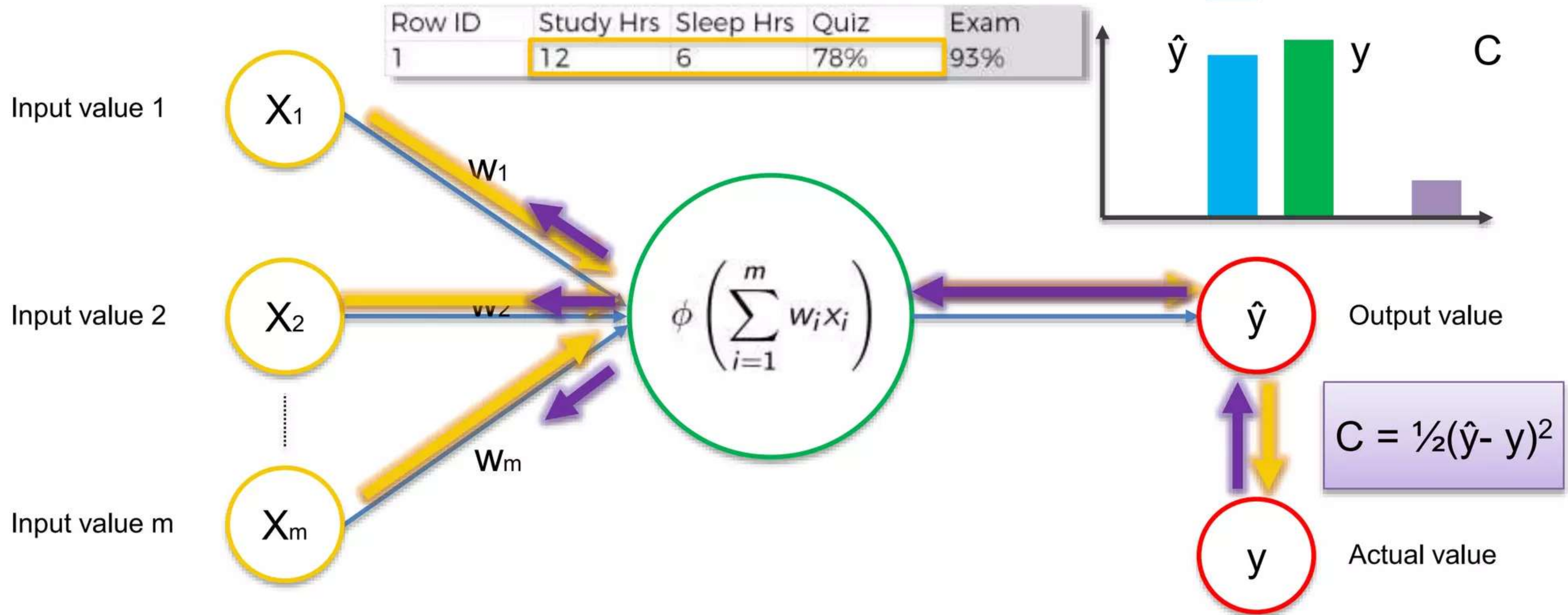
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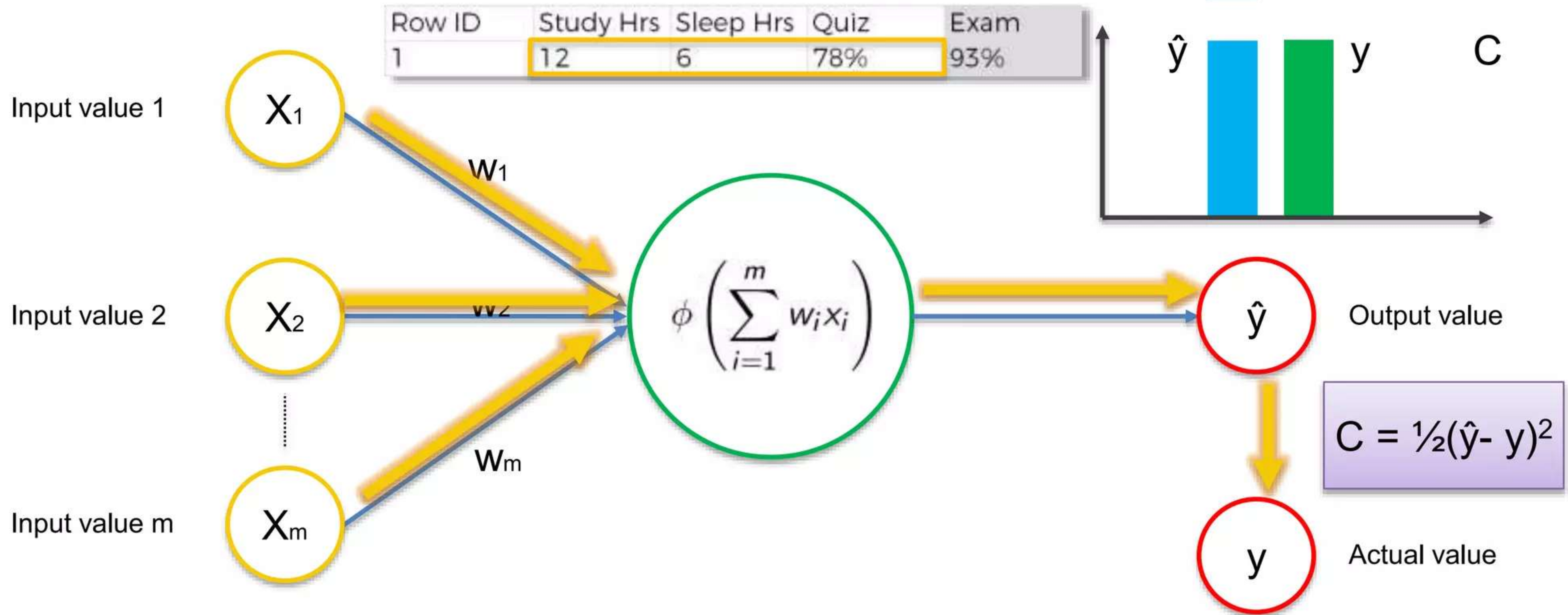
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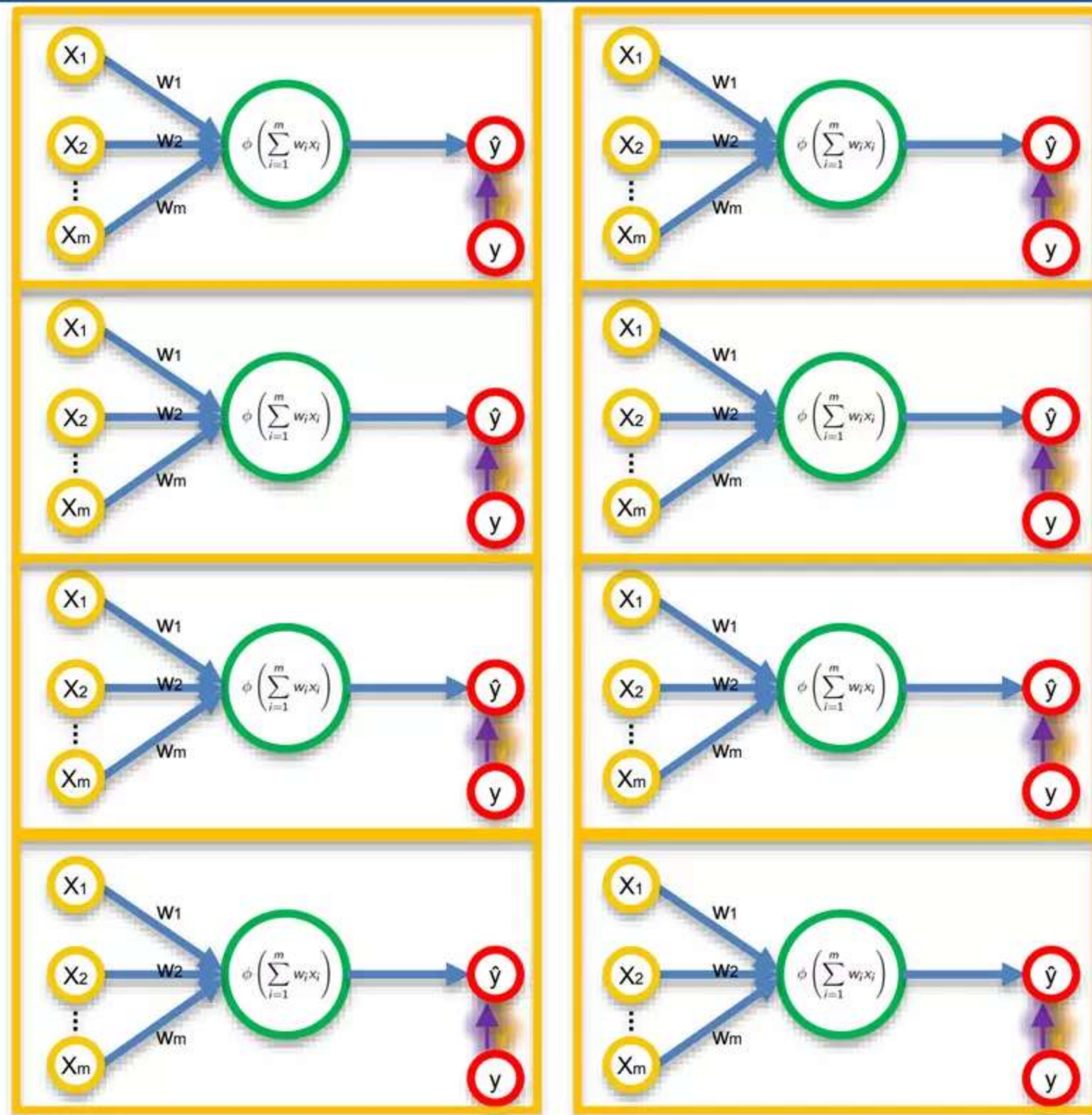
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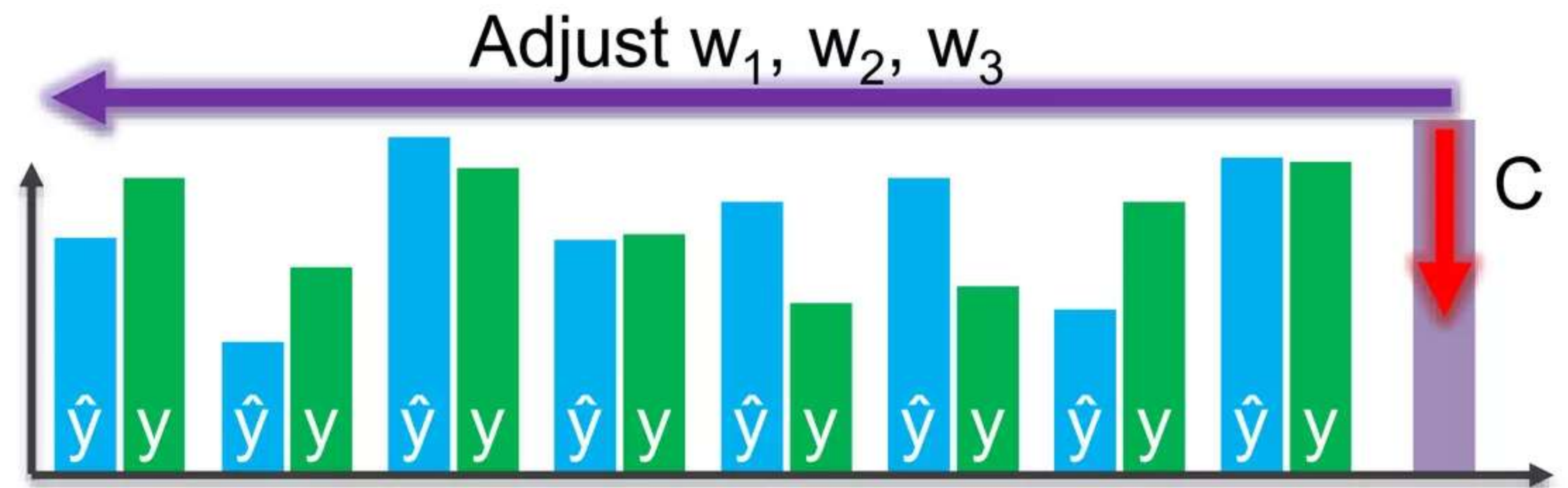


How do Neural Networks learn?



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$$C = \sum \frac{1}{2}(\hat{y} - y)^2$$



How do Neural Networks learn?

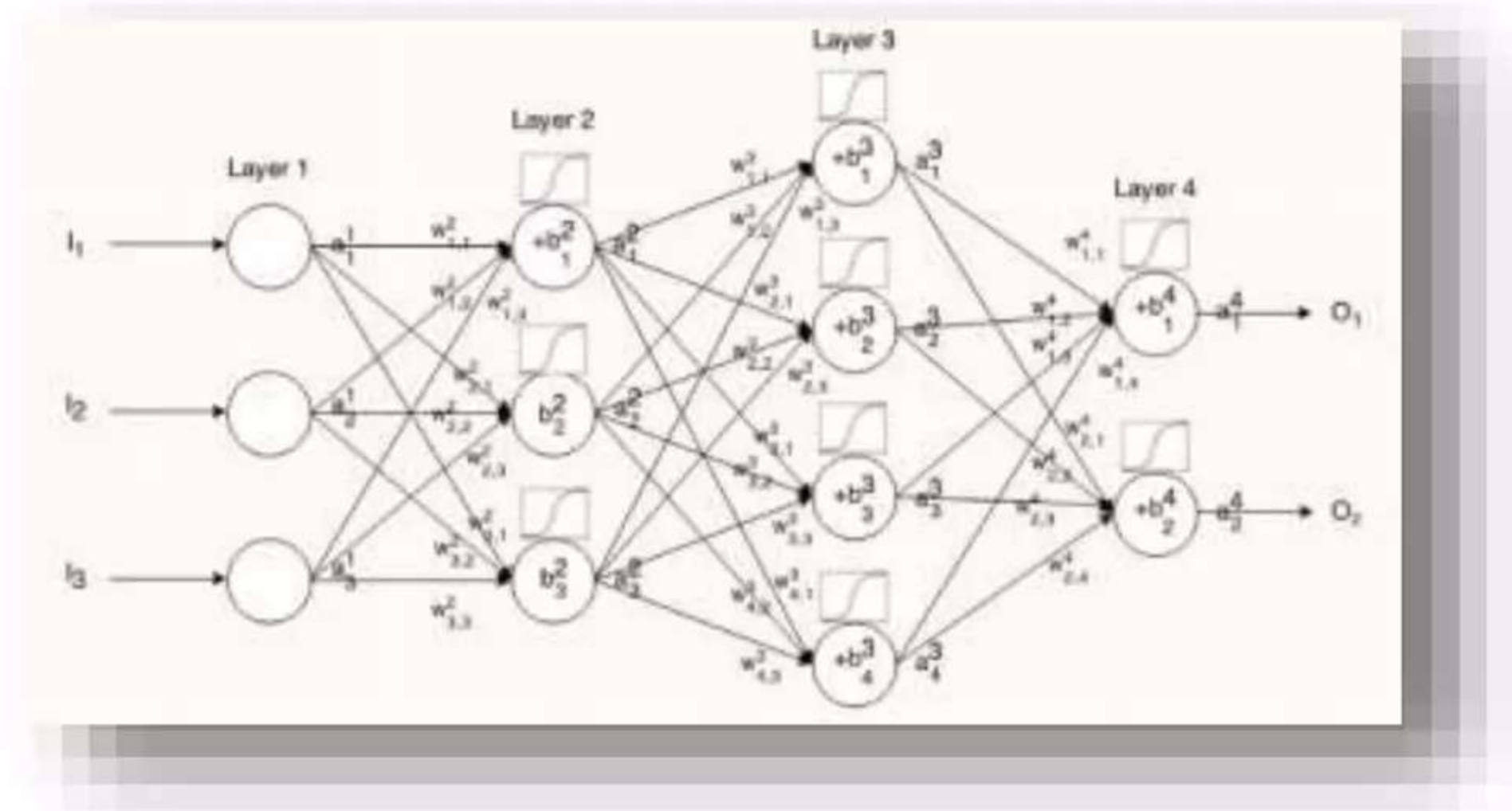
Additional Reading:

A list of cost functions used in neural networks, alongside applications

CrossValidated (2015)

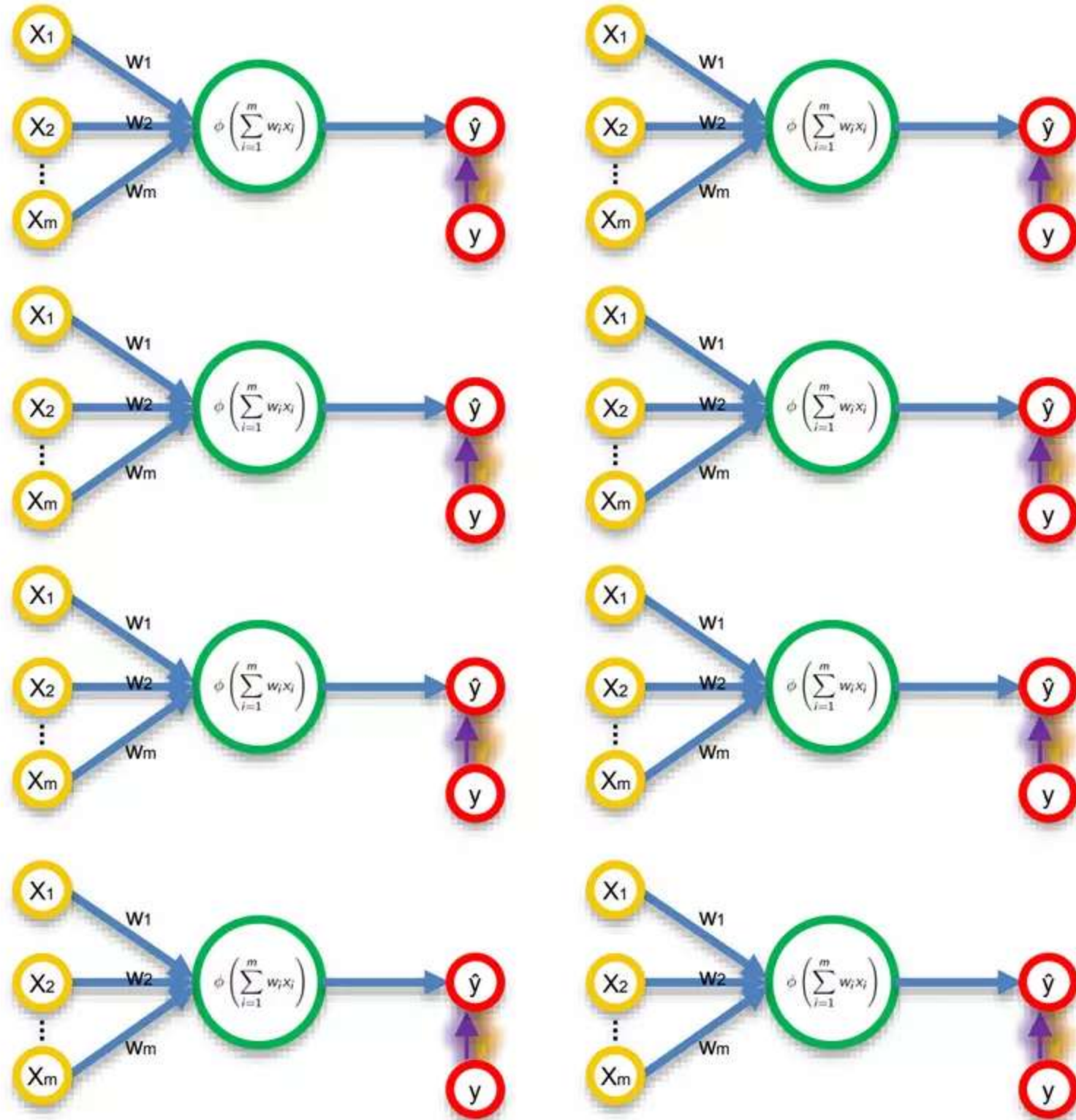
Link:

<http://stats.stackexchange.com/questions/154879/a-list-of-cost-functions-used-in-neural-networks-alongside-applications>



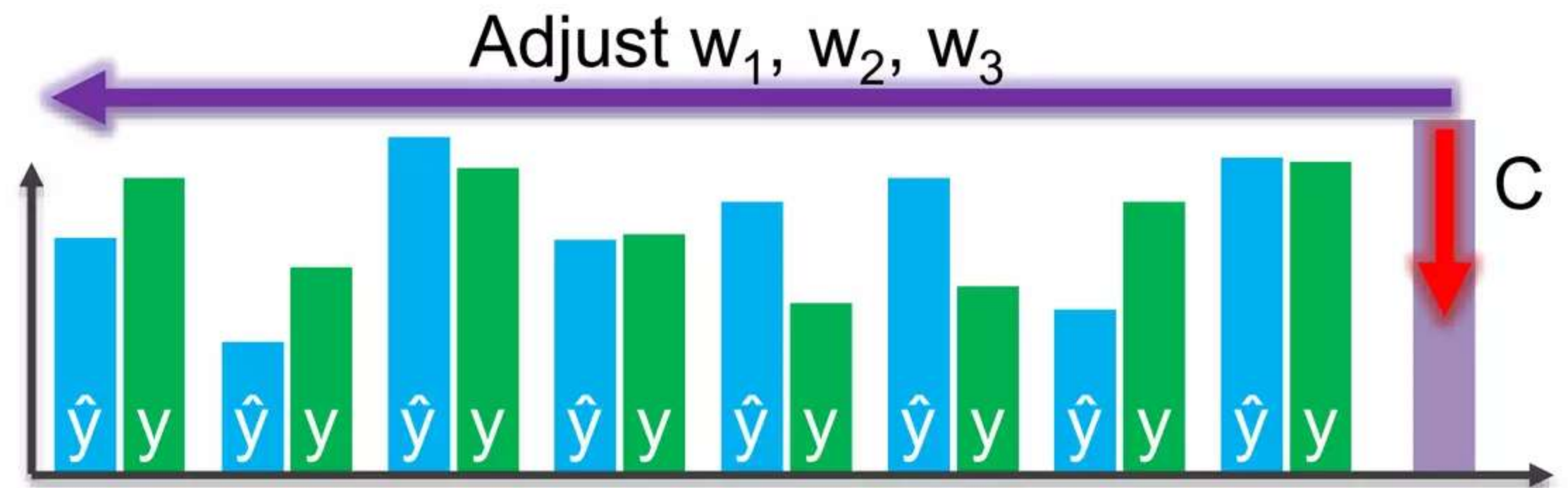
Gradient Descent

Gradient Descent

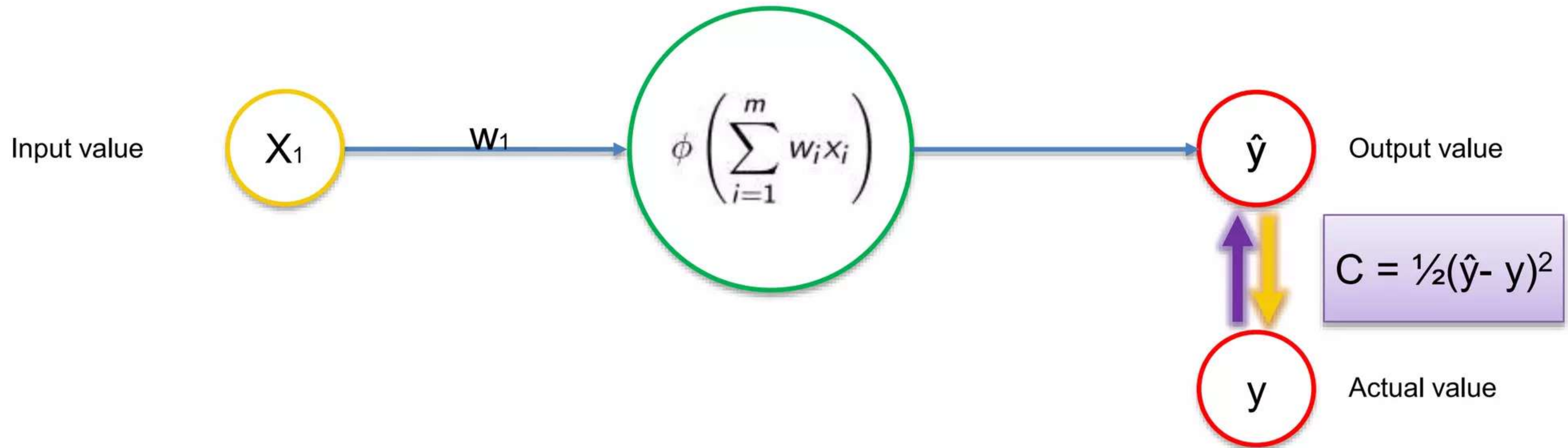


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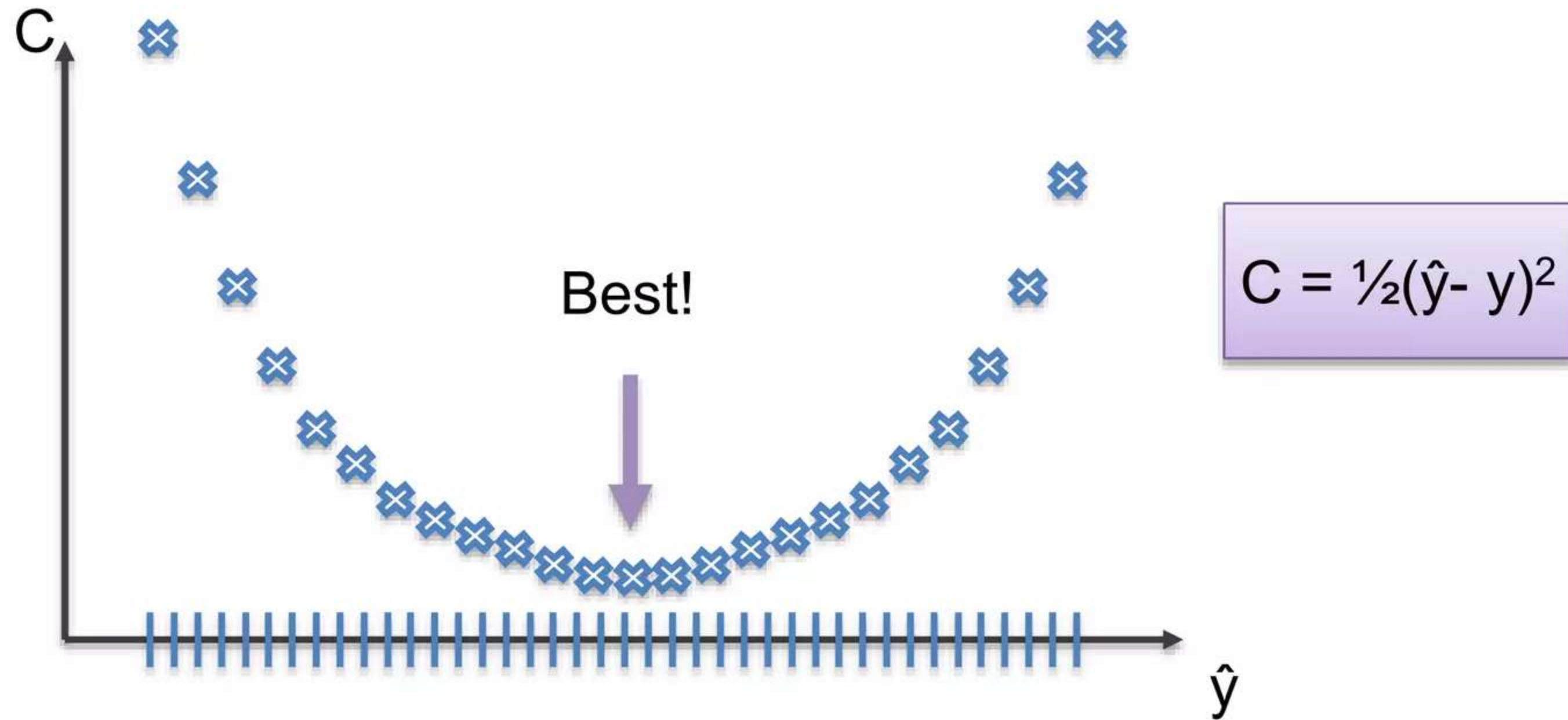
$$C = \sum \frac{1}{2}(\hat{y} - y)^2$$



Gradient Descent



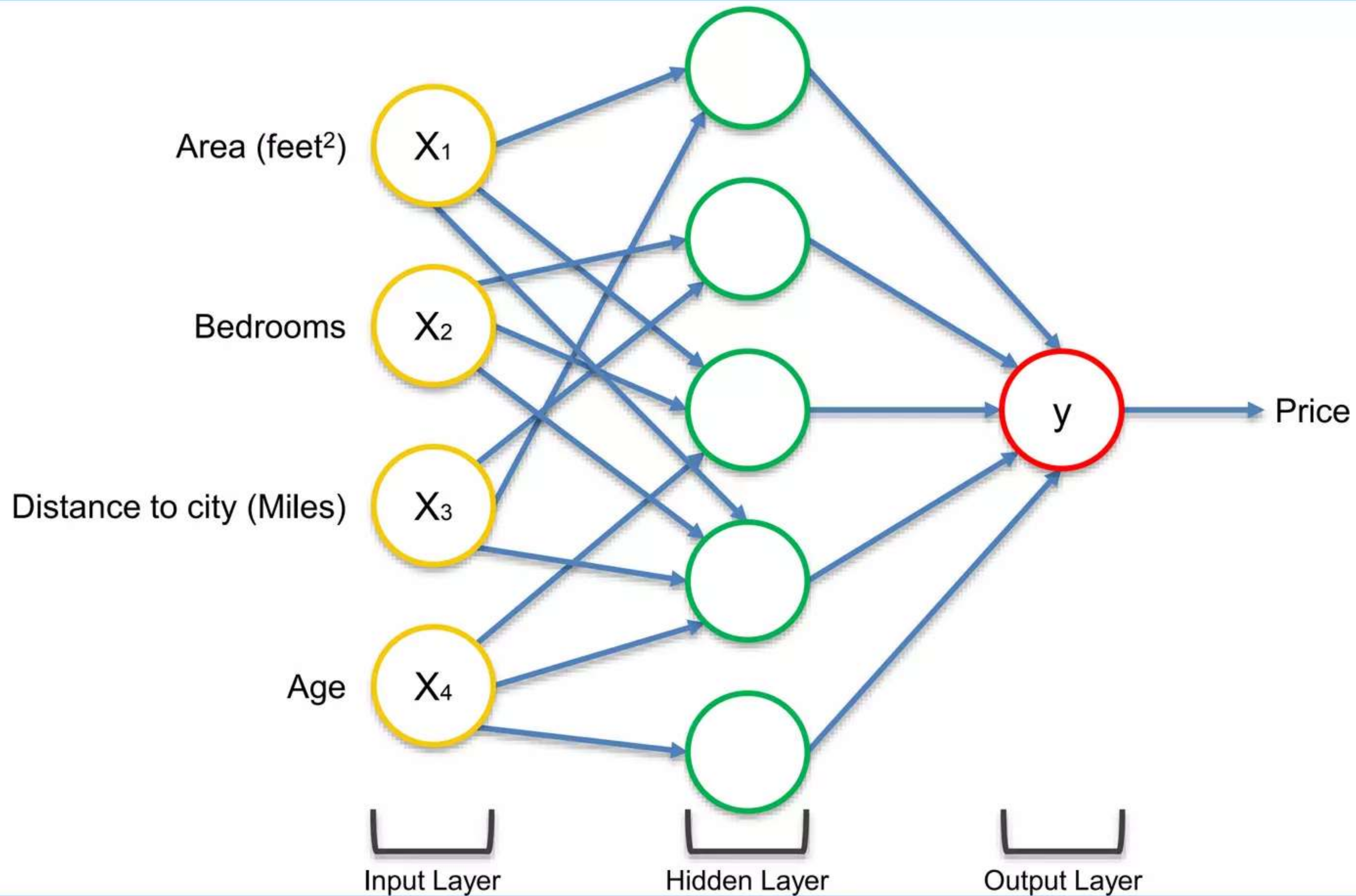
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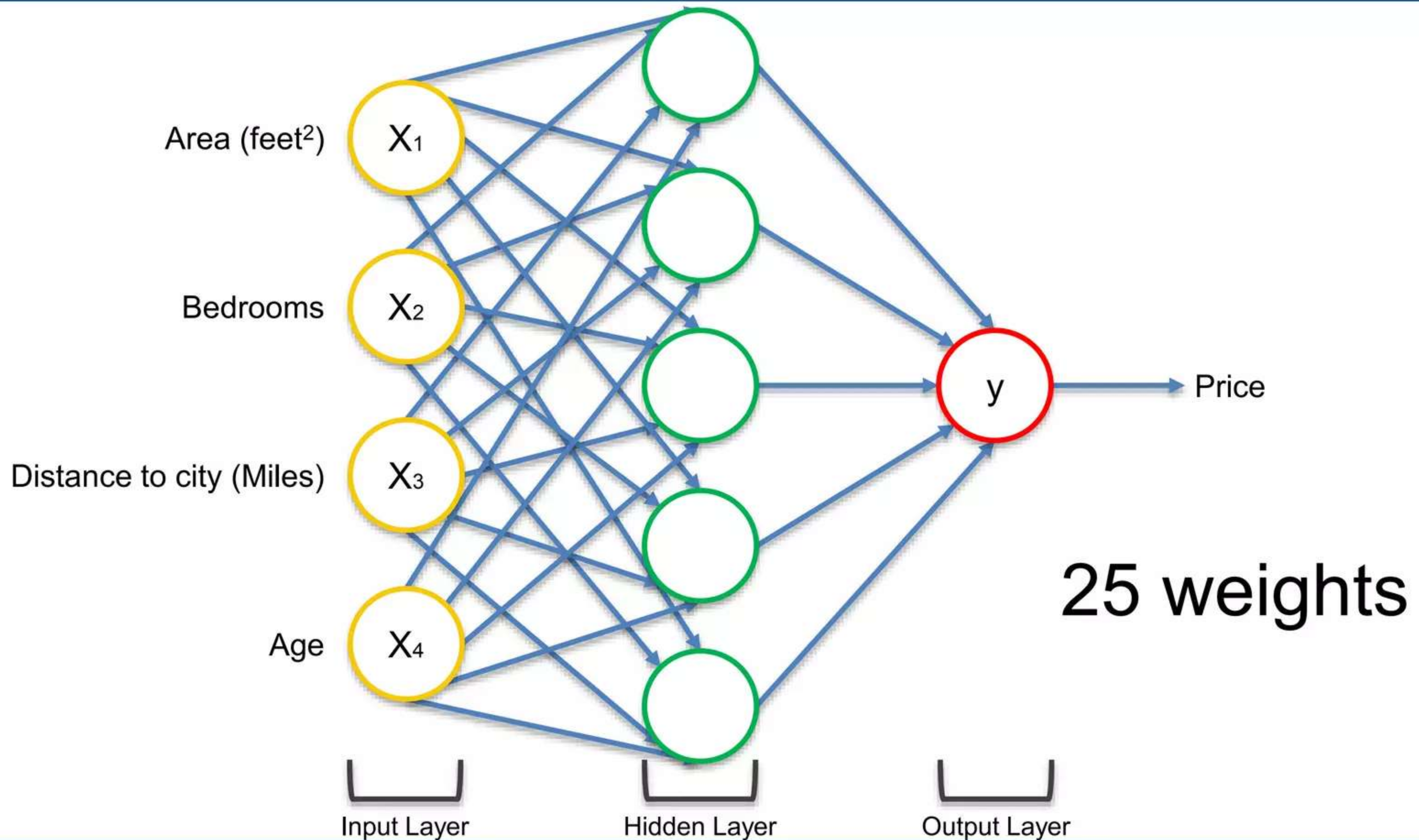
Gradient Descent

Curse of Dimensionality

Gradient Descent



Gradient Descent



Gradient Descent

$1,000 \times 1,000 \times \dots \times 1,000 = 1,000^{25} = 10^{75}$ combinations

Sunway TaihuLight: World's fastest Super Computer

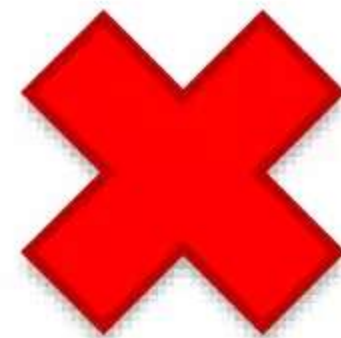
93 PFLOPS

93×10^{15}

$10^{75} / (93 \times 10^{15})$

$= 1.08 \times 10^{58}$ seconds

$= 3.42 \times 10^{50}$ years



Gradient Descent

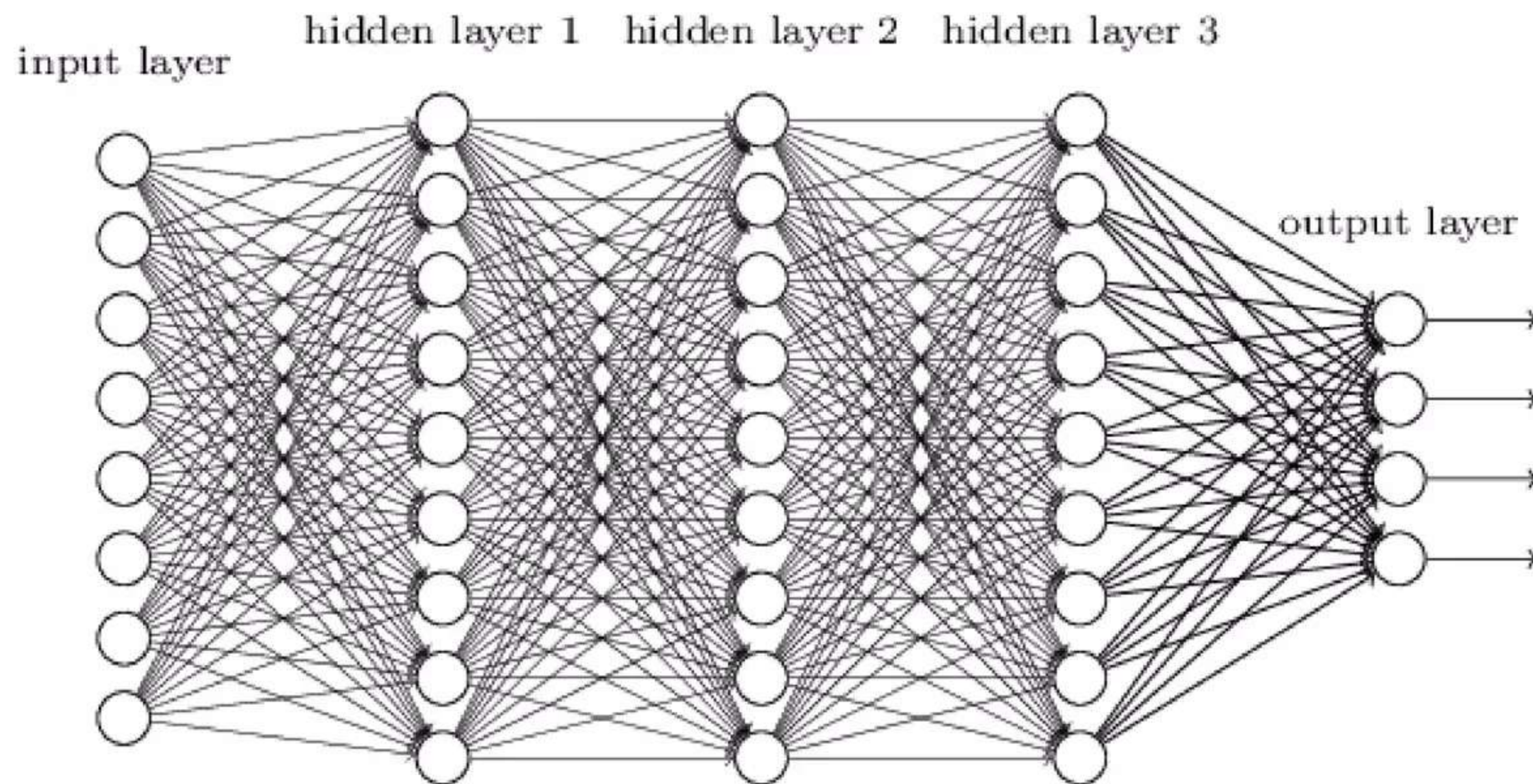
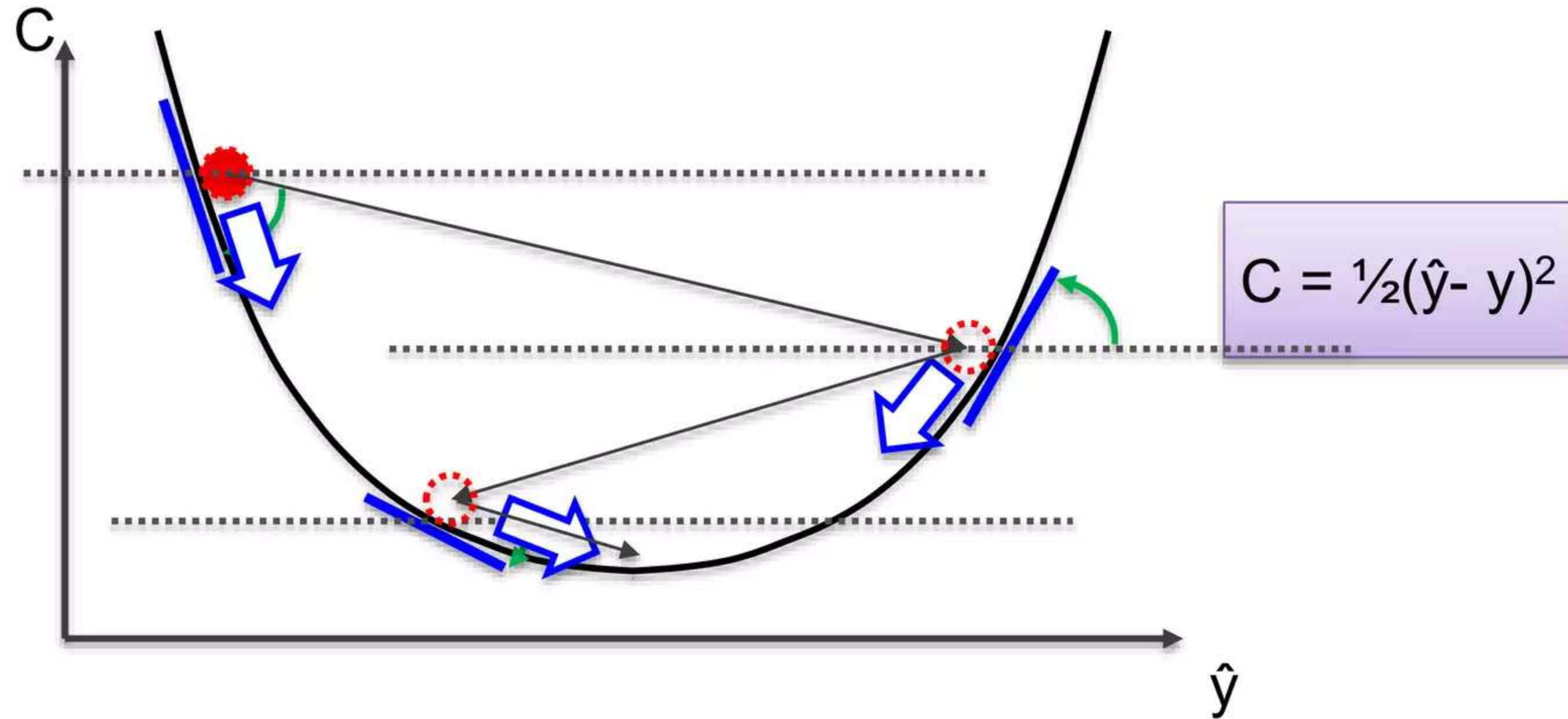


Image Source: neuralnetworksanddeeplearning.com

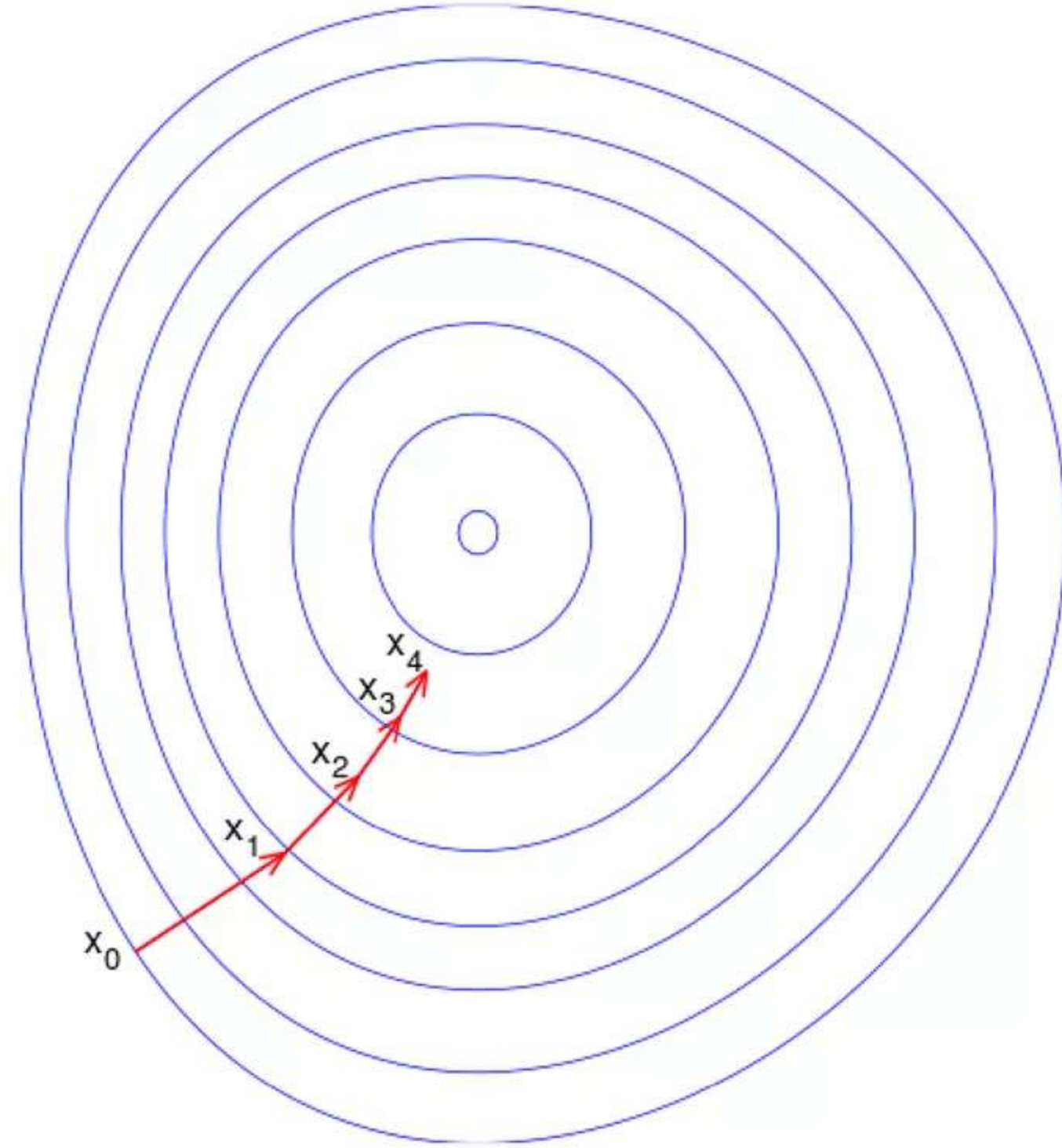
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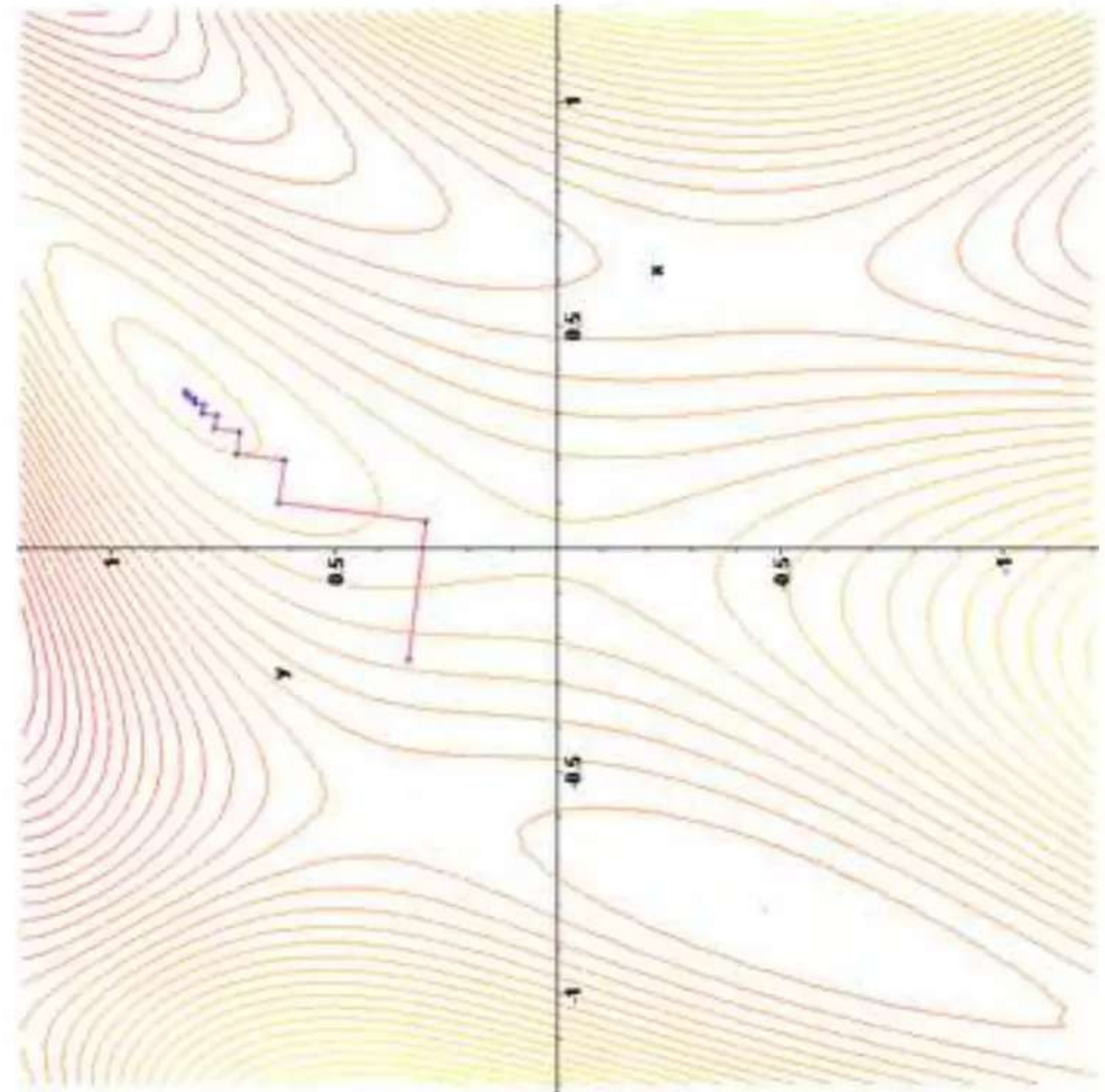
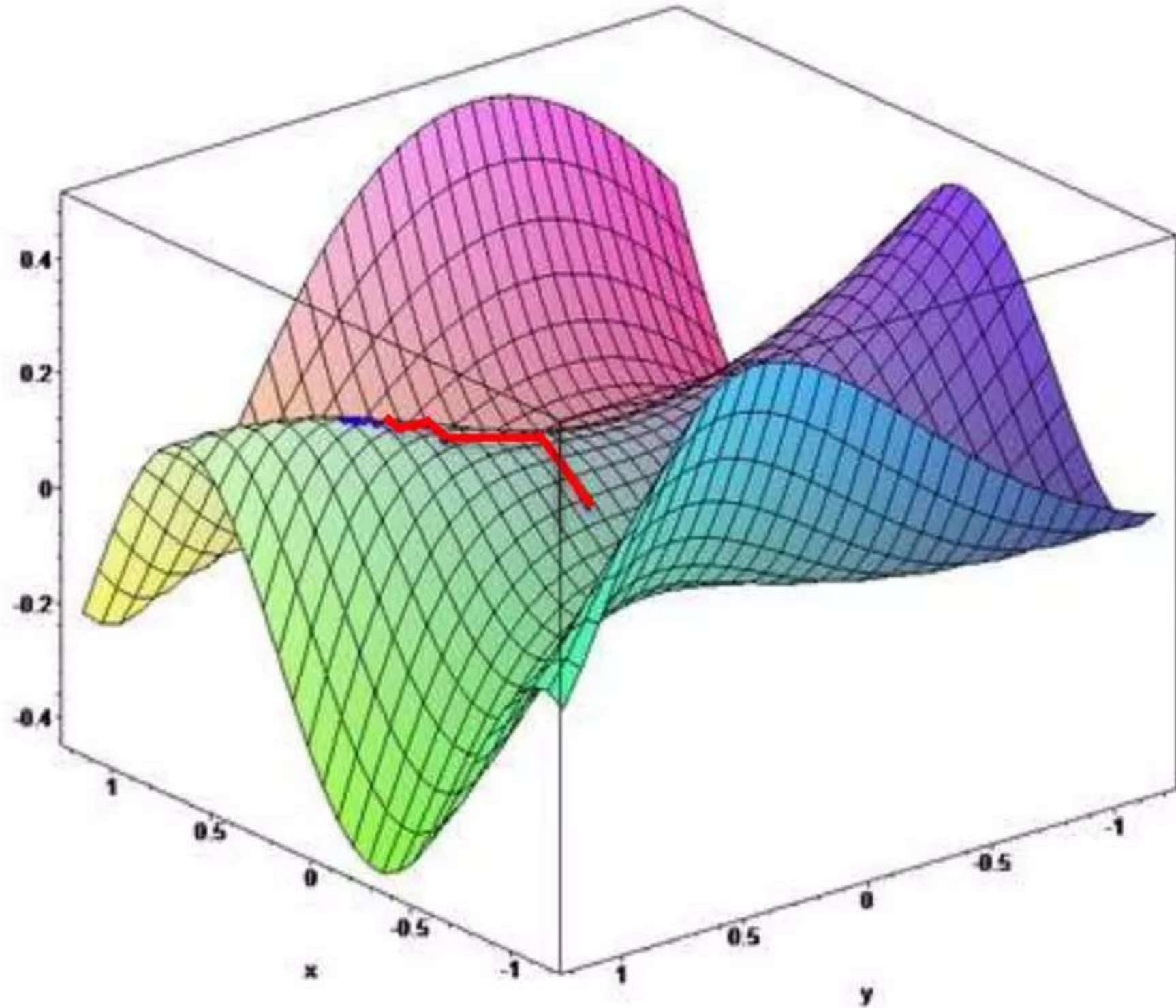
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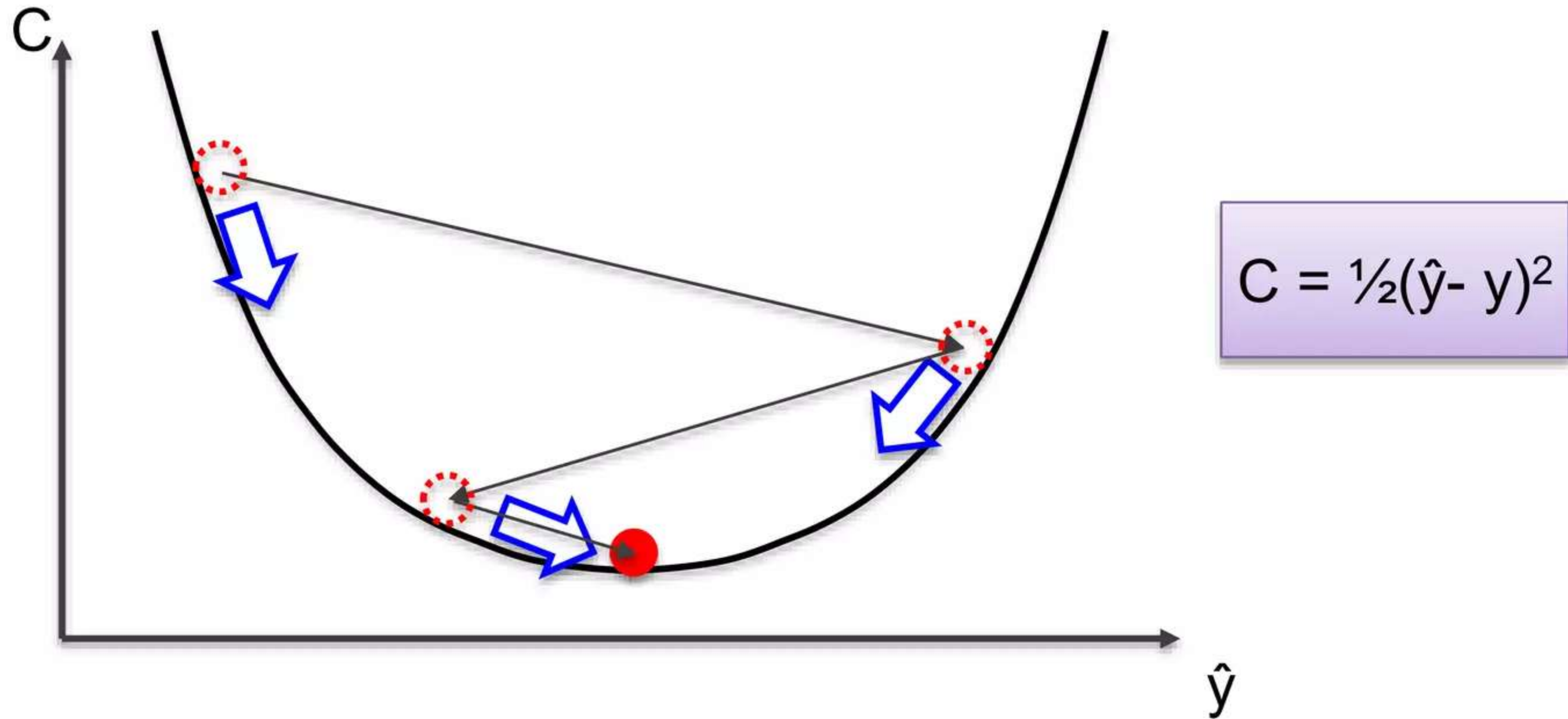


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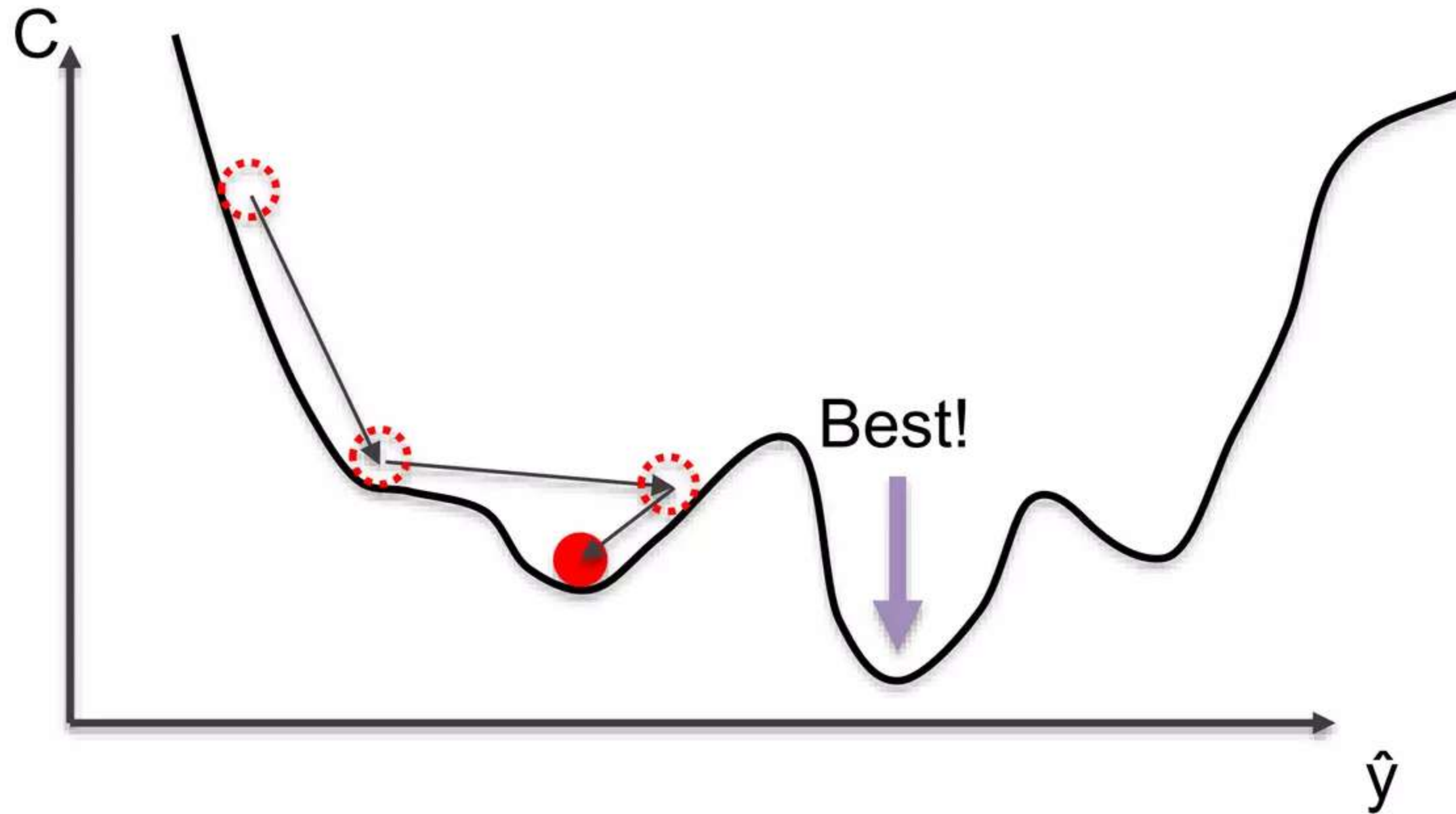


Stochastic Gradient Descent

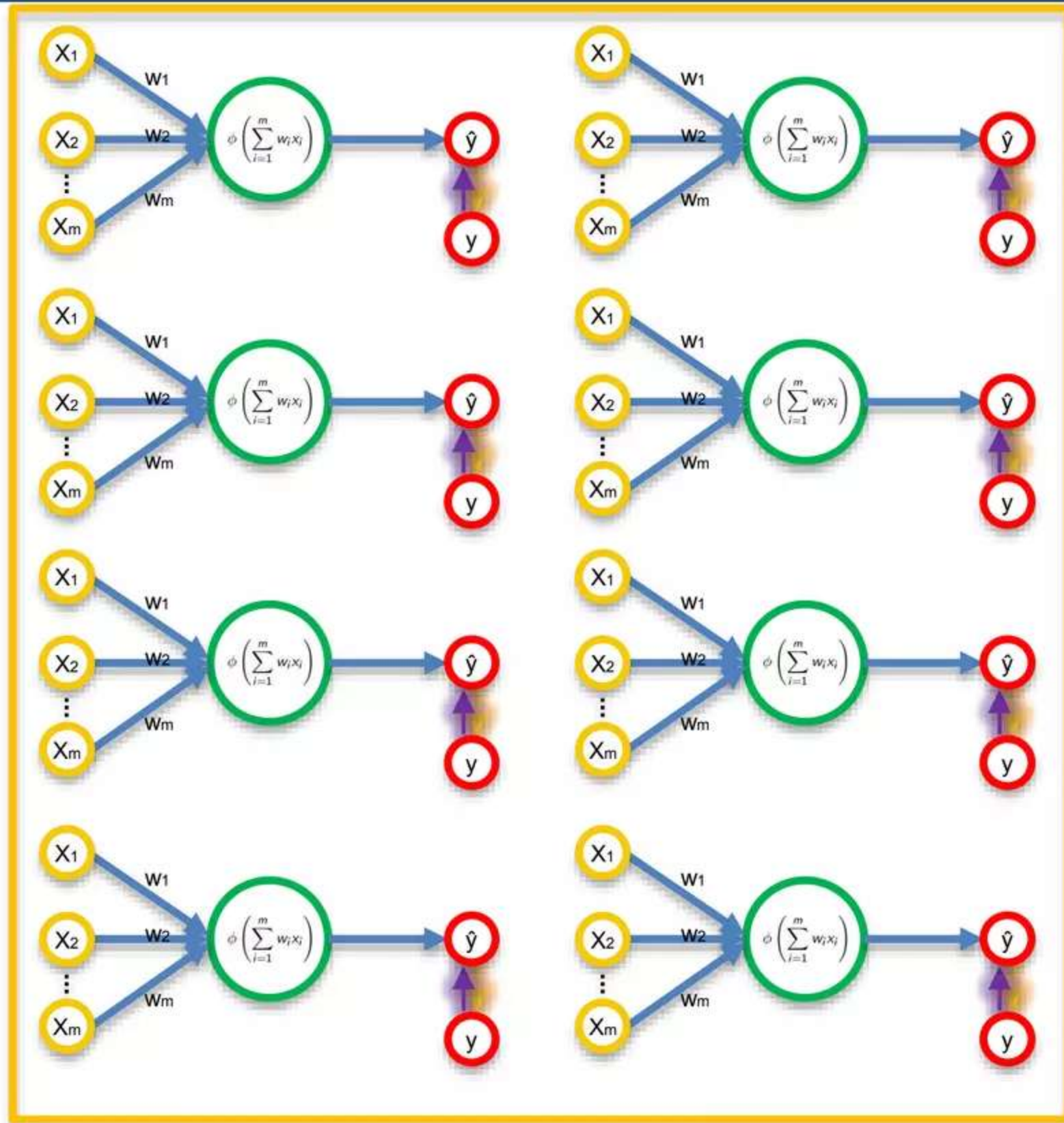
Stochastic Gradient Descent



Stochastic Gradient Descent

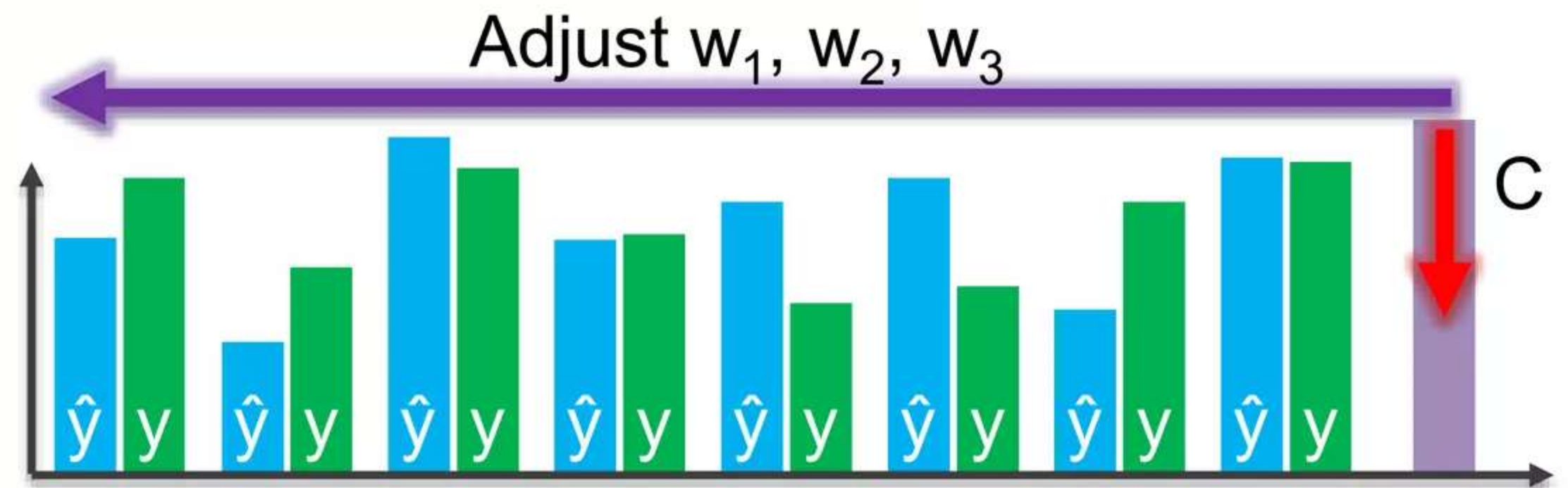


Stochastic Gradient Descent

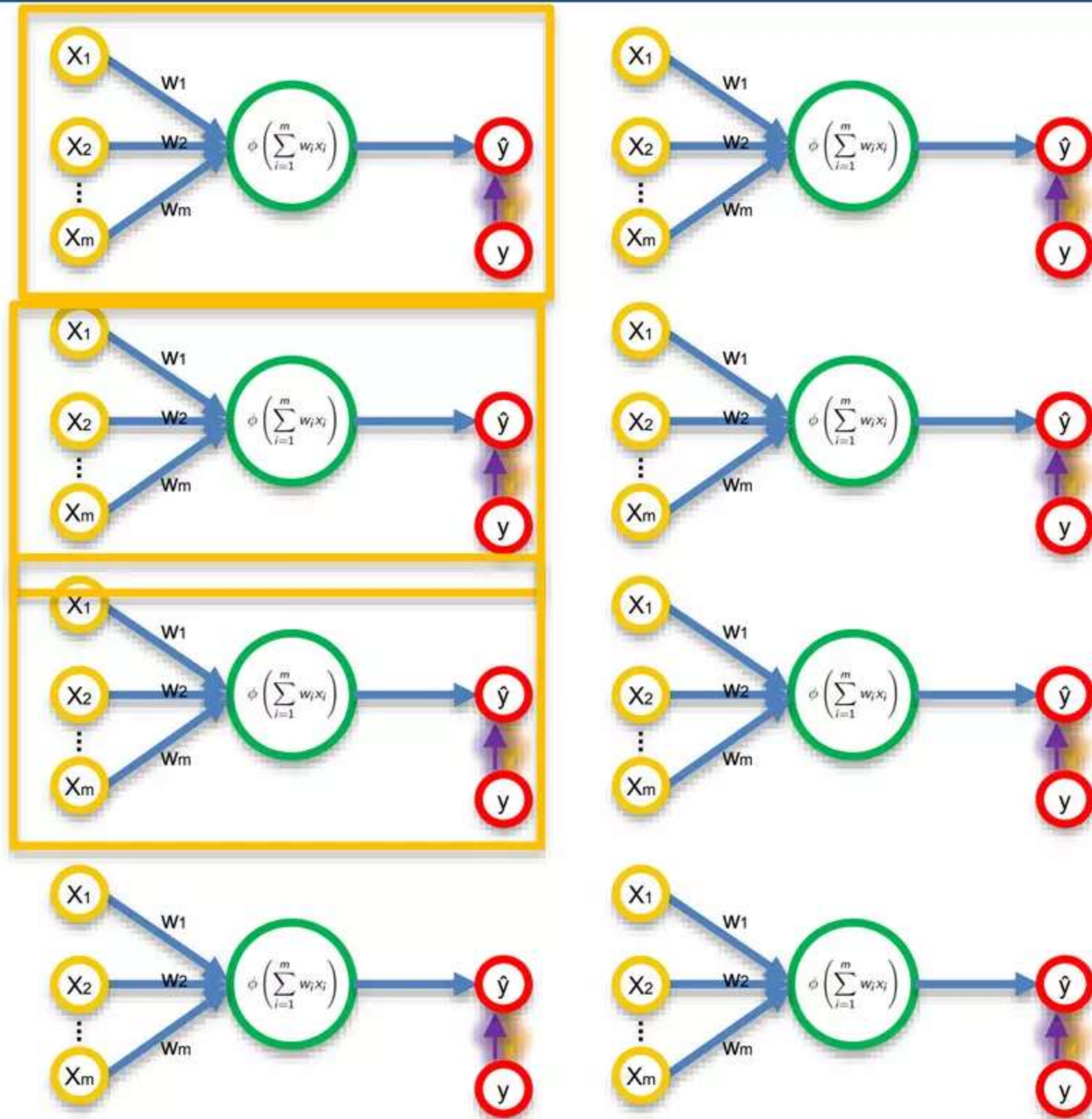


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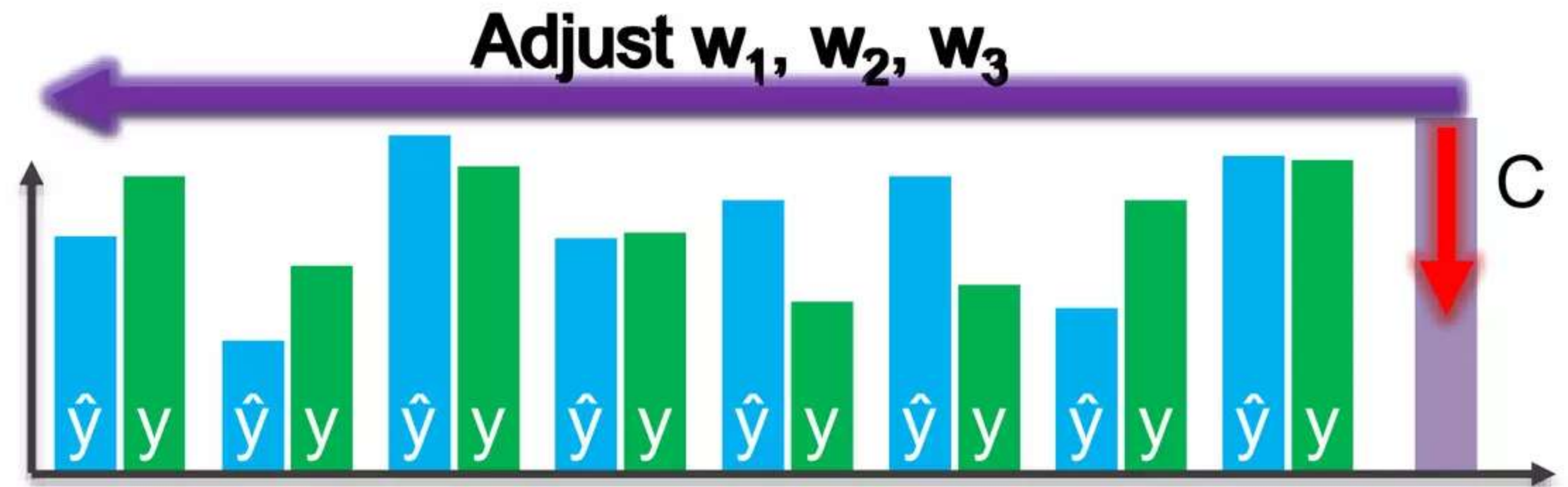


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Stochastic Gradient Descent

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Upd w's

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Upd w's
Upd w's
Upd w's
Upd w's
Upd w's
Upd w's
Upd w's
Upd w's

Batch
Gradient
Descent

Stochastic
Gradient
Descent

Stochastic Gradient Descent

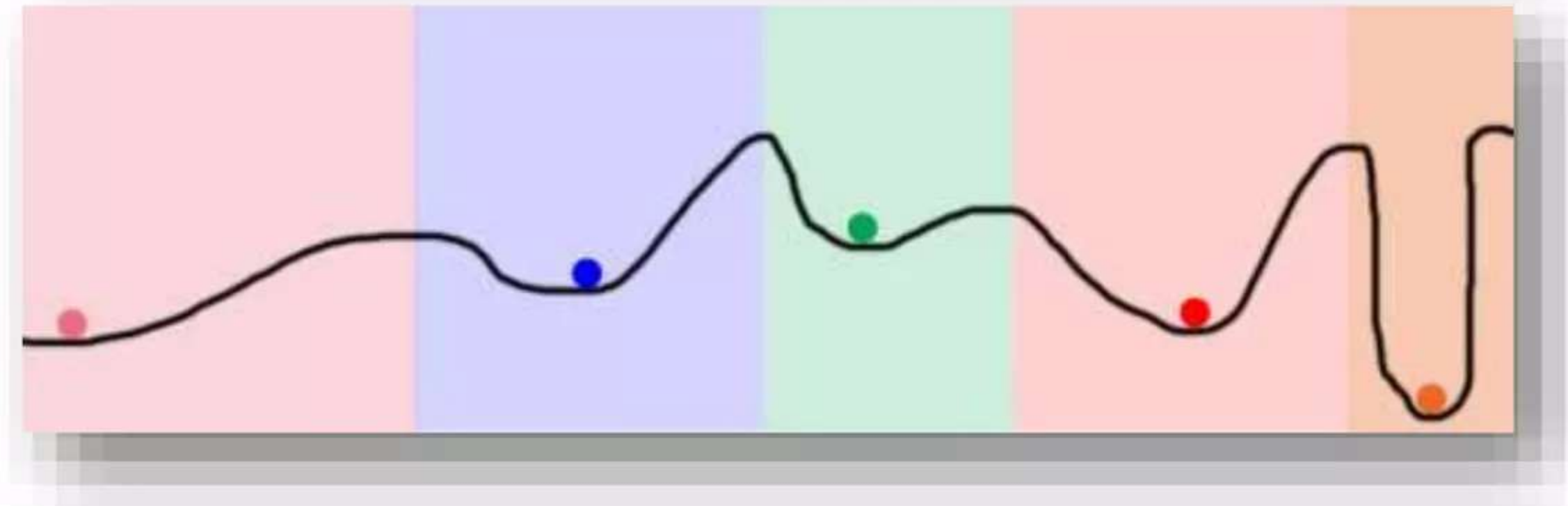
Additional Reading:

A Neural Network in 13 lines of Python (Part 2 - Gradient Descent)

Andrew Trask (2015)

Link:

<https://iamtrask.github.io/2015/07/27/python-network-part2/>



Stochastic Gradient Descent

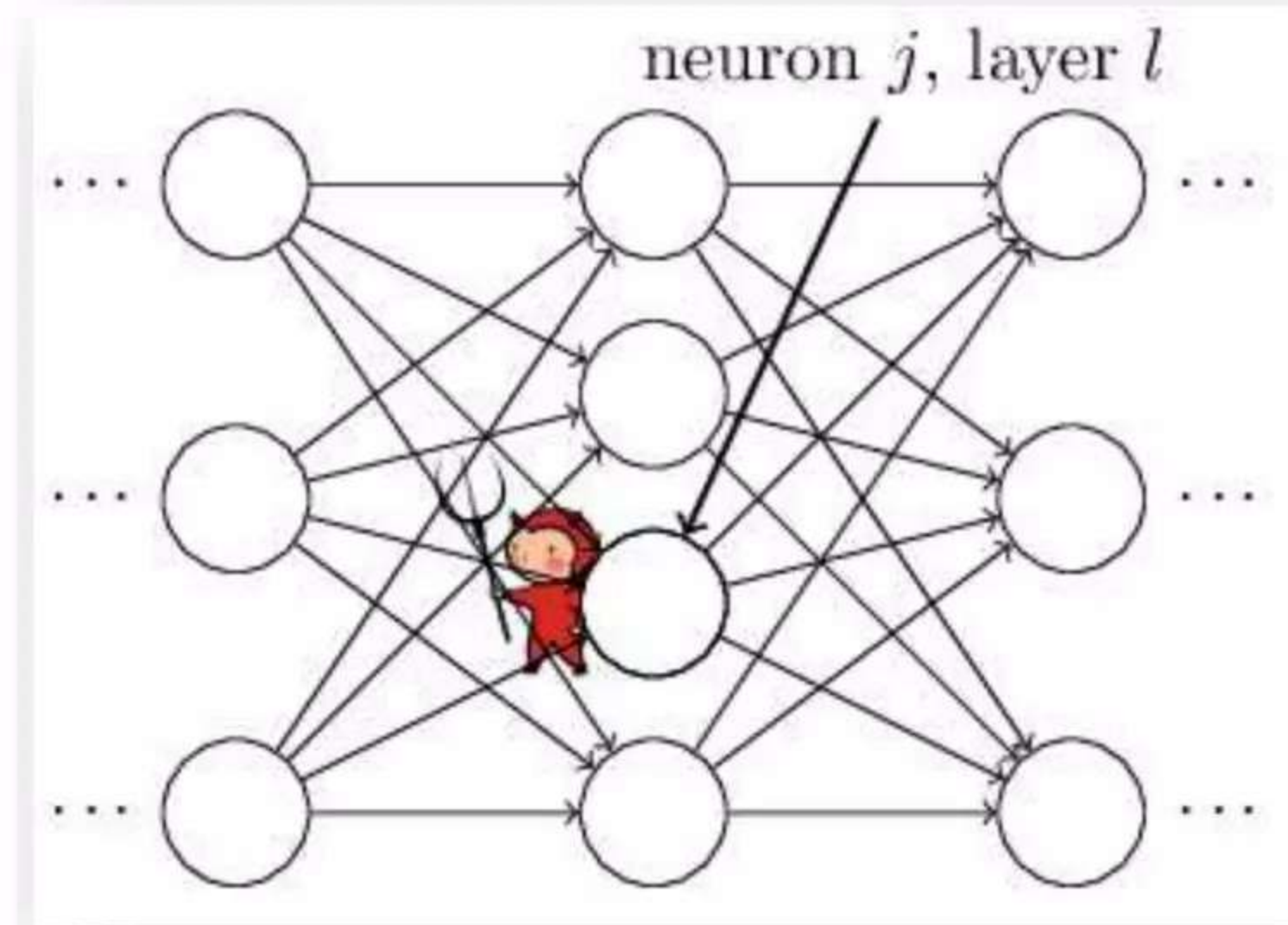
Additional Reading:

Neural Networks and Deep Learning

Michael Nielsen (2015)

Link:

<http://neuralnetworksanddeeplearning.com/chap2.html>



Backpropagation

Gradient Descent

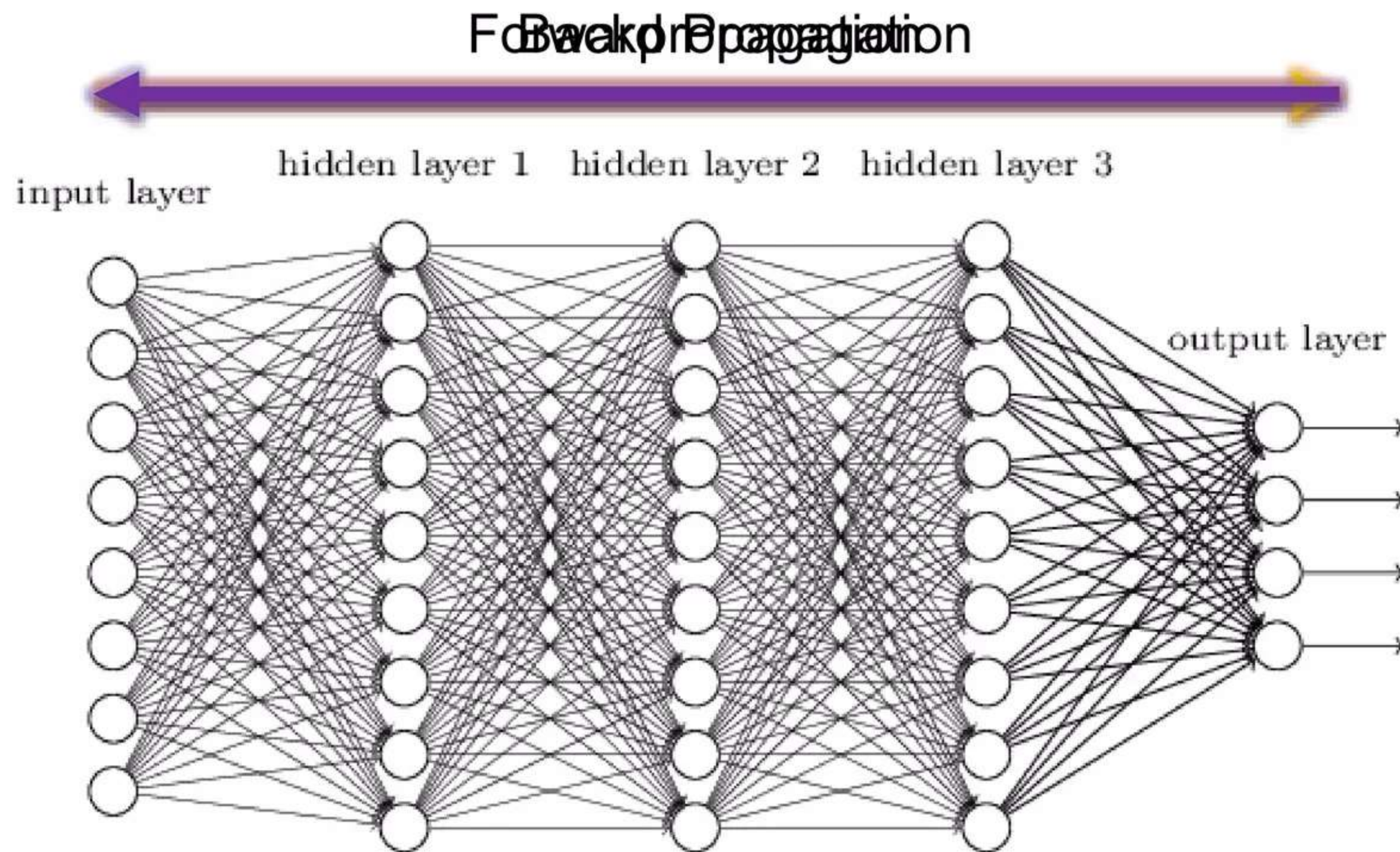


Image Source: neuralnetworksanddeeplearning.com

Stochastic Gradient Descent

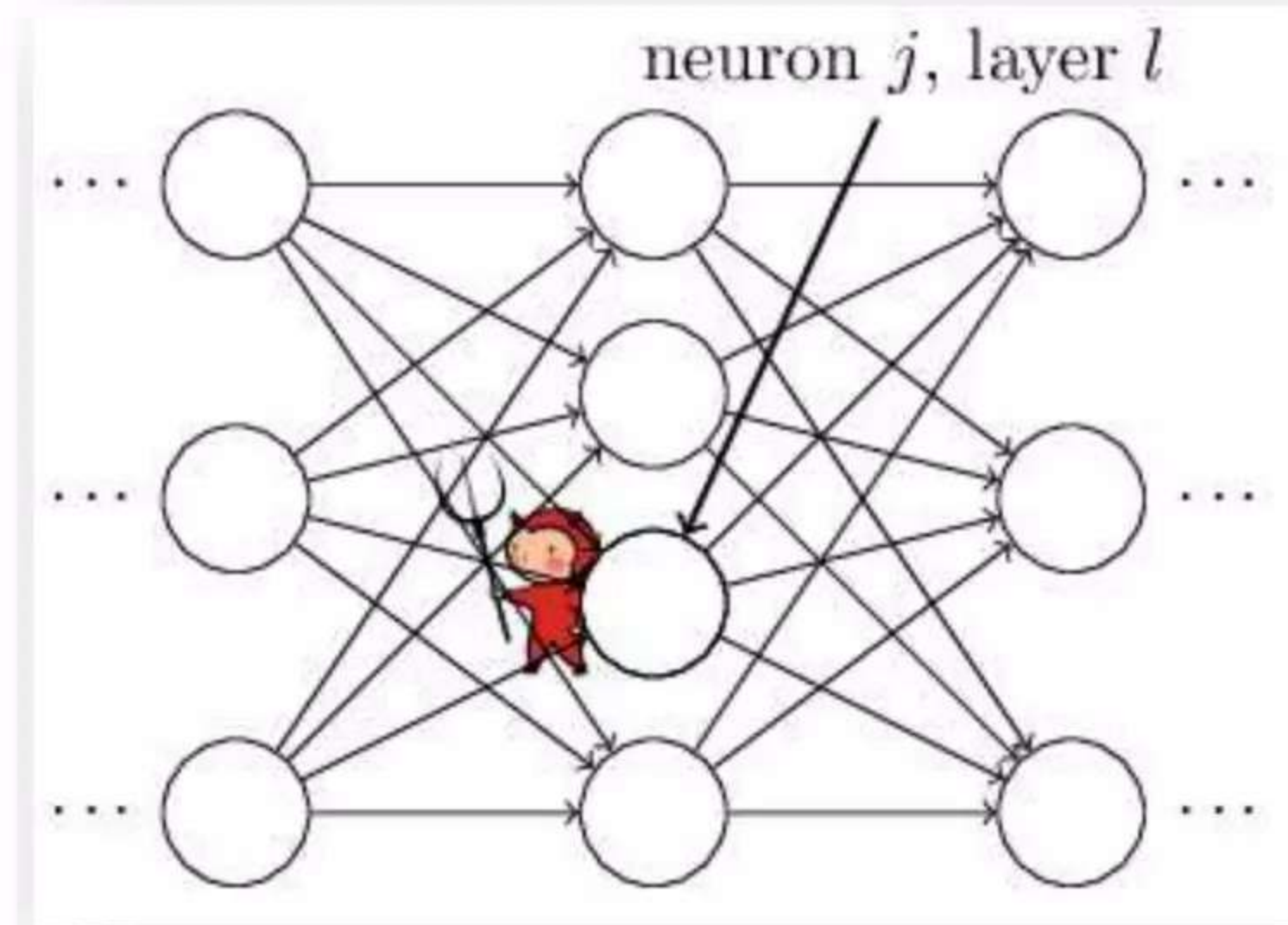
Additional Reading:

Neural Networks and Deep Learning

Michael Nielsen (2015)

Link:

<http://neuralnetworksanddeeplearning.com/chap2.html>



Training the ANN with Stochastic Gradient Descent

STEP 1: Randomly initialise the weights to small numbers close to 0 (but not 0).



STEP 2: Input the first observation of your dataset in the input layer, each feature in one input node.



STEP 3: Forward-Propagation: from left to right, the neurons are activated in a way that the impact of each neuron's activation is limited by the weights. Propagate the activations until getting the predicted result y .



STEP 4: Compare the predicted result to the actual result. Measure the generated error.



STEP 5: Back-Propagation: from right to left, the error is back-propagated. Update the weights according to how much they are responsible for the error. The learning rate decides by how much we update the weights.



STEP 6: Repeat Steps 1 to 5 and update the weights after each observation (Reinforcement Learning). Or:
Repeat Steps 1 to 5 but update the weights only after a batch of observations (Batch Learning).



STEP 7: When the whole training set passed through the ANN, that makes an epoch. Redo more epochs.